Clinical Approach to Headache

Christina Kinnevey, MD
Greg Gayer, PhD

Introduction

Video 1: Dr. Kinnevey

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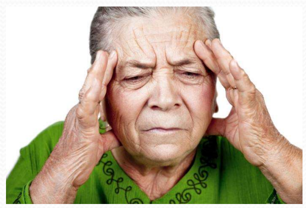


Objectives

1. Identify	Identify Signs, Symptoms and classification (IHS) of the primary and secondary headaches and trigeminal neuralgia
2. Describe	Describe treatments- both preventative and acute for primary and secondary headaches and trigeminal neuralgia • Nonpharmacologic • pharmacologic
3. Delineate	Delineate appropriate work-up including history, physical exam, tests and imaging for the patient with a headache
4. Understand	Understand the basic pathophysiology of both primary and secondary headaches
5. Define	Define signs and symptoms that are red flags for headache
6. Outline	Outline important anatomy pertaining to patients with headache
7. Consider	osteopathic implications in the treatment of primary and secondary headaches

1. Diagnosis: a. Differentiate between primary headaches (e.g., tension-type headaches, migraines, cluster headaches) and secondary headaches (e.g., elevated intracranial pressure, infection, hemorrhage) based on clinical presentation and history. b. Recognize the clinical features and diagnostic criteria of migraines, including both migraine with aura and migraine without aura. c. Identify the clinical features of other primary headaches and secondary headaches that may require urgent evaluation, such as thunderclap headaches or headaches associated with neurological deficits. d. Conduct a focused neurological examination to assess for signs of

elevated intracranial pressure, focal neurological deficits, or other concerning neurological symptoms.

2. Treatment: a. Understand the stepwise approach to managing acute migraines, including the use of nonsteroidal anti-inflammatory drugs (NSAIDs), triptans, antiemetics, and rescue medications. b. Learn the principles of managing chronic migraines and the role of preventive medications, lifestyle modifications, and behavioral therapies. c. Recognize the importance of prompt and appropriate treatment for secondary headaches caused by underlying conditions such as infections or bleeding. d. Understand the use of analgesics, osmotic agents, and other medications in managing elevated intracranial pressure.
2. Management: a. Develop skills in creating a patient-centered management plan, considering the frequency and severity of headaches, impact on daily activities, and patient preferences. b. Learn to counsel patients on self-management strategies, including stress reduction, sleep hygiene, dietary adjustments, and trigger avoidance. c. Understand the indications for neuroimaging and other diagnostic tests in patients with severe or atypical headaches to rule out serious underlying causes. d. Recognize the importance of interdisciplinary collaboration in managing complex headache cases, involving neurologists, neurosurgeons, radiologists,

and other specialists when necessary.

2. Follow-up: a. Learn to monitor treatment efficacy through regular follow-up appointments and patient feedback. b. Understand the importance of medication adherence and potential side effect management during follow-up visits. c. Develop skills in recognizing treatment failure, medication overuse headaches, and medication side effects that may require treatment adjustments.
3. Emergent Issues: a. Identify red flag symptoms and neurological signs that may indicate elevated intracranial pressure or other serious underlying conditions requiring immediate intervention. b. Learn to differentiate between benign headaches and those requiring urgent evaluation and intervention. c. Understand the steps to manage emergent headaches, including the use of intravenous medications, neurosurgical interventions, and critical care support.
4. Patient Communication: a. Enhance communication skills to effectively elicit the patient's headache history and understand the impact of headaches on their daily life. b. Learn to provide clear and empathetic explanations of diagnoses, treatment plans, and potential outcomes to patients and their families.
5. Ethical Considerations: a. Gain awareness of ethical considerations in pain management and end-of-life

care for patients with severe or chronic headaches.
b. Understand the importance of respecting patient autonomy and informed consent in headache management decisions.

Headaches
A Real Pain

- Highly prevalent- 52% of the adult world suffers from headaches in general[1]
  - Tension headache: 26%
  - Migraine: 14%
  - Chronic daily headache: 4.6%
- Each day, 15.8% of the world's population had headache
- Severe headaches account for 3.5 million ED visits per year in the U.S. (2.4% of all visits)
- Affects quality of life, disability, work productivity
- Barriers in treatment include improper diagnosis, underestimation of morbidity of conditions and noncompliance with treatment recommendations

1.. Stovner, L.J., Hagen, K., Linde, M. *et al.* The global prevalence of headache: an update, with analysis of the influences of methodological factors on prevalence estimates. *J Headache Pain* **23**, 34 (2022). https://doi.org/10.1186/s10194-022-01402-2

2. Yang S, Orlova Y, Lipe A, Boren M, Hincapie-Castillo JM, Park H, Chang CY, Wilson DL, Adkins L, Lo-Ciganic WH. Trends in the Management of Headache Disorders in US Emergency Departments: Analysis of 2007-2018 National Hospital Ambulatory Medical Care Survey Data. J Clin Med. 2022 Mar 3;11(5):1401. doi: 10.3390/jcm11051401. PMID: 35268492; PMCID: PMC89108682.

Taking a Headache History

- Age, Gender, Ethnicity
- PMHX (comorbid diseases), PSHX, Medications, Allergies
- Family history of headaches, stroke, cardiovascular disease, blood clots
- Social history (drugs, etoh, smoking, work, **stress,** diet- food triggers, sleeping patterns)
- Trauma history
- Onset, Location, radiation, **unilateral or bilateral**, pain quality, pain rating
- Is this the WORST headache ever? How quickly did it come on? Is it getting worse or staying the same?
- Description of pain: throbbing, pressure, sharp, tight band
- How often and how long do they last?
- Associated symptoms**: nausea**, vomiting , neck pain, tinnitus, nasal congestion, rhinnorhea, fever, **photophobia**, phonophobia, osmophobia, polyuria
- Vision: Blurred vision, double vision, eye strain, floaters, eye tearing, due for new glasses prescription?
- Neuro symptoms: numbness, tingling, weakness, gait disturbance, confusion
- Worse with valsalva (bending, lifting, coughing, straining)
- Association with exercise, strenuous activities or sex
- For women – where are they in their **menstrual cycle**
- **Triggers:** associated with stress, menses, certain foods, events preceding headache, changes in sleep, exercise, weight or diet
- Aura: Any symptoms that come before the onset of the headache that indicate the headache is coming
- If there are chronic headaches- are they getting worse in severity or frequency
- What has been trialed before and does it help?
  - What medications (OTC or prescribed)- and how often/how much are they taking. Do they help?
  - Heat, ice, manipulation, injections, physical therapy
- Have they had imaging performed in the past

Focused Physical Exam for a Headache

- Vitals (HR, BP, Temperature)
- HEENT- palpation, sinus palpation, ears, nose, throat, lymph nodes, Chapman's points
- Neuro-
  - Alert and oriented
  - Cranial nerve exam
  - Fundoscopic exam
  - Strength, sensation, reflexes
  - Pronator drift, coordination, gait, romberg
- Vascular
  - Heart, carotids (bruits), temporal artery palpation
- Musculoskeletal- head, neck, thorax, sacrum
  - **Autonomics-**
    - T1-4, superior cervical ganglia (sympathetic)
    - Parasympathetics with CN 5,7
    - Consider viscerosomatic referral pain- vagus nerve
  - **Biomechanics**
    - Cranial motion, Cervical spine motion, Thoracic spine motion, muscular attachments, fascial connections
  - **Circulation**
    - Thoracic inlet, Diaphragm
  - **Screening**
    - Mitchell Model (short leg- sacral dysfunction, postural considerations, upper crossed)
    - Mood, stress

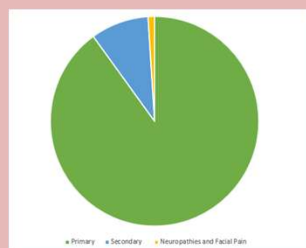
The International Classification of Headache Disorders

Primary Headache

- Headaches due to intrinsic dysfunction in the nervous system
- 90% of total headache cases
- NOT life threatening
- Examples:
  - Tension Type Headache
  - Migraine
  - Cluster Headache

Neuropathies and Facial Pain

- Facial neuropathic pain from activation of the **trigeminal**, intermedius, glossopharyngeal, or vagus nerves or upper cervical roots via the occipital nerves



Secondary Headache

- Headache is a sign of an underlying medical condition or injury
- 10% of headaches
- Examples:
  - Head trauma
  - Whiplash
  - Stroke (ischemic or hemorrhagic)
  - Meningitis
  - Substance withdrawal
  - Idiopathic intracranial HTN
  - Intracranial neoplasia
  - Seizure

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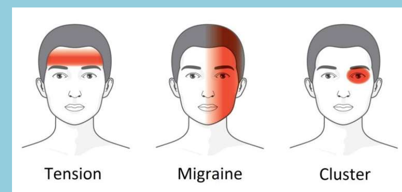
Primary Headaches Overview

Video 2: Dr. Kinnevey

Primary Headache

- **Migraine**
  - **Without Aura**
  - **With Aura (typical/ brainstem/ hemiplegic/retinal)**
  - **Chronic**
- **Tension Type Headache**
  - **Episodic** (Infrequent or frequent)
  - **Chronic**
  - Probable tension-type headache



- **Trigeminal Autonomic Cephalgia** (clinical features of unilateral headache and prominent cranial parasympathetic autonomic features)
  - **Cluster Headache** (episodic or chronic)
  - Paroxysmal Hemicrania (episodic or chronic)
  - Short-Lasting, unilateral, neuralgiform headache (episodic or chronic)
  - Hemicrania continua (remitting or unremitting)
  - Probable trigeminal autonomic cephalgia
- **Other Primary Headache Disorder**

- Cough
- Exercise
- Sexual activity
- Thunderclap*
- Cold-stimulus

- External pressure
- Stabbing
- Nummular
- Hypnic
- New Daily Persistent Headache

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You should know the general guidelines for diagnosis of each of the above listed **BOLDED** headaches

KNOW the DESCRIPTION and the DIAGNOSTIC CRITERIA

If the type of headache is not **bolded** here- you are NOT expected to know it for exam material

*Note – if you see the word thunderclap, in general, do NOT think primary headache disorder. Think intracranial bleed.

Migraines

- **Migraine without Aura (Common)**
  - Cause of migraine unknown
  - Throbbing, periodic attacks (4-72 hours), unilateral, nausea, vomiting, photophobia, phonophobia, exacerbated by head movement, relief by rest, +family history
    - 80-90% + family history, 3:1 female:male ratio, triggers
- **Migraine with Aura (Classic)- 20%**
  - **Typical aura:** Flashes of light, zigzag lines, enlarging scintillating scotoma (shimmering blind spot), unilateral numbness, no motor, brainstem or retinal symptoms
  - **Brainstem aura:** dysarthria, vertigo, tinnitus, diplopia, ataxia, decreased LOC, no motor no retinal symptoms
  - **Hemiplegic** - fully reversible motor weakness and visual, sensory and or speech/language symptoms
  - **Retinal** - Repeated attacks of **monocular visual disturbance**, including scintillations, scotomata or blindness, associated with migraine headache.
  - Lasts for 10-30 minutes- typically precedes headache by 30 minutes to 1 hour
  - Aura is thought to be caused by hyperexcitability (? MG, Ca?)
  - Increased risk of stroke
- **Chronic Migraine**
  - Intermittant attacks at increasing frequency- typically **15 days or more per month**
  - Comorbid conditions include: depression, axiety, panic, bipoloar, OCD
- **Status Migrainosus**
  - Debilitatating migraine headache lasting more than 72 hours

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Consider the mneumonic **POUND**
Pulsatile quality
h**O**urs duration: 4-72
Unilateral Location
Nausea and vomiting
Disabling intensity

Migraine Without Aura- Classification

A) At least 5 attacks fulfilling criteria B through D

B) Headache attacks lasting 4-72 hours (untreated)

C) Headache has at least 2 of the following:

Unilateral location
Pulsating quality
Moderate or severe pain intensity
Aggravation by or causing avoidance of routine physical activity

D) During headache, at least one of the following:

Nausea and/or vomiting
Photophobia and phonophobia

E) Not better accounted for by another ICHD-3 diagnosis

Migraine with Aura- Classification

A. At least two attacks fulfilling criteria B and C
B. One or more of the following fully reversible aura symptoms:
   - Visual
   - Sensory
   - Speech and/or language
   - Motor
   - Brainstem
   - Retinal

C. At least two of the following four characteristics:
   - At least one aura symptom spreads gradually over > or =5 minutes, and/or two or more symptoms occur in succession
   - Each individual aura symptom lasts 5-60 minutes
   - At least one aura symptom is unilateral
   - The aura is accompanied, or followed within 60 minutes, by headache

D. Not better accounted for by another ICHD-3 diagnosis and TIA has been excluded

<u>Migraine with typical aura</u>

A. At least two attacks fulfilling criteria B through D
B. Aura consisting of visual, sensory and/or speech/language symptoms, each fully reversible, but no motor, brainstem, or retinal symptoms
C. At least two of the following four characteristics:
   At least one aura symptom spreads gradually over > or =5 minutes, and/or two or more symptoms occur in succession
   Each individual aura symptom lasts 5-60 minutes
   At least one aura symptom is unilateral
   The aura is accompanied, or followed within 60 minutes, by headache
D. Not better accounted for by another ICHD-3 diagnosis and TIA has been excluded

Typical aura without headache 1.2.1.2- can have aura without the headache too

Migraine symptoms can really occur anywhere V3 distributes- there is even a midface migraine that can be confused with sinusitis

Chronic Migraine- Classification

A. Headache (TTH-like and/or migraine-like) on ≥15 d/mo for >3 mo and fulfilling criteria B and C

B. In a patient who has had ≥5 attacks fulfilling criteria B-D for 1.1 *Migraine without aura* and/or criteria B and C for 1.2 *Migraine with aura*

C. On ≥8 d/mo for >3 mo fulfilling any of the following:

   1. criteria C and D for 1.1 *Migraine without aura*
   2. criteria B and C for 1.2 *Migraine with aura*
   3. believed by the patient to be migraine at onset and relieved by a triptan or ergot derivative

D. Not better accounted for by another

ICHD-3 diagnosis

Note:

Patients meeting criteria for 1.3 *Chronic migraine* and for 8.2 *Medication-overuse headache* should be given both diagnoses After drug withdrawal, migraine will either revert to the episodic subtype or remain chronic, and be re-diagnosed accordingly; in the latter case, the diagnosis of 8.2 *Medication-overuse headache* may be rescinded

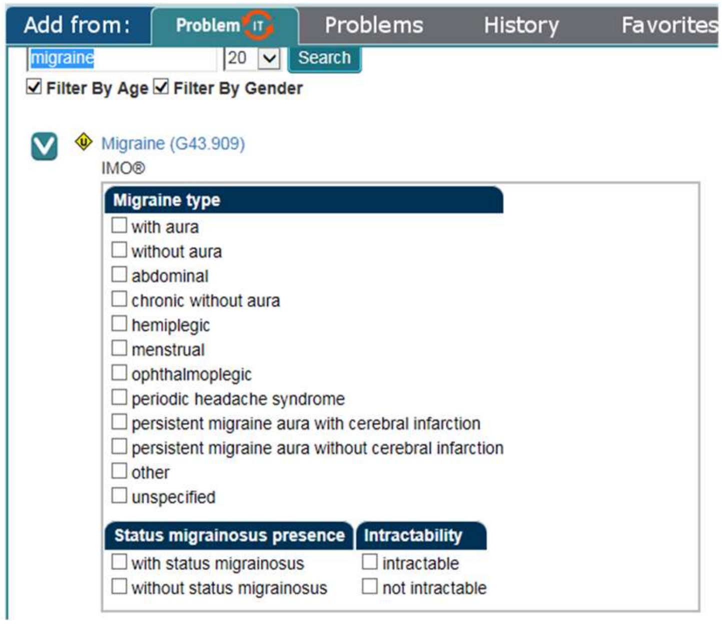
There are also “Exertional Migraines” which can occur when a patient is doing heavy work, lifting weights or sexual climax- however other forms of headache need to be ruled out.

Status Migrainosus:

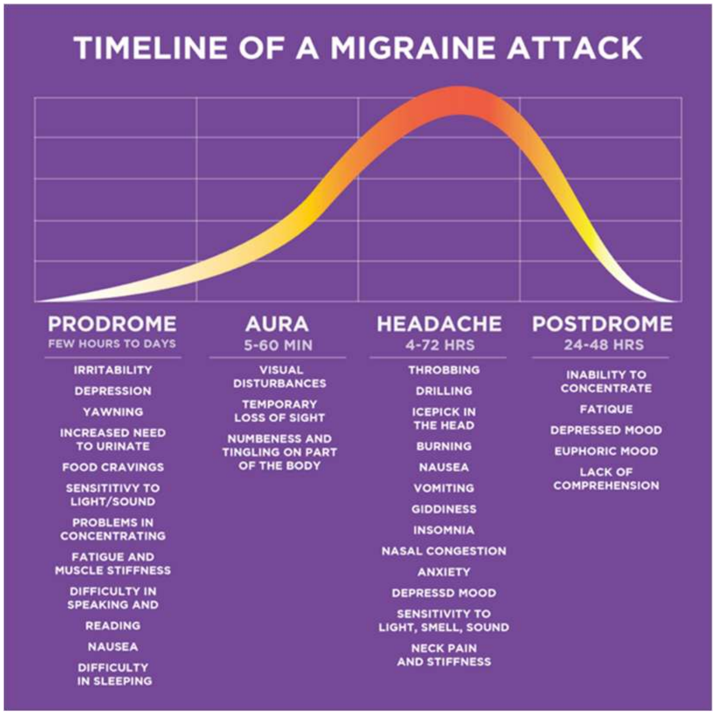
A debilitating migraine attack lasting for more than 72 hours.

Diagnostic criteria:
A headache attack fulfilling criteria B and C
Occurring in a patient with 1.1 Migraine without aura and/or 1.2 Migraine with aura, and typical of previous attacks except for its duration and severity
Both of the following characteristics:
unremitting for >72 hours1
pain and/or associated symptoms are debilitating2
Not better accounted for by another ICHD-3 diagnosis.
Notes:
Remissions of up to 12 hours due to medication or sleep are accepted.

Specifying the Migraine Type



ICD-10 diagnosis codes require distinguishing between status presence and intractability. However, intractable migraine and status migrainosus terms are often used interchangeably. Intractability though specifically refers to whether it is refractory to medication / treatment. Status migrainosus refers to duration being longer than 72 hours.



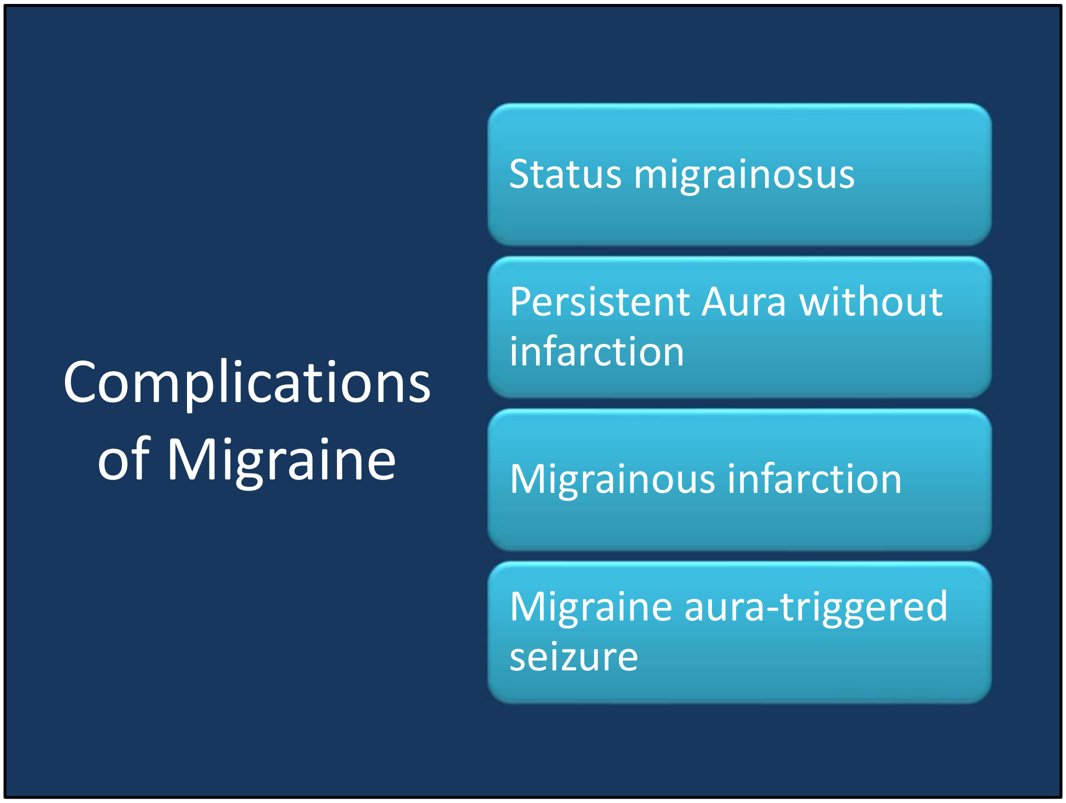
https://americanmigrainefoundation.org/understanding-migraine/timeline-migraine-attack/

Migraine is a complex neurological event
Different phases may overlap or be highly variable between patients

Aura phase-
wave of cortical neural excitation accompanied by hyperemia
Then an electrical wave of spreading neural depression and oligemia (this is considered a neural- not vascular phenomena)
Aura usually precedes the headache- but sometimes they arrive at the same time or sometimes there is aura with no headache

Information obtained from Dr. Stacey Pierce-Talsma’s 2018 lecture:

Also check out this article: https://americanmigrainefoundation.org/understanding-migraine/the-sphenopalatine-ganglion-spg-and-headache/
Remember when we learned the importance of this ganglia (parasymp) in the sphenoid lecture? We talk about trying to treat it with manipulation- they talk about trying to block it with an injection
(this information will not be on the exam)



Complications of migraine — Complications of migraine are characterized by attacks associated with prolonged symptoms or, rarely, with infarction or seizures. Prolonged symptoms may last for the entire headache, for several days or weeks, or in some cases leave a permanent neurologic deficit.

- **Status migrainosus** is a debilitating migraine attack lasting for more than 72 hours
- **Persistent aura without infarction** is defined by aura symptoms persisting for one week or more with no evidence of infarction on neuroimaging
- **Migrainous infarction** is defined by a migraine attack, occurring in a patient with migraine with aura, in which one or more aura symptoms persist for more than one hour and neuroimaging shows an infarction in a relevant brain area
- **Migraine aura-triggered seizure** is a seizure triggered by an attack of migraine with aura

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Migraines and Birth Control

- A study published in 2016 in *Neurology* found a significant association between migraine with visual aura and ischemic stroke compared with controls without headache (hazard ratio [HR] 1.7; 95% CI, 1.2-2.6, P =.008)[1]
- The use of combined hormonal contraceptives (CHC) is independently associated with increased risk of stroke
- A 2016 meta-analysis of 7 studies found a **2-4 fold increased risk of stroke among people who have migraines and use CHCs** vs those not using CHCs.[2]
- A review published in 2017 in the *Cleveland Clinic Journal of Medicine* stated that "this contraindication is based on data from the 1960s and 1970s, when oral contraceptives contained much higher doses of estrogen. Stroke risk is not significantly increased with today's preparations, many of which contain less than 30 µg of ethinyl estradiol." [3]

Migraine with aura is not an absolute contraindication to prescribing combined hormonal birth control, but most providers would err on the side of caution when there are so many good alternative progesterone forms of contraception

1. Androulakis, X. Michelle, et al. "Ischemic stroke subtypes and migraine with visual aura in the ARIC study." *Neurology* 87.24 (2016): 2527-2532. https://www.ncbi.nlm.nih.gov/pubmed/27956563

2. Tepper, Naomi K., et al. "Safety of hormonal contraceptives among women with migraine: a systematic review." *Contraception* 94.6 (2016): 630-640. https://www.contraceptionjournal.org/article/S0010-7824(16)30051-8/fulltext

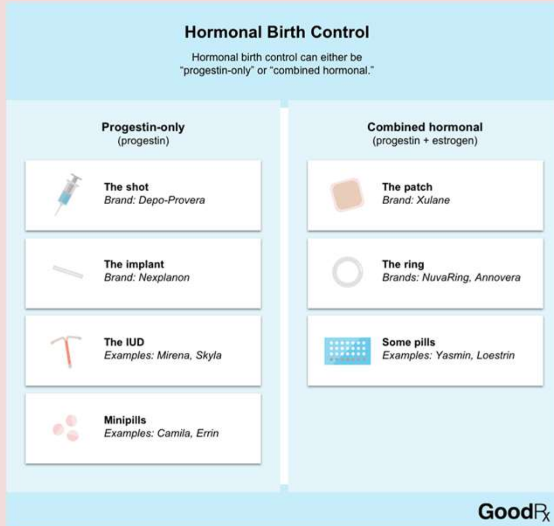
3. Hill, Chapel. "Combined hormonal contraceptives and migraine: an update on the evidence." *Cleveland Clinic journal of medicine* 84.8 (2017): 631. https://www.ncbi.nlm.nih.gov/pubmed/28806162

Sprintec

norgestimate/ethinyl estradiol

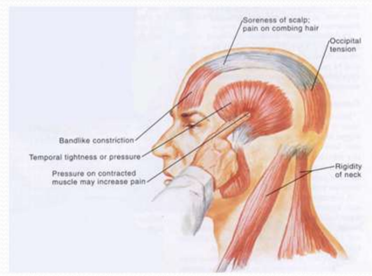
Contraindications/Cautions

- benign or malignant hepatic tumors
- active hepatic dz
- cholestatic jaundice hx, OCP or pregnancy-assoc.
- thrombophlebitis or hx
- thromboembolism or hx
- surgery, major w/ prolonged immobilization
- smokers >35 yo or heavy smokers
- thrombogenic valvular or rhythm heart dz
- CAD
- cerebrovascular dz
- thrombophilias, acquired or hereditary
- persistent SBP >160, DBP >100
- HTN w/ vascular dz
- diabetes mellitus w/ vascular dz
- migraine headaches w/ aura
- migraine headaches w/o aura (pts >35 yo)
- undiagnosed vaginal bleeding
- SLE, antibody positive or unknown
- w/in 21 days postpartum or 6wk postpartum if breastfeeding



Tension-Type Headache

- Can be infrequent episodic, frequent episodic or chronic
- Bilateral, pressing/tightness, bandlike, absence of nausea/vomiting but can can have either photo/phonophobia
- Patients often call these headaches “migraines”
- Typically duller, less intense, less localized lasting several hours to a day
- Pathophysiology
  - Still debated. Likely multifactorial
  - Thought to be largely due to activation of pericranial myofascial nociceptors
- Treatment:
  - Non pharmacologic- OMT, acupuncture, stress reduction, treating sleeping disorders
  - Abortive
    - NSAIDs (many OTC preparations)
  - Preventive
    - TCA, Beta blocker, anticonvulsant



Tension headaches can be infrequent episodic, frequent episodic, Chronic tension

Infrequent episodic tension:
At least 10 episodes of headache occurring on <1 day/month on average (<12 days/year)

Frequent episodic tension:
At least 10 episodes of headache occurring on 1-14 days/month on average for >3 months (≥12 and <180 days/year)

Chronic tension
Headache occurring on ≥15 days/month on average for >3 months (≥180 days/year),

Frequent episodic Tension Classification
Description:
Frequent episodes of headache, typically bilateral, pressing or tightening in quality and of mild to moderate intensity, lasting minutes to days. The pain does not worsen with routine physical activity and is not associated with nausea, although photophobia or phonophobia may be present.
Diagnostic criteria:
At least 10 episodes of headache occurring on 1-14 days/month on average for >3

months (≥12 and <180 days/year) and fulfilling criteria B-D
Lasting from 30 minutes to 7 days
At least two of the following four characteristics:

- bilateral location
- pressing or tightening (non-pulsating) quality
- mild or moderate intensity
- not aggravated by routine physical activity such as walking or climbing stairs

Both of the following:

- no nausea or vomiting
- no more than one of photophobia or phonophobia

Not better accounted for by another ICHD-3 diagnosis[1].

Note:

When headache fulfils criteria for both 1.5 *Probable migraine* and 2.2 *Frequent episodic tension-type headache*, code as 2.2 *Frequent episodic tension-type headache* (or as either subtype of it for which the criteria are fulfilled) under the general rule that definite diagnoses always trump probable diagnoses.

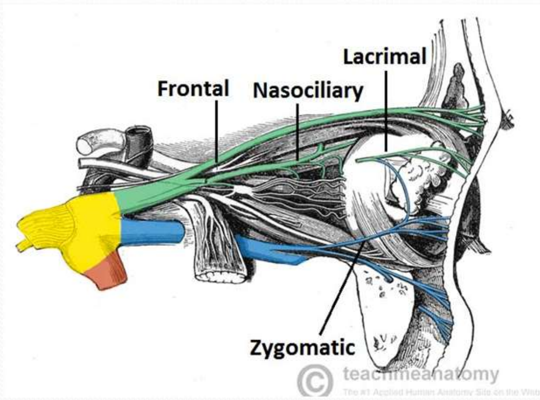
Comment:

2.2 *Frequent episodic tension-type headache* often coexists with 1.1 *Migraine without aura*. Both disorders need to be identified, preferably through use of a diagnostic headache diary, because the treatments of each differ considerably. It is important to educate patients to distinguish between these headache types if they are to select the right treatment for each while avoiding medication overuse and its adverse consequence of 8.2 *Medication-overuse headache*

https://www.uptodate.com/contents/tension-type-headache-in-adults-pathophysiology-clinical-features-and-diagnosis?search=tension%20headache&source=search_result&selectedTitle=2~107&usage_type=default&display_rank=2

Trigeminal Autonomic Cephalalgias

Characterized by pain in the distribution of the first division of the trigeminal nerve (V1 Ophthlamic) and accompanied by autonomic features in the same distribution



Includes Cluster, Paroxysmal hemicrania, hemicrania continua, and SUNCT headaches but for this exam, just focus on learning about cluster headaches

Cluster Headaches

Cluster headaches may involve pain around one eye, along with drooping of the lid, tearing and congestion on the same side as the pain

ADAM

- Rare, men:women 3:1, appears btwn ages 20-40
- Clusters or bouts lasting weeks to months
  - Can also be episodic or chronic
  - Circadian periodicity- may only occur yearly
- Unilateral pain, typically around the eye, often during sleeping times- longest attacks of the TAC's
- **Often history of smoking and etoh**
- Associated with:
  - Conjunctival injection, Lacrimation, Nasal congestion, Rhinorrhea, Forehead facial sweating, Miosis, Ptosis, Restlessness/agitation
- Lasts 15-180 minutes, can have 1-6 per day
- Pathophysiology:
  - Possible disturbance of the hypothalmus
    - Autonomic and melatonin disturbance
- Treatment may include **high flow oxygen** (12L minute for 15 minutes), triptans/ergot, sometimes indomethacin or corticosteroids
- Prophylactic/preemptive treatment is important
  - **CA blockers** (verapamil), anticonvulsants, lithium carbonate, TCA's, beta blockers
- Chronic cluster- occipital nerve block or sphenopalatine ganglion block

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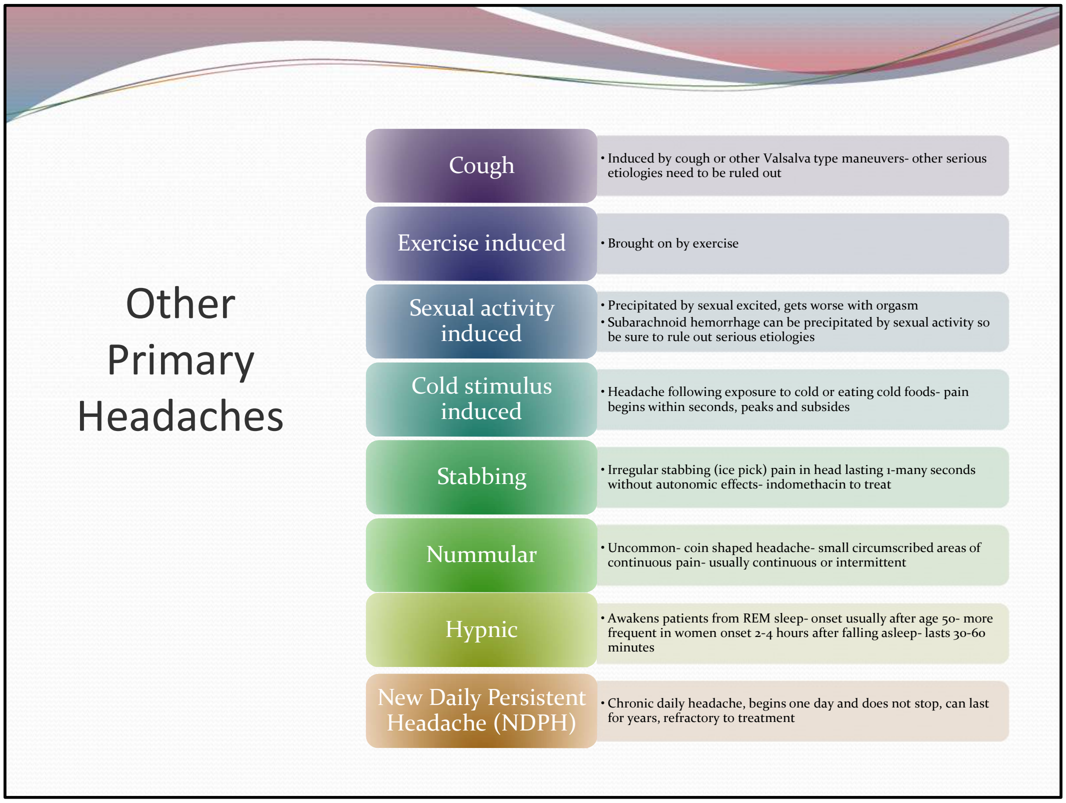
Description:
Attacks of severe, strictly unilateral pain which is orbital, supraorbital, temporal or in any combination of these sites, lasting 15-180 minutes and occurring from once every other day to eight times a day. The pain is associated with ipsilateral conjunctival injection, lacrimation, nasal congestion, rhinorrhoea, forehead and facial sweating, miosis, ptosis and/or eyelid oedema, and/or with restlessness or agitation.
Diagnostic criteria:
At least five attacks fulfilling criteria B-D
Severe or very severe unilateral orbital, supraorbital and/or temporal pain lasting 15-180 minutes (when untreated)[1]
Either or both of the following:
   at least one of the following symptoms or signs, ipsilateral to the headache:
      – conjunctival injection and/or lacrimation
      – nasal congestion and/or rhinorrhoea
      – eyelid oedema
      – forehead and facial sweating
      – miosis and/or ptosis
   a sense of restlessness or agitation
Occurring with a frequency between one every other day and 8 per day[2]
Not better accounted for by another ICHD-3 diagnosis.

Notes:
During part, but less than half, of the active time-course of 3.1 *Cluster headache*, attacks may be less severe and/or of shorter or longer duration.
During part, but less than half, of the active time-course of 3.1 *Cluster headache*, attacks may be less frequent.
Comments:
Attacks occur in series lasting for weeks or months (so-called cluster periods or bouts) separated by remission periods usually lasting months or years. About 10-15% of patients have 3.1.2 *Chronic cluster headache*, without such remission periods. In a large series with good follow-up, one quarter of patients had only a single cluster period. Such patients meet the criteria for and should be coded as 3.1 *Cluster headache*.
During a cluster period in 3.1.1 *Episodic cluster headache*, and at any time in 3.1.2 *Chronic cluster headache*, attacks occur regularly and may be provoked by alcohol, histamine or nitroglycerin.
The pain of 3.1 *Cluster headache* is maximal orbitally, supraorbitally, temporally or in any combination of these sites, but may spread to other regions. During the worst attacks, the intensity of pain is excruciating. Patients are usually unable to lie down, and characteristically pace the floor. Pain usually recurs on the same side of the head during a single cluster period.
Age at onset is usually 20-40 years. For unknown reasons, men are afflicted three times more often than are women.
Acute attacks involve activation in the region of the posterior hypothalamic grey matter. 3.1 *Cluster headache* may be autosomal dominant in about 5% of cases.
Some patients have been described who have both 3.1 *Cluster headache* and 13.1.1 *Trigeminal neuralgia* (sometimes referred to as *cluster-tic syndrome*). They should receive both diagnoses. The importance of this observation is that both conditions must be treated for the patient to become headache free.

(Information obtained from Dr. Stacey Pierce-Talsma's 2018 lecture)



- You are only responsible for knowing about the first 4 types of headaches on the slide

<u>Primary Cough Headache</u>
Primary cough headache is a generalized headache that **begins suddenly, lasts for several minutes, and is precipitated by coughing;** it is preventable by avoiding coughing or other precipitating events, which can include sneezing, straining, laughing, or stooping. In all patients with this syndrome serious etiologies must be excluded before a diagnosis of "benign" primary cough headache can be established. A Chiari malformation or any lesion causing obstruction of CSF pathways or displacing cerebral structures can be the cause of the head pain. Other conditions that can present with cough or exertional headache as the initial symptom include cerebral aneurysm, carotid stenosis, and vertebrobasilar disease. Benign cough headache can resemble benign exertional headache (below), but patients with the former condition are typically older.
Primary Cough Headache: Treatment
Indomethacin 25–50 mg two to three times daily is the treatment of choice. Some patients with cough headache obtain pain relief with LP; this is a simple option when compared to prolonged use of indomethacin, and it is effective in about one-third of patients. The mechanism of this response is unclear.

<u>Primary Exertional Headache</u>
Primary exertional headache has features resembling both cough headache and migraine. It may be **precipitated by any form of exercise; it often has the pulsatile quality of migraine.** The pain, which can last from 5 min to 24 h, is bilateral and

throbbing at onset; migrainous features may develop in patients susceptible to migraine. Primary exertional headache can be prevented by avoiding excessive exertion, particularly in hot weather or at high altitude.
The mechanism of primary exertional headache is unclear. **Acute venous distension likely explains one syndrome**, the acute onset of headache with straining and breath holding, as in weightlifter's headache. As exertion can result in headache in a number of serious underlying conditions, these must be considered in patients with exertional headache. Pain from angina may be referred to the head, probably by central connections of vagal afferents, and may present as exertional headache (cardiac cephalgia). The link to exercise is the main clinical clue that headache is of cardiac origin. Pheochromocytoma may occasionally cause exertional headache. Intracranial lesions and stenosis of the carotid arteries are other possible etiologies.
Primary Exertional Headache: Treatment
Exercise regimens should begin modestly and progress gradually to higher levels of intensity. Indomethacin at daily doses from 25 to 150 mg is generally effective in benign exertional headache. Indomethacin (50 mg), ergotamine (1 mg orally), dihydroergotamine (2 mg by nasal spray), or methysergide (1–2 mg orally given 30–45 min before exercise) are useful prophylactic measures.

Primary Sex Headache
Sex headache is precipitated by sexual excitement. The pain usually begins as a **dull bilateral headache which suddenly becomes intense at orgasm.** The headache can be prevented or eased by ceasing sexual activity before orgasm. Three types of sex headache are reported: a dull ache in the head and neck that intensifies as sexual excitement increases; a sudden, severe, explosive headache occurring at orgasm; and a postural headache developing after coitus that resembles the headache of low CSF pressure. The latter arises from vigorous sexual activity and is a form of low CSF pressure headache. Headaches developing at the time of orgasm are not always benign; **5–12% of cases of subarachnoid hemorrhage are precipitated by sexual intercourse.** Sex headache is reported by men more often than women and may occur at any time during the years of sexual activity. It may develop on several occasions in succession and then not trouble the patient again, even without an obvious change in sexual activity. In patients who stop sexual activity when headache is first noticed, the pain may subside within a period of 5 min to 2 h. In about half of patients, sex headache will subside within 6 months. About half of patients with sex headache have a history of exertional headaches, but there is no excess of cough headache. Migraine is probably more common in patients with sex headache.
Primary Sex Headache: Treatment
Benign sex headaches recur irregularly and infrequently. Management can often be limited to reassurance and advice about ceasing sexual activity if a mild, warning headache develops. Propranolol can be used to prevent headache that recurs regularly or frequently, but the dosage required varies from 40 to 200 mg/d. An alternative is the calcium channel-blocking agent diltiazem, 60 mg tid. Ergotamine (1 mg) or indomethacin (25–50 mg) taken about 30–45 min prior to sexual activity can also be helpful.

Primary Stabbing Headache
The essential features of primary stabbing headache are stabbing pain confined to the head or, rarely, the face, lasting from 1 to many seconds or minutes and occurring as a single stab or a series of stabs; absence of associated cranial autonomic features; absence of cutaneous triggering of attacks; and a pattern of recurrence at irregular intervals (hours to days). The pains have been variously described as "ice-pick pains"

or "jabs and jolts." They are more common in patients with other primary headaches, such as migraine, the TACs, and hemicrania continua.

Primary Stabbing Headache: Treatment

The response of primary stabbing headache to indomethacin (25–50 mg two to three times daily) is usually excellent. As a general rule the symptoms wax and wane, and after a period of control on indomethacin, it is appropriate to withdraw treatment and observe the outcome.

Hypnic Headache

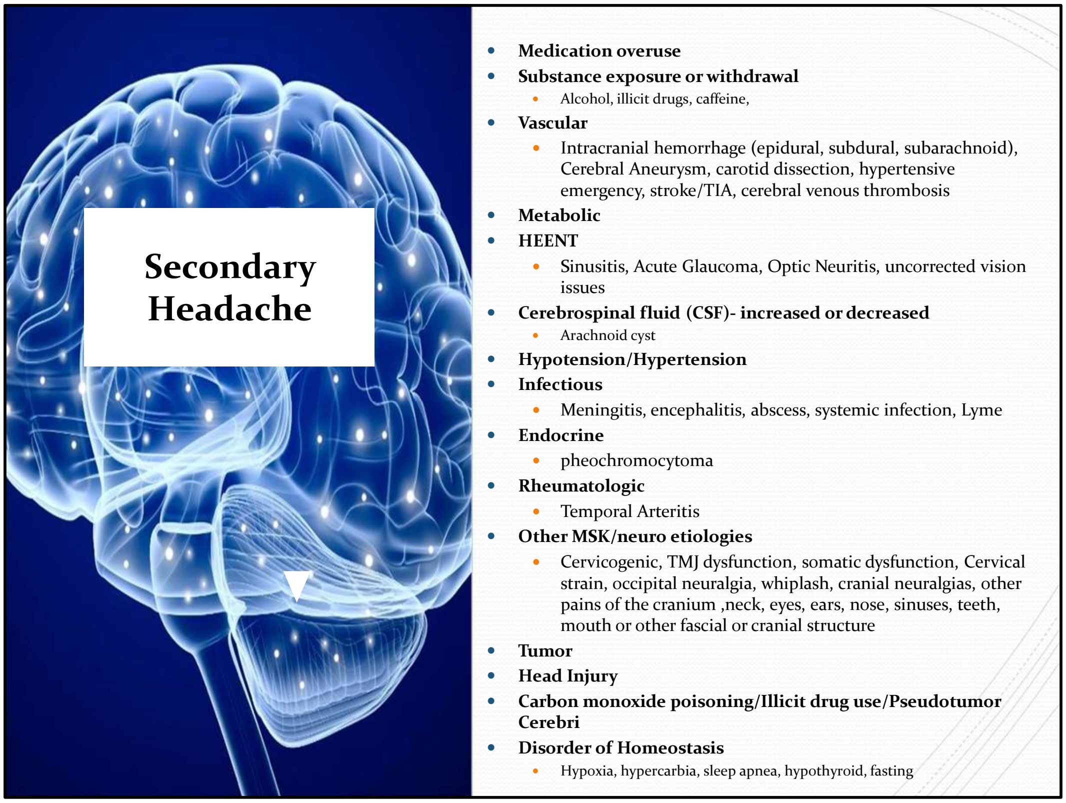
This headache syndrome typically begins a few hours after sleep onset. The headaches last from 15 to 30 min and are typically moderately severe and generalized, although they may be unilateral and can be throbbing. Patients may report falling back to sleep only to be awakened by a further attack a few hours later; up to three repetitions of this pattern occur through the night. Daytime naps can also precipitate head pain. Most patients are female, and the onset is usually after age 60. Headaches are bilateral in most, but may be unilateral. Photophobia or phonophobia and nausea are usually absent. The major secondary consideration in this headache type is poorly controlled hypertension; 24-h blood pressure monitoring is recommended to detect this treatable condition.

Hypnic Headache: Treatment

Patients with hypnic headache generally respond to a bedtime dose of lithium carbonate (200–600 mg). For those intolerant of lithium, verapamil (160 mg) or methysergide (1–4 mg at bedtime) may be alternative strategies. One to two cups of coffee or caffeine, 60 mg orally, at bedtime may be effective in approximately one-third of patients. Case reports suggest that flunarizine, 5 mg nightly, can be effective.

Secondary Headaches Overview

Video 3: Dr. Kinnevey



The vast majority of patients without danger signs do not have a secondary cause of headache [24,25]. As an example, in a study of 373 patients with chronic headache at a tertiary referral center, all had one or more of the following characteristics that prompted referral for head CT scan: increased severity of symptoms or resistance to appropriate drug therapy, change in characteristics or pattern of headache, or family history of an intracranial structural lesion [26]. Only two exams (less than 1 percent) showed potentially significant lesions (one low grade glioma and one aneurysm); only the aneurysm was treated.

Part I: primary headaches
Migraine
Tension
Cluster and TAC
Other primary

Part II: secondary headaches
Attributed to trauma or injury of the head or neck
Attributed to cranial or cervical vascular disorder
Attributed to nonvascular intracranial disorder
Attributed to a substance or it's withdrawal
Attributed to disorder of homeostasis
Attributed to disorder of the cranium neck eyes ears nose sinuses teeth mouth or

other facial or cranial structure
Attributed to psychiatric disorder

Part III: painful cranial neuropathies, other facial pains and other headaches
Painful cranial neuropathies and other facial pains
Other headache disorders

Red Flags

- **May indicate Secondary Headache**
  - Valsalva induced pain
  - Positional pain
  - Onset in middle age or later
  - Recent head trauma
  - Chronic systemic illness or immunocompromised/HIV/cancer
  - Pain that disturbs sleep
  - Anticoagulation
  - New headache
  - Changes in pattern, severity or frequency of headache
  - Associated seizures
  - Worst headache ever
  - Subacute- worsening over days
- **Red flag physical exam findings**
  - Fever
  - Nuchal rigidity
  - Changes in mental status
  - Optic disc edema(papilledema)
  - Cranial Nerve palsy
  - Weakness
  - Visual-field deficit
  - Ataxia
  - Horner's syndrome
  - Rash

Use your H &P to determine if any imaging studies are required
Differentiate between primary headaches and secondary headaches and use the red flags as a guidepost
Bad things have other signs and symptoms including the red flags
Very few of your patients with headaches will have something like a tumor- but you don't want to miss it- so look for red flags

Also people with primary headaches can develop secondary headaches so watch for changing headache patterns/frequency/severity

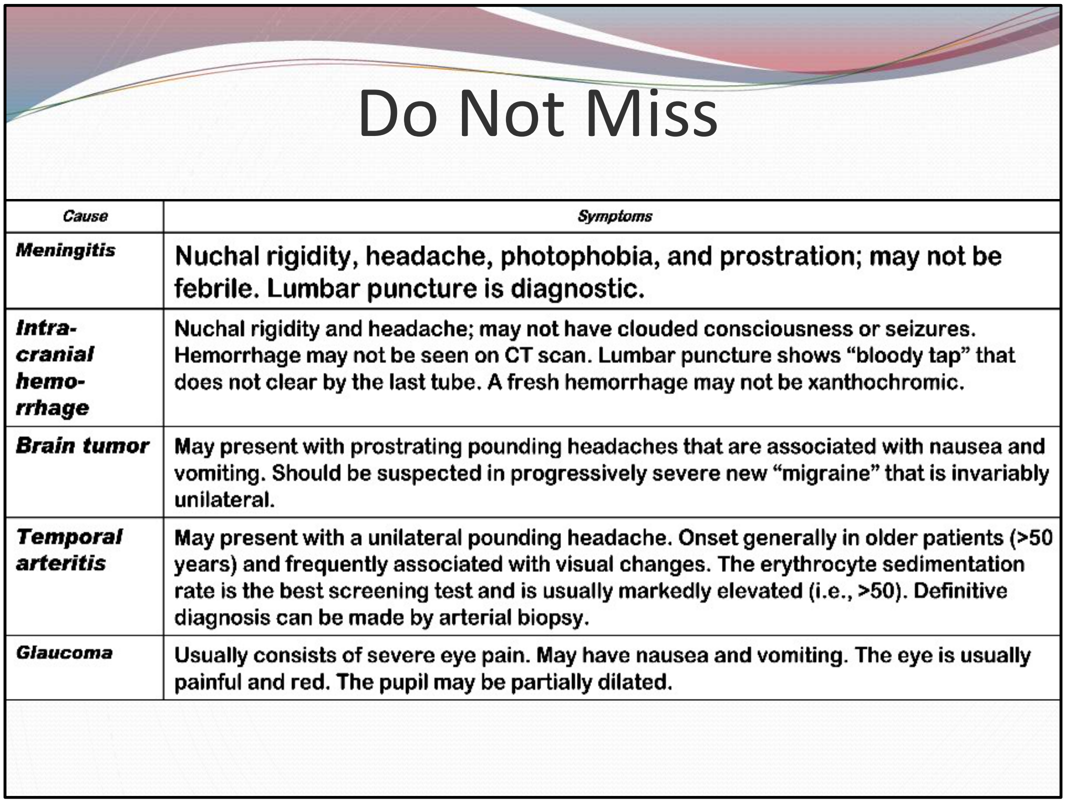
An imaging study is not necessary in the vast majority of patients with headache- but imaging (usually CT) is warrented in patients with red flags or suspicion of intracranial pathology

New headache in patients over 50 should also be considered suspicious

SNOOP Mnemonic for Red Flags

	Stands for…	Example	Think of…
S	**S**ystemic Symptoms **S**econdary risk factors	Fever, weight loss, fatigue HIV, cancer, immune suppression	Infection, metastatic cancer, meningitis
N	**N**eurologic Symptoms / signs	Altered level of consciousness, focal deficits	Encephalitis, mass lesion, stroke
O	**O**nset	Thunderclap, abrupt	Subarachnoid hemorrhage, intraparenchymal hemorrhage, reversible cerebral vasoconstriction syndrome
O	**O**lder	New after age 50	Temporal arteritis
P	**P**ositional **P**revious Headache **P**apilledema	Change upright vs. laying down Change with neck position Different in quality than prior headaches Visual disturbances	Intracranial hypotension or hypertension, posterior fossa pathology

https://americanheadachesociety.org/wp-content/uploads/2018/07/Primary_or_Secondary_Headache.pdf
http://image.slidesharecdn.com/haepilepsywrobelgoldberg2014-141103094608-conversion-gate02/95/headaches-epilepsy-presentation-from-epilepsy-education-exchange-2014-4-638.jpg?cb=1415008187



Do Not Miss

Cause	Symptoms
Meningitis	**Nuchal rigidity, headache, photophobia, and prostration; may not be febrile. Lumbar puncture is diagnostic.**
Intra-cranial hemo-rrhage	**Nuchal rigidity and headache; may not have clouded consciousness or seizures. Hemorrhage may not be seen on CT scan. Lumbar puncture shows “bloody tap” that does not clear by the last tube. A fresh hemorrhage may not be xanthochromic.**
Brain tumor	**May present with prostrating pounding headaches that are associated with nausea and vomiting. Should be suspected in progressively severe new “migraine” that is invariably unilateral.**
Temporal arteritis	**May present with a unilateral pounding headache. Onset generally in older patients (>50 years) and frequently associated with visual changes. The erythrocyte sedimentation rate is the best screening test and is usually markedly elevated (i.e., >50). Definitive diagnosis can be made by arterial biopsy.**
Glaucoma	**Usually consists of severe eye pain. May have nausea and vomiting. The eye is usually painful and red. The pupil may be partially dilated.**

Specific features suggesting a secondary headache source — Other features that suggest a specific source of headache pain include the following:

●Impaired vision or seeing halos around light suggests the presence of glaucoma. Suspicion for subacute angle closure glaucoma should be raised by relatively short duration (often less than one hour) unilateral headaches that do not meet criteria for migraine arising after age 50 [16].

●Visual field defects suggest the presence of a lesion of the optic pathway (eg, due to a pituitary mass).

●Sudden, severe, unilateral vision loss suggests the presence of optic neuritis. Optic neuritis typically presents with painful, monocular visual loss that evolves over several hours to a few days. One-third of patients have visible optic nerve inflammation (papillitis) on funduscopic examination. (See "Optic neuritis: Pathophysiology, clinical features, and diagnosis".)

●Blurring of vision on forward bending of the head, headaches upon waking early in the morning that improve with sitting up, and double vision or loss of coordination and balance should raise the suspicion of raised intracranial pressure (ICP); this should also be considered in patients with chronic, daily, progressively worsening headaches associated with chronic nausea. Idiopathic intracranial hypertension (pseudotumor cerebri) typically affects obese women of child-bearing age. Characteristic features are headache, papilledema, vision loss or diplopia, elevated lumbar puncture (LP) opening pressure with normal cerebrospinal fluid (CSF) composition. (See "Evaluation and management of elevated intracranial pressure in adults" and "Idiopathic intracranial hypertension (pseudotumor cerebri): Clinical features and diagnosis".)

●In patients who present with headache that is relieved with recumbency and exacerbated with upright posture, the diagnosis of headache attributed to spontaneous intracranial hypotension should be considered. An additional major feature of this headache syndrome is diffuse meningeal enhancement on brain MRI. The accepted etiology is CSF leakage, which may occur in the context of rupture of an arachnoid membrane. (See "Spontaneous intracranial hypotension: Pathophysiology, clinical features, and diagnosis".)

●The presence of nausea, vomiting, worsening of headache with changes in body position (particularly bending over), an abnormal neurologic examination, and/or a significant change in prior headache pattern suggests the headache was caused by a tumor. (See "Overview of the clinical features and diagnosis of brain tumors in adults".)

●Intermittent headache with generalized sweating, tachycardia, and/or sustained or paroxysmal hypertension is suggestive of pheochromocytoma. (See "Clinical presentation and diagnosis of pheochromocytoma".)

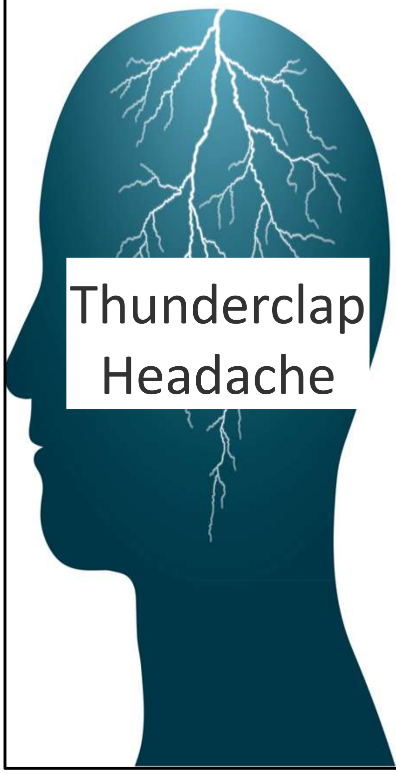
●Morning headache is nonspecific and can occur as part of a primary headache syndrome or may be secondary to a number of disorders including sleep apnea, chronic obstructive pulmonary disease, and the obesity hypoventilation syndrome. (See "Clinical presentation and diagnosis of obstructive sleep apnea in adults" and "Chronic obstructive pulmonary disease: Definition, clinical manifestations, diagnosis, and staging" and "Clinical manifestations and diagnosis of obesity hypoventilation syndrome".)

Medication Overuse Headache

- Persistent and recurring headache with regular use of analgesic, ergotamines or triptans
  - Weeks to months of overuse- usually takings meds more than 4 days each week
  - Insidious increase in headache frequency
  - Escalation use of offending agents
  - Evidence of physical dependency
  - Reliable onset of headache within hours to days of the last treatment
- Treatment is discontinuation of all analgesics
  - Physiologic washout period is 2 weeks
    - Worse headaches during withdrawal
    - Antiemetics and hydration
    - Controlled taper if narcotics, benzos or barbituates
  - Should continue to refrain for 10-12 weeks

Previously known as rebound headaches

Description:
Headache occurring on 15 or more days/month in a patient with a pre-existing primary headache and developing as a consequence of regular overuse of acute or symptomatic headache medication (on 10 or more or 15 or more days/month, depending on the medication) for more than 3 months. It usually, but not invariably, resolves after the overuse is stopped.
Diagnostic criteria:
Headache occurring on ≥15 days/month in a patient with a pre-existing headache disorder
Regular overuse for >3 months of one or more drugs that can be taken for acute and/or symptomatic treatment of headache[1;2;3]
Not better accounted for by another ICHD-3 diagnosis.

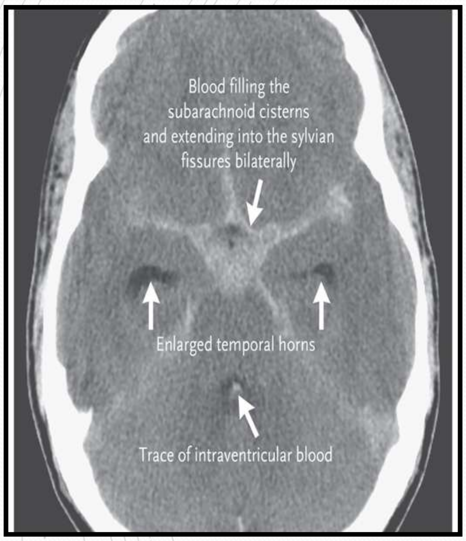


Thunderclap Headache

- Abrupt sudden onset (less than one minute)
- A medical emergency
- Differential Diagnosis
  - **Subarachnoid hemorrhage**
  - **Intracerebral hemorrhage**
  - Reversible cerebral vasoconstriction syndrome (RCVS)
  - Cerebral vein thrombosis
    - Usually a more gradual subacute onset- MRI with venography
  - Pituitary apoplexy
    - Hemorrhage or infarction of the pituitary gland in the setting of pituitary adenoma
  - Sudden CSF hypotension/leak
    - Loss of CSF causes brain to shift downward- tension on vessels and nerves- Diffuse meningeal enhancement on MRI, lumbar puncture
  - Cervical artery dissection
    - Headache, neck pain, tinnitus, bruit or cranial neuropathy
    - CT angiography, US, MR angiography
  - Acute cervical disk herniation
    - Non contrast MRI- unless previous instrumentation- then with contrast
  - Acute occipital/Cervical junction events
  - Hypertensive crisis
- Management
  - Noncontrast head CT
  - Lumbar puncture if normal imaging of cerebral circulation

https://www.uptodate.com/contents/overview-of-thunderclap-headache?search=thunderclap%20headache&source=search_result&selectedTitle=1~30&usage_type=default&display_rank=1

Subarachnoid Hemorrhage



- Worst headache of life- **thunderclap**
- Caused by rupture of intracranial aneurysm in 80% of cases (other vascular malformations and vasculitis)
  - About half occur during nonstrenuous activity
- Some people have sentinel headaches 2-8 weeks before overt subarachnoid hemorrhage
- Usually age >40
- High mortality rate
- Signs and symptoms of elevated ICP, nausea, vomiting, neck stiffness, focal neurologic deficits, brief loc, elevated BP, history of smoking
- Immediate CT scan without contrast
  - Lumbar puncture if that is negative
    - RBC count, xanthochromia
  - CT angiography if doubt still remains

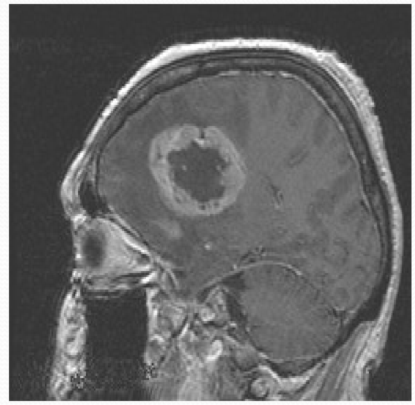
From the NEJM article (https://www.nejm.org/doi/full/10.1056/NEJMcp1605827?linkId=40680854&page=11&sort=oldest)

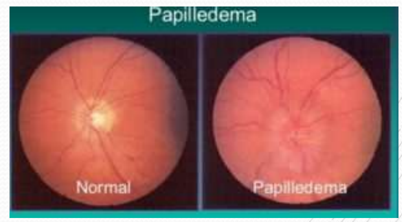
"When an aneurysm ruptures, it is an intracranial catastrophe. Blood pushes into the subarachnoid space at arterial pressure until the intracranial pressure equalizes across the rupture site and stops the bleeding, with thrombus formation at the bleeding site. The reported case fatality rate is 25 to 50%,[15] owing to consequences of either the original bleeding or rerupture; this estimate does not fully account for patients who die before receiving medical attention.[16]

Aneurysmal subarachnoid hemorrhage accounts for only 1% of all headaches evaluated in the emergency department.[21] Consequently, a sentinel headache may be dismissed as a migraine headache or other headache without further evaluation; the likelihood of death or disability is four times as high among patients in whom a sentinel headache is misdiagnosed as it is among patients in whom a sentinel headache is correctly diagnosed.[22] Therefore, a high index of suspicion for aneurysmal subarachnoid hemorrhage from the patient's history is warranted and potentially lifesaving"

Brain Tumor

- May or may not have signs of increased intracranial pressure:
  - Papilledema (optic disc edema), headache, nausea/vomiting
- Usually has focalizing signs (other neurologic signs)
- Posterior fossa tumors are more likely to produce headache as they are more rapidly growing tumors
- Diagnosis by CT or MRI- (MRI a better choice for soft tissue)





From uptodate: https://www.uptodate.com/contents/brain-tumor-headache

Tumors and other mass lesions can cause headache by direct pressure on any of the above structures. Pain arising from the intracranial cavity is a form of visceral pain, and therefore referred to more superficial structures; it is not perceived as coming directly from the compressed region [4]. In general, unilateral lesions refer pain ipsilaterally, supratentorial lesions refer pain to the frontal regions, and posterior fossa masses refer pain to the occiput [4]. However, these associations have poor localizing value in clinical practice. As an example, in a study that evaluated 37 patients with infratentorial tumors, 73 percent of patients had headache in the frontal, temporal, and/or parietal regions, while nuchal and occipital pain occurred in only 27 percent [5].
Acute or severe headache from a brain tumor can potentially be caused by elevated intracranial pressure resulting from hydrocephalus, mass effect of the tumor itself, or hemorrhage into or around the tumor [6]. A tumor can distort or displace structures distant from its locale, resulting in referred pain that does not correspond with its location. In such cases of "distant traction," elevated intracranial pressure may be the cause of the headache [7]. However, the relationship between elevated intracranial pressure and headache in brain tumor remains uncertain

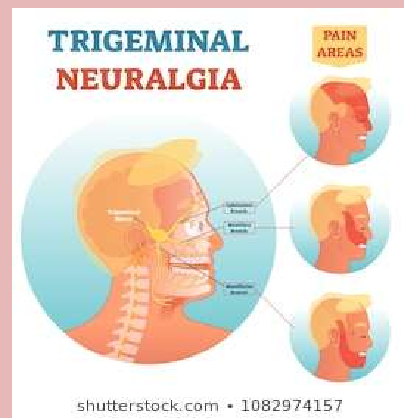
Neuropathies & Facial Pains

Video 4: Dr Kinnevey

The International Classification of Headache Disorders

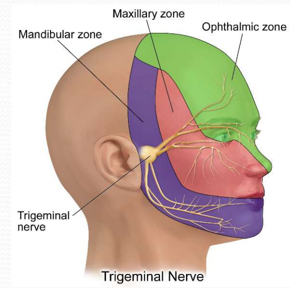
Neuropathies & Facial Pains and Other Headaches

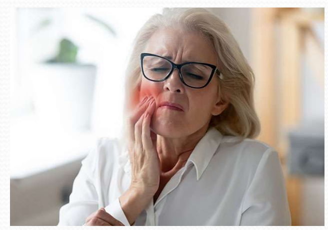
- Facial neuropathic pain from activation of the **trigeminal**, intermedius, glossopharyngeal, or vagus nerves or upper cervical roots via the occipital nerves
- May be caused from vascular compression of the respective nerve, infection (e.g. Varicella zoster virus) or structural abnormality (e.g. multiple sclerosis plaque)



Trigeminal Neuralgia

- Recurrent paroxysms of unilateral facial pain in the distribution(s) of one or more divisions of the trigeminal nerve
  - lasts from a fraction of a second to 2 minutes
  - severe intensity
  - electric shock-like, shooting, stabbing or sharp in quality
- Most cases are caused by compression of the nerve root within a few mm of entry into the pons (usually by an aberrant loop of an artery or vein)
- Tx:
  - Pharmacotherapy is first line for most cases (carbamazepine, gabapentin, lamotrigine or baclofen)
  - Surgical therapy using microvascular decompression

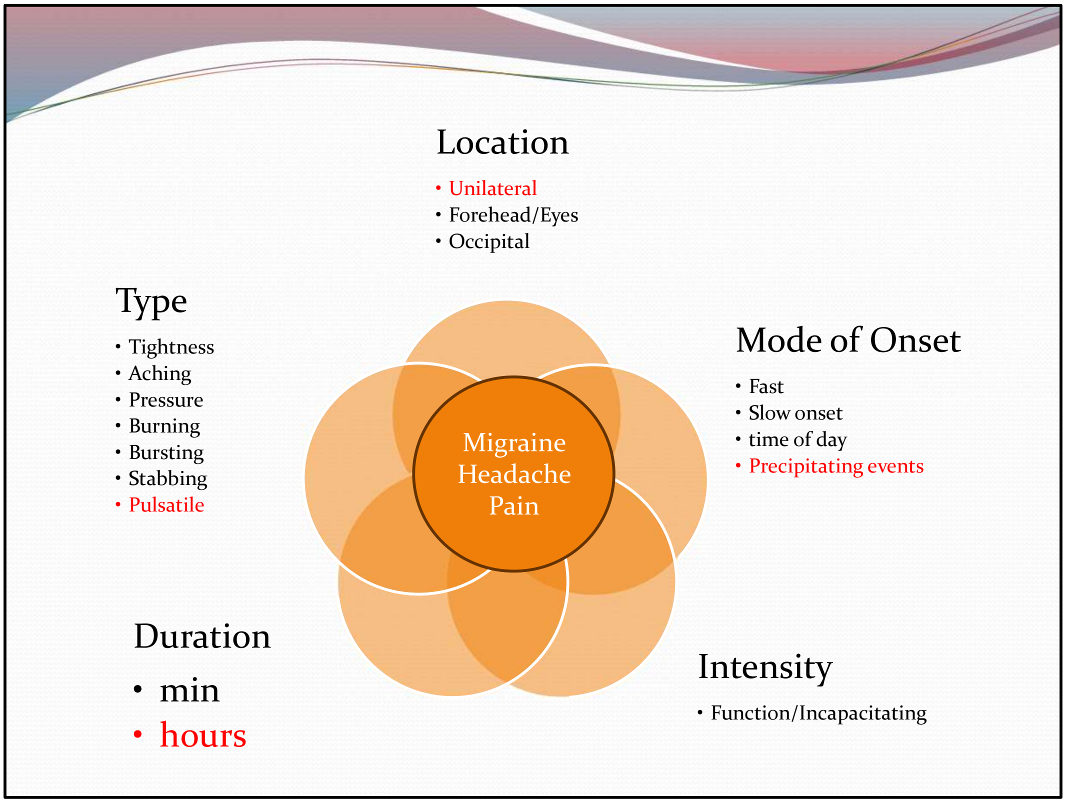




Recurrent paroxysms of unilateral facial pain in the distribution(s) of one or more divisions of the trigeminal nerve, with no radiation beyond[1], and fulfilling criteria B and C
Pain has all of the following characteristics:
Precipitated by innocuous stimuli within the affected trigeminal distribution[4]
Not better accounted for by another ICHD-3 diagnosis.

Migraine Pathophysiology

- Video 5: Dr. Gayer



Headache and Other Craniofacial Pains. In: Ropper AH, Samuels MA, Klein JP, Prasad S. eds. *Adams and Victor's Principles of Neurology, 12e*. McGraw Hill; 2023. Accessed September 08, 2023.

In the introductory chapter on pain, reference was made to the necessity of determining the quality, severity, location, duration, and time course of any pain as well as the conditions that produce or relieve it. In the case of headache, a detailed history following these lines will determine the diagnosis more often than will the physical examination or imaging. Nevertheless, a few aspects of the examination are worth emphasizing. For example, auscultation of the skull may rarely disclose a bruit (with large arteriovenous malformations); palpation may disclose the tender, hardened or elevated arteries of temporal arteritis; sensitive areas overlying a cranial metastasis or an inflamed paranasal sinus may be apparent; or there may be a tender occipital nerve. Examination of neck flexion may reveal meningitis; however, apart from these special instances, examination of the head itself, although necessary, seldom discloses the diagnosis.
The *quality* of cephalic pain is essential to diagnosis but the sensation may be difficult for the patient to describe. When asked to compare the pain to some other sensory experience, the patient may allude to tightness, aching, pressure, burning, bursting, sharpness, or stabbing. Among the most important aspects is whether the headache is pulsatile, usually implying migraine, but one must keep in mind that patients sometimes use the word *throbbing* to refer to a waxing and waning of the headache

without any relation to the pulse, or use the term to transmit the severity of pain. Similarly, statements about the *intensity* of the pain reflect to some extent the patient's temperament, attitudes, and customary ways of experiencing and reacting to pain. One useful index of severity is the degree to which the pain has incapacitated the patient. A severe migraine attack seldom allows the migraineur to perform the day's work. Other rough indices of the severity of headache are its propensity to awaken the patient from sleep or to prevent sleep, and autonomic reactions to the pain such as sweating and tachycardia. The most intense cranial pains are those associated with meningitis and subarachnoid hemorrhage, which can have grave consequences, but also migraine, cluster headache, or tic douloureux that do not have the same implications.

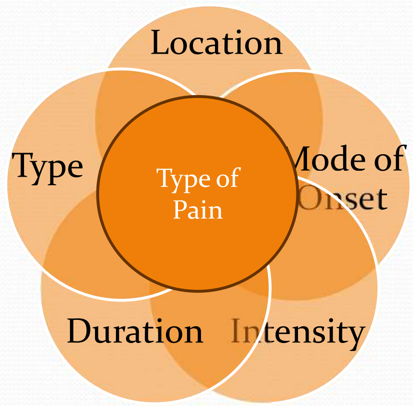
Information regarding the *location* of a headache is informative. Migraine headache is unilateral in two-thirds of attacks and is commonly associated with nausea, vomiting, and sensitivity to light, sound, and smells. Inflammation of an extracranial artery from temporal arteritis causes pain localized to the site of the vessel. Lesions of the paranasal sinuses, teeth, eyes, and upper cervical vertebrae induce a less sharply localized pain but still one that is referred to a certain region, usually the forehead or maxilla or around the eyes. Intracranial lesions in the posterior fossa generally cause pain in the occipitonuchal region and usually are ipsilateral if the lesion is one-sided. Supratentorial lesions induce frontotemporal pain that approximates the site of an intracranial lesion. Localization, however, may also be deceiving. Pain in the frontal regions may be caused by such diverse disorders as glaucoma, sinusitis, thrombosis of the vertebral or basilar artery, pressure on the tentorium, and increased intracranial pressure. Similarly, ear pain may signify disease of the ear itself, but as often, it is referred from other regions, such as the throat, cervical muscles, spine, or structures in the posterior fossa. Periorbital and supraorbital pain, while usually indicative of local ocular disease, may reflect dissection of the cervical portion of the internal carotid artery. Headaches localized to the vertex or biparietal regions are infrequent and should raise the suspicion of sphenoid or ethmoid sinus disease or thrombosis of the superior sagittal venous sinus.

The *mode of onset,* the *variation* of the pain over time, and the *duration* of the headache, with respect both to a single attack and to the profile of a series of headaches over a period of months or years, are also useful data. At one extreme, the headache of subarachnoid hemorrhage (caused by a ruptured aneurysm) occurs as an abrupt attack that attains its maximal severity in a matter of seconds or, in the case of meningitis, it may come on more gradually, over several hours or days. Simulating the rapid onset, severe headaches of subarachnoid hemorrhage are a group of "thunderclap headaches" of diverse causes but principally cerebral venous thrombosis and vasospasm syndromes. Brief sharp pain, lasting a few seconds, in the eyeball (ophthalmodynia) or cranium ("ice-pick" pain) and "ice-cream headache" caused by pharyngeal cooling are more common in migraineurs and are significant only by reason of their benignity.

With regard to characteristic temporal patterns, migraine of the typical type usually has its onset in the early morning hours or in the daytime, reaches its peak of severity over several to 30 min, and lasts, unless treated, for 4 to 24 h, occasionally longer. Often, it is terminated by sleep. A migrainous patient having several attacks per week usually proves to have a chronic form of migraine or a combination of migraine and analgesic "overuse headache," meaning that the headache returns when the effect of the drug has worn off or rarely, this pattern is associated with some unexpected intracranial lesion. By contrast, cluster headache is characterized by the occurrence of unbearably severe unilateral orbitotemporal pain coming on within 1 or 2 h after falling asleep or at predictable times during the day and recurring nightly or daily for a period of several weeks to months; usually an individual attack of "cluster" dissipates in 30 to 45 min but may occasionally last several hours. The headache of intracranial tumor may appear at any time of the day or night; it may interrupt sleep, vary in intensity, and last a few minutes to hours as the tumor raises intracranial pressure. With posterior fossa masses, the headache tends to be worse in the morning, on awakening. Tension headaches (a vague category now called "tension-type headache") can persist with varying intensity for weeks to months or even longer; when such headaches are protracted, there is often an associated depressive illness. In general, headaches that have recurred regularly for many years prove to be migraine or tension in type.

The relationship of headache to certain biologic events and to *precipitating, aggravating, or relieving* factors can be of significance in diagnosis. Headaches that occur regularly in the premenstrual period are usually generalized and mild in degree, but attacks of migraine may also occur at this time (catamenial migraine). Headaches that have their origin in cervical spine disease are most typically intense after a period of inactivity, such as a night's sleep, and the first movements of the neck are stiff and painful. Headache, or more often face ache, from infection of the nasal sinuses may appear on awakening or in midmorning and is characteristically worsened by stooping and changes in atmospheric pressure; there is associated midfrontal or maxillary tenderness. Eyestrain after long-sustained periods of reading, or exposure to the glare of video displays, may be associated with head pain but it is transient and not an important cause of headache. In certain individuals alcohol, intense exercise (such as weight lifting), stooping, straining, coughing, and sexual intercourse are known to initiate a bursting (thunderclap) headache, lasting a few seconds to minutes. If a headache is made worse by sudden movement or by coughing or straining, an intracranial source is tentatively suggested. Migraine often occurs several hours or a day following a period of intense activity and stress ("weekend," or "letdown" migraine). Some patients have discovered that their migraine is relieved momentarily by gentle compression of the carotid or superficial temporal artery on the painful side, and others report that the carotid near the angle of the jaw is tender during the headache. Pain that is noticed when the scalp is stroked in combing or fixing the hair (allodynia) is common in migraine but could be a symptom of inflammation of the temporal arteries (temporal arteritis). Certain medications, most vasodilators such as nitroglycerin and dipyridamole but also monosodium glutamate, are apt to cause headaches. The headaches that follow a period or excessive alcohol (hangover) or a concussion are well known. In many of these instances, a propensity for migraine may create a susceptibility to headache that is induced by these precipitants.

The Brain tissue does not have nociceptors but rather interprets referred signals from other areas that can.



Where does the pain perception originate?

Headache and Other Craniofacial Pains. In: Ropper AH, Samuels MA, Klein JP, Prasad S. eds. *Adams and Victor's Principles of Neurology, 12e*. McGraw Hill; 2023. Accessed September 08, 2023.

In the introductory chapter on pain, reference was made to the necessity of determining the quality, severity, location, duration, and time course of any pain as well as the conditions that produce or relieve it. In the case of headache, a detailed history following these lines will determine the diagnosis more often than will the physical examination or imaging. Nevertheless, a few aspects of the examination are worth emphasizing. For example, auscultation of the skull may rarely disclose a bruit (with large arteriovenous malformations); palpation may disclose the tender, hardened or elevated arteries of temporal arteritis; sensitive areas overlying a cranial metastasis or an inflamed paranasal sinus may be apparent; or there may be a tender occipital nerve. Examination of neck flexion may reveal meningitis; however, apart from these special instances, examination of the head itself, although necessary, seldom discloses the diagnosis.
The *quality* of cephalic pain is essential to diagnosis but the sensation may be difficult for the patient to describe. When asked to compare the pain to some other sensory experience, the patient may allude to tightness, aching, pressure, burning, bursting, sharpness, or stabbing. Among the most important aspects is whether the headache is pulsatile, usually implying migraine, but one must keep in mind that patients sometimes use the word *throbbing* to refer to a waxing and waning of the headache

without any relation to the pulse, or use the term to transmit the severity of pain. Similarly, statements about the *intensity* of the pain reflect to some extent the patient's temperament, attitudes, and customary ways of experiencing and reacting to pain. One useful index of severity is the degree to which the pain has incapacitated the patient. A severe migraine attack seldom allows the migraineur to perform the day's work. Other rough indices of the severity of headache are its propensity to awaken the patient from sleep or to prevent sleep, and autonomic reactions to the pain such as sweating and tachycardia. The most intense cranial pains are those associated with meningitis and subarachnoid hemorrhage, which can have grave consequences, but also migraine, cluster headache, or tic douloureux that do not have the same implications.

Information regarding the *location* of a headache is informative. Migraine headache is unilateral in two-thirds of attacks and is commonly associated with nausea, vomiting, and sensitivity to light, sound, and smells. Inflammation of an extracranial artery from temporal arteritis causes pain localized to the site of the vessel. Lesions of the paranasal sinuses, teeth, eyes, and upper cervical vertebrae induce a less sharply localized pain but still one that is referred to a certain region, usually the forehead or maxilla or around the eyes. Intracranial lesions in the posterior fossa generally cause pain in the occipitonuchal region and usually are ipsilateral if the lesion is one-sided. Supratentorial lesions induce frontotemporal pain that approximates the site of an intracranial lesion. Localization, however, may also be deceiving. Pain in the frontal regions may be caused by such diverse disorders as glaucoma, sinusitis, thrombosis of the vertebral or basilar artery, pressure on the tentorium, and increased intracranial pressure. Similarly, ear pain may signify disease of the ear itself, but as often, it is referred from other regions, such as the throat, cervical muscles, spine, or structures in the posterior fossa. Periorbital and supraorbital pain, while usually indicative of local ocular disease, may reflect dissection of the cervical portion of the internal carotid artery. Headaches localized to the vertex or biparietal regions are infrequent and should raise the suspicion of sphenoid or ethmoid sinus disease or thrombosis of the superior sagittal venous sinus.

The *mode of onset,* the *variation* of the pain over time, and the *duration* of the headache, with respect both to a single attack and to the profile of a series of headaches over a period of months or years, are also useful data. At one extreme, the headache of subarachnoid hemorrhage (caused by a ruptured aneurysm) occurs as an abrupt attack that attains its maximal severity in a matter of seconds or, in the case of meningitis, it may come on more gradually, over several hours or days. Simulating the rapid onset, severe headaches of subarachnoid hemorrhage are a group of "thunderclap headaches" of diverse causes but principally cerebral venous thrombosis and vasospasm syndromes. Brief sharp pain, lasting a few seconds, in the eyeball (ophthalmodynia) or cranium ("ice-pick" pain) and "ice-cream headache" caused by pharyngeal cooling are more common in migraineurs and are significant only by reason of their benignity.

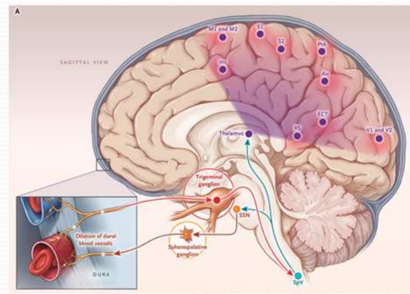
With regard to characteristic temporal patterns, migraine of the typical type usually has its onset in the early morning hours or in the daytime, reaches its peak of severity over several to 30 min, and lasts, unless treated, for 4 to 24 h, occasionally longer. Often, it is terminated by sleep. A migrainous patient having several attacks per week usually proves to have a chronic form of migraine or a combination of migraine and analgesic "overuse headache," meaning that the headache returns when the effect of the drug has worn off or rarely, this pattern is associated with some unexpected intracranial lesion. By contrast, cluster headache is characterized by the occurrence of unbearably severe unilateral orbitotemporal pain coming on within 1 or 2 h after falling asleep or at predictable times during the day and recurring nightly or daily for a period of several weeks to months; usually an individual attack of "cluster" dissipates in 30 to 45 min but may occasionally last several hours. The headache of intracranial tumor may appear at any time of the day or night; it may interrupt sleep, vary in intensity, and last a few minutes to hours as the tumor raises intracranial pressure. With posterior fossa masses, the headache tends to be worse in the morning, on awakening. Tension headaches (a vague category now called "tension-type headache") can persist with varying intensity for weeks to months or even longer; when such headaches are protracted, there is often an associated depressive illness. In general, headaches that have recurred regularly for many years prove to be migraine or tension in type.

The relationship of headache to certain biologic events and to *precipitating, aggravating, or relieving* factors can be of significance in diagnosis. Headaches that occur regularly in the premenstrual period are usually generalized and mild in degree, but attacks of migraine may also occur at this time (catamenial migraine). Headaches that have their origin in cervical spine disease are most typically intense after a period of inactivity, such as a night's sleep, and the first movements of the neck are stiff and painful. Headache, or more often face ache, from infection of the nasal sinuses may appear on awakening or in midmorning and is characteristically worsened by stooping and changes in atmospheric pressure; there is associated midfrontal or maxillary tenderness. Eyestrain after long-sustained periods of reading, or exposure to the glare of video displays, may be associated with head pain but it is transient and not an important cause of headache. In certain individuals alcohol, intense exercise (such as weight lifting), stooping, straining, coughing, and sexual intercourse are known to initiate a bursting (thunderclap) headache, lasting a few seconds to minutes. If a headache is made worse by sudden movement or by coughing or straining, an intracranial source is tentatively suggested. Migraine often occurs several hours or a day following a period of intense activity and stress ("weekend," or "letdown" migraine). Some patients have discovered that their migraine is relieved momentarily by gentle compression of the carotid or superficial temporal artery on the painful side, and others report that the carotid near the angle of the jaw is tender during the headache. Pain that is noticed when the scalp is stroked in combing or fixing the hair (allodynia) is common in migraine but could be a symptom of inflammation of the temporal arteries (temporal arteritis). Certain medications, most vasodilators such as nitroglycerin and dipyridamole but also monosodium glutamate, are apt to cause headaches. The headaches that follow a period or excessive alcohol (hangover) or a concussion are well known. In many of these instances, a propensity for migraine may create a susceptibility to headache that is induced by these precipitants.

Nociception and Pain Features

Nociceptors receive sensory information located:

- Skin, subcutaneous tissue, muscles, extracranial arteries, periosteum
- Structures of eye, ear, nasal cavities, paranasal sinus
- Venus Sinuses
- Dura and arteries within the dura
- Middle meningeal and superficial temporal arteries
- First three cervical nerves and cranial nerves passing through dura
- Cervical areas Discs, ligaments, facets, paracervical muscles, carotid and vertebral arteries, irritation of C1,2,3 (innervating neck muscles, scalp, meninges and arteries of posterior fossa) as well as pathology around the foramen magnum



Trigeminovascular system relay nociceptive signals to produce the perception of migraine pain.

N Engl J Med 2020; 383:1866-1876

Adams and Victor's Principles of Neurology, 12e. McGraw Hill; 2023.

The Journal of Headache and Pain **volume 22**, Article number: 138 (2021)

Headache and Other Craniofacial Pains. In: Ropper AH, Samuels MA, Klein JP, Prasad S. eds. *Adams and Victor's Principles of Neurology, 12e*. McGraw Hill; 2023. Accessed September 08, 2023.

These observations inform us that only certain cranial structures are sensitive to noxious stimuli: (1) skin, subcutaneous tissue, muscles, extracranial arteries, and external periosteum of the skull; (2) the delicate structures of the eye, ear, nasal cavities, and paranasal sinuses; (3) intracranial venous sinuses and their large tributaries because they are intradural; (4) parts of the dura at the base of the brain and the arteries within the dura, particularly the proximal parts of the anterior and middle cerebral arteries and the intracranial segment of the internal carotid artery; (5) the middle meningeal and superficial temporal arteries; and (6) the first three cervical nerves and cranial nerves as they pass through the dura. Interestingly, pain is practically the only sensation produced by the stimulation of these structures. Much of the pia-arachnoid, the parenchyma of the brain, and the ependyma and choroid plexuses lack sensitivity.

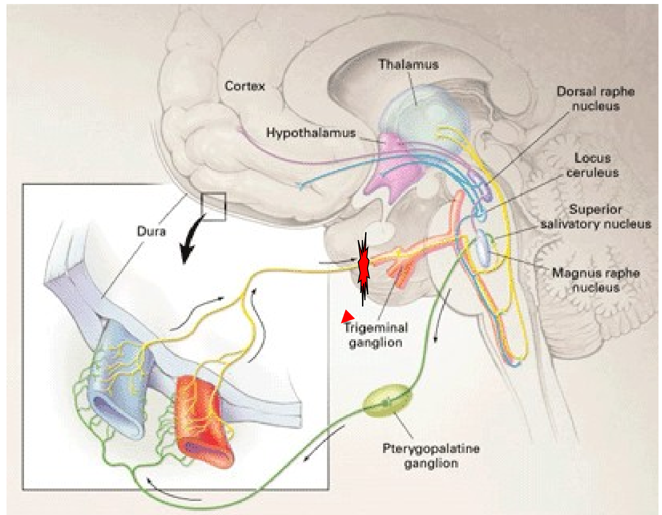
The sites of referred pain from the aforementioned structures are important in understanding the genesis of cranial pain. Pain that arises from distention of the middle meningeal artery is projected to the back of the eye and temporal area. Pain from the intracranial segment of the internal carotid artery and proximal parts of the middle and anterior cerebral arteries is felt in the eye and orbitotemporal regions.

The pathways whereby cephalic sensory stimuli are transmitted to the central nervous system (CNS) are the trigeminal nerves, particularly their first and, to some extent, second divisions, which convey impulses from the forehead, orbit, anterior and middle fossae of the skull, and the upper surface of the tentorium. The sphenopalatine branches of the facial nerve convey impulses from the nasoorbital region. The ninth and tenth cranial nerves and the first three cervical nerves transmit impulses from the inferior surface of the tentorium and all of the posterior fossa. The tentorium roughly demarcates the trigeminal from the cervical–vagal–glossopharyngeal innervation zones. The central sensory connections, which ascend through the brainstem or the cervical spinal cord and brainstem to the thalamus, are described in Chaps. 7 and 8. Sympathetic fibers from the three cervical ganglia and parasympathetic fibers from the sphenopalatine and otic ganglia are mixed with the trigeminal and other sensory fibers. These assume importance in certain headache syndromes considered further on.

Trigeminovascular relay system

Cerebral vasodilation, perivascular tissue inflammation produces activation of trigeminal complex resulting in pain.

Release of vasoactive neuropeptides including substance P, neurokinin A, serotonin, dopamine and calcitonin gene-related peptide (CGRP) lead to reciprocal vasodilation and activation of sensory neurons



Goadsby P et al.
N Engl J Med 2002;346:257-270

Stimulation of the trigeminal ganglion results in release of vasoactive neuropeptides, including serotonin, dopamine, substance P, calcitonin gene-related peptide, and neurokinin A.

Release of these neuropeptides results in the process of neurogenic inflammation. The two main components of this sterile inflammatory process are vasodilation and plasma protein extravasation.

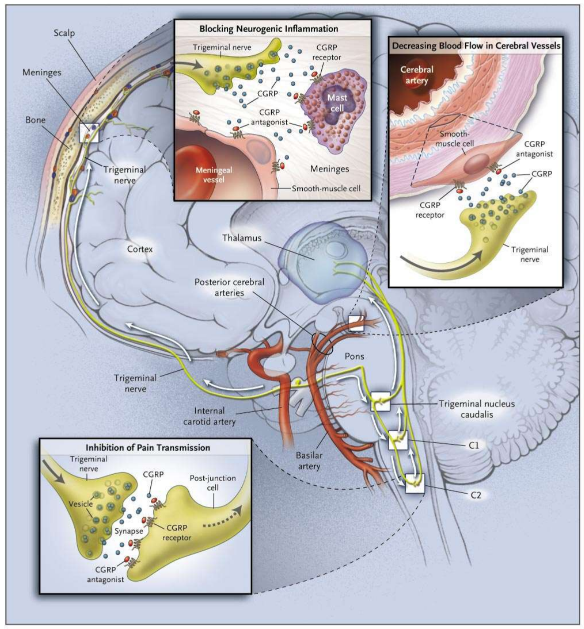
Neurogenic inflammation is responsible for the pain of migraine. Elevated levels of vasoactive neuropeptides have been found in the cerebrospinal fluid of patients with chronic migraine, suggesting chronic activation of the trigeminovascular system in these patients. Neurogenic inflammation may lead to the process of sensitization.

Dr. Pierce-Talsma's OMM notes:
- OMM hat- OMG can you totally relate why in OCMM we focused on location of the pterygopalatine ganglia- what bone restrictions could effect it? What is the pathway of the trigeminal nerve- specifically the ophthalmic division? Also how important might that trigeminal/cervical complex be?

CGRP's Role in neurovascular inflammation

CGRP is involved in vasodilation and pain transmission

CGRP Antagonist for prevention



Durham P. N Engl J Med 2004;350:1073-1075

Figure. Possible Sites of Action of the Nonpeptide CGRP-Receptor Antagonist BIBN 4096 BS. The binding of the calcitonin gene-related peptide (CGRP) antagonist to receptors on major cerebral vessels would decrease blood flow to the brain. Binding to meningeal blood vessels and mast cells would inhibit neurogenic inflammation, and binding to receptors on second-order neurons would inhibit the transmission of pain.

UpToDate

CGRP antagonists — The calcitonin gene-related peptide (CGRP) is a therapeutic target in migraine because of its hypothesized role in mediating trigeminovascular pain transmission and the vasodilatory component of neurogenic inflammation (see "Pathophysiology, clinical manifestations, and diagnosis of migraine in adults", section on 'Role of calcitonin gene-related peptide'). In 2018, the US Food and Drug Administration (FDA) approved the CGRP antagonists erenumab [36], fremanezumab [37], and galcanezumab [38] for migraine prevention.

Although probably unrelated to treatment, there have been three deaths in clinical trial subjects who received CGRP monoclonal antibodies [39-41]. Additional concerns have been raised about the potential for adverse cardiovascular, pulmonary, and psychiatric effects with CGRP antagonists [42], even though these were not observed in randomized trials.

There is little evidence to guide the use of the CGRP antagonists in specific populations (eg, children, older adults, and women during pregnancy and lactation). Some experts recommend avoiding the use of CGRP antagonists for women who are pregnant or likely to become pregnant, given the potential risks of reducing the action of CGRP during pregnancy, and avoiding CGRP antagonists for individuals with or at high risk for cardiovascular disease, since CGRP has theoretical cardioprotective and vasodilatory effects [11,43]. More data are needed regarding the long-term safety of this class of medications [42].

Erenumab — Erenumab, a human monoclonal antibody that binds to and inhibits the CGRP receptor, was modestly effective for migraine prevention in a placebo-controlled trial of over 900 adults with episodic migraine [44]. The trial randomly assigned subjects in a 1:1:1 ratio to subcutaneous injections of erenumab 70 mg, erenumab 140 mg, or placebo monthly for six months.

At baseline, the mean number of migraine days per month was 8.3 in the overall population. At months four through six, the number of migraine days per month was reduced by 3.2 in the 70 mg erenumab group, 3.7 in the 140 mg erenumab group, and 1.8 in the placebo group. The rates of adverse events were similar between the erenumab and placebo groups.
Another trial evaluated 246 subjects with treatment-resistant episodic migraine who had failed to respond to or did not tolerate between two and four previous migraine preventive medications [12]. Subjects were randomly assigned to receive erenumab 140 mg or placebo by subcutaneous injection every four weeks. At 12 weeks, the proportion of patients who had a 50 percent or more reduction in the mean number of monthly migraine days was greater for the group assigned to erenumab compared with the group assigned to placebo (30 versus 17 percent, odds ratio 2.7, 95% CI 1.4-5.2, absolute risk difference 13 percent).
The recommended dose of erenumab is 70 mg subcutaneous injection once monthly in the abdomen, thigh, or upper arm. The dose can be increased to 140 mg once monthly if needed. Injection site reactions and constipation are the most common adverse effects.
Fremanezumab — Fremanezumab, a human monoclonal antibody that binds to both isoforms of the CGRP ligand, appears to be modestly effective for prevention of episodic migraine [13,39]. One trial randomly assigned 875 adults with episodic migraine in a 1:1:1 ratio to subcutaneous fremanezumab 225 mg monthly for three months, a single dose of fremanezumab 675 mg (intended to support a quarterly dose regimen), or placebo [39]. At three months compared with placebo, the mean number of migraine days per month

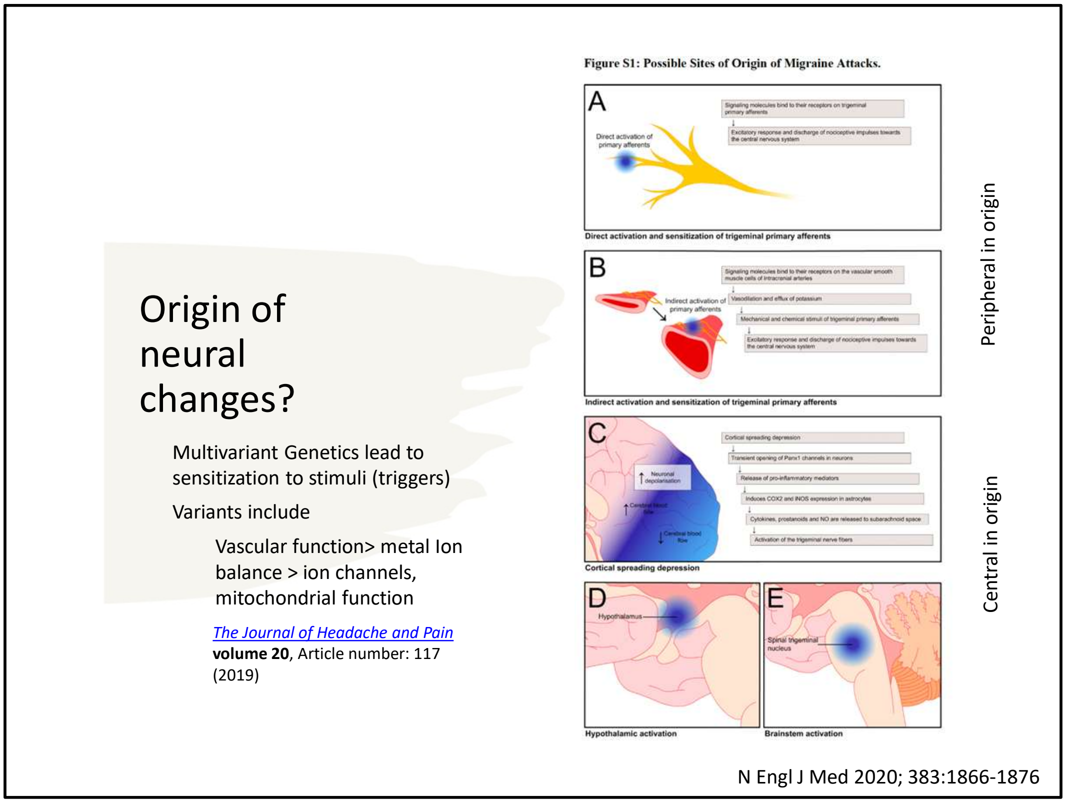
decreased by 1.5 in the fremanezumab monthly dose group and by 1.3 days in the single higher dose group. The proportion of patients with at least a 50 percent reduction in the mean number of monthly migraine days was 48 percent in the fremanezumab monthly dose group and 44 percent in the fremanezumab single higher dose group, compared with 28 percent for the placebo group.

The recommended dose of fremanezumab is 225 mg once monthly or 675 mg (given as three consecutive injections of 225 mg each) every three months administered subcutaneously in the abdomen, thigh, or upper arm. The most common adverse effects are injection site reactions.

Galcanezumab — Galcanezumab, a human monoclonal antibody that binds to the CGRP ligand, is also modestly effective for episodic migraine prevention, as demonstrated in several placebo-controlled trials [45,46]. As an example, one of these trials randomly assigned 858 patients with episodic migraine to monthly subcutaneous galcanezumab 120 mg, galcanezumab 240 mg, or placebo in a 1:1:2 ratio [47]. At six months, the mean number of migraine days per month decreased by 4.7 and 4.6 for the galcanezumab 120 and 240 mg groups, respectively, compared with 2.8 days for the placebo group.

Galcanezumab treatment is started with a loading dose of 240 mg, given as two consecutive doses of 120 mg each, followed by monthly doses of 120 mg, administered subcutaneously in the thigh, upper arm, or buttocks. Injection site reactions were the most common adverse events in clinical trials.

Other agents — A number of other agents have been studied for migraine prevention. Guidelines issued in 2012 from the AAN concluded that Petasites (butterbur) is effective for migraine prevention [48]. In addition, the AAN noted that feverfew, magnesium, certain nonsteroidal anti-inflammatory drugs, and riboflavin are probably effective.



The current state of knowledge in this field suggests that a primary neuronal dysfunction leads to a sequence of changes intracranially and extracranially that account for migraine, including the three phases of premonitory symptoms, aura, and headache.

Patients with migraine are likely to be vulnerable to outside triggers due to **mutations in the regulation of ion balance** (e.g. mutations in calcium channels subtypes and Na+/K+ ATPase subunits) in neurons. The primary neuronal dysfunction leads to a sequence of changes intracranially and extracranially that account for migrainous symptoms, including the three phases of premonitory (prodrome), aura, and headache symptoms.

Migraine headaches are typically activated upon **environmental triggers** that produces changes deep within the brain.

Prodrome: Once initiated a cascade of **central cortical depression and reduced blood flow (oligemia) spreads slowly throughout the cortex**.

Aura: Changes in nerve cell activity and blood flow, associated with a brief wave of hyperemia followed by prolonged oligemia may (18%-31% of patients with migraine) result in producing visual disturbance, numbness, or tingling, and dizziness depending on which region of the cortex is most affected.

Headache Pain: Pain results from a combination of cranial blood vessel dilation, trigeminal innervation of the blood vessels, and the reflex connections from the trigeminal system with cranial parasympathetic outflow. **Central activation of parasympathetic outflow** (see green neuron and pterygopalantine ganglion below) produces **vascular inflammation in blood vessels of the dura**. The inflammation, in turn, stimulates the trigeminal nerve (see yellow and trigeminal ganglion below) that relays nociceptive inputs to the spinal cord resulting in a severe or throbbing pain sensation. In particular, the **involvement of the ophthalmic division of the trigeminal nerve and its overlap with layers in the C1 and C2 in the spinal cord produce a type of referred pain that is perceived to be in the frontal and temporal region commonly found in the migrainous headache** (see inset). This process can cycle for 4-72 hours.

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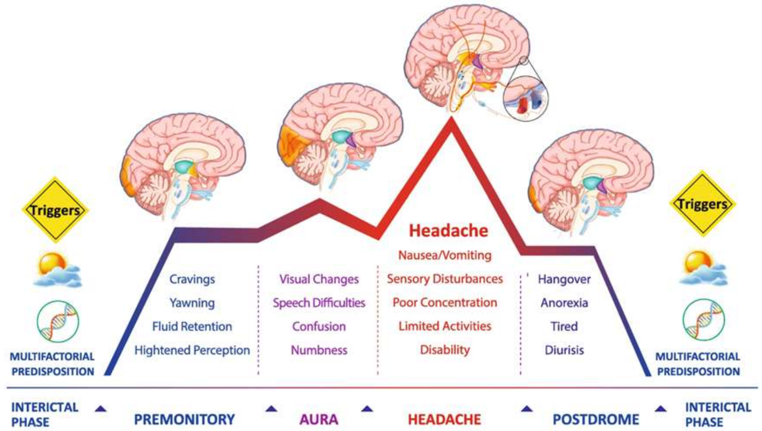
- Neurovascular-Reflex key neurotransmitters: **Stimulation of the trigeminal ganglion** results in release of vasoactive neuropeptides, including **serotonin, dopamine, substance P, calcitonin gene-related peptide, and neurokinin A.** Release of these neuropeptides results in the process of neurogenic inflammation. The two main components of this sterile inflammatory process are vasodilation and plasma protein extravasation. Neurogenic inflammation is responsible for the pain of migraine. Elevated levels of vasoactive neuropeptides have been found in the cerebrospinal fluid of patients with chronic migraine, suggesting chronic activation of the trigeminovascular system in these patients. (Neurogenic inflammation may lead to the process of sensitization. Goadsby, PJ, Lipton RB, Ferrari, MD. Migraine. Current Understanding and Treatment. N Engl J Med2002;346:257-270).

- Central Neurotransmitter Pathways involved in migraine features :The symptoms of the migraineur are likely to involve both the serotonergic and dopaminergic pathways in the brain. Peripheral serotonergic pathways that involve the vasculature are important in the cascade of events that produce migraine attacks. In addition, serotonin, released from brainstem nuclei may underlie the migraine symptoms that involve mood and appetite. Data also support a role for dopamine in the pathophysiology of certain subtypes of migraine. For example, migrainous attacks can be induced by dopaminergic stimulation. Moreover, there is **dopamine receptor hypersensitivity in migraineurs that may be exhibited via yawning, nausea and vomiting, and hypotension**. Dopamine receptor antagonists are effective therapeutic agents in migraine management, especially when given parenterally or concurrently with other antimigraine agents. (Funct Neurol. 2000;15 Suppl 3:171-81)

Pathogenesis of Migraine



1–**Vulnerability**

Genetics (vascular function, ion channels, mitochondrial function, metal ion transport)

2–**Cortical spread Depression, Oligemia**

3--**Reduced blood flow and changes in electrical impulses may result in visual disturbance, numbness or tingling, and dizziness depending on areas involved**

4–**Trigeminal Nerve/Vascular loop--*blood vessel dilation and inflammation* resulting in severe or throbbing pain**

The Journal of Headache and Pain **volume 20**, Article number: 117 (2019)

What causes migraine H/As?

The current state of knowledge in this field suggests that a primary neuronal dysfunction leads to a sequence of changes intracranially and extracranially that account for migraine, including the three phases of premonitory symptoms, aura, and headache.

Patients with migraine are likely to be vulnerable to outside triggers due to mutations in the regulation of ion balance (e.g. mutations in calcium channels subtypes and Na+/K+ ATPase subunits) in neurons. The primary neuronal dysfunction leads to a sequence of changes intracranially and extracranially that account for migrainous symptoms, including the three phases of premonitory (prodrome), aura, and headache symptoms.

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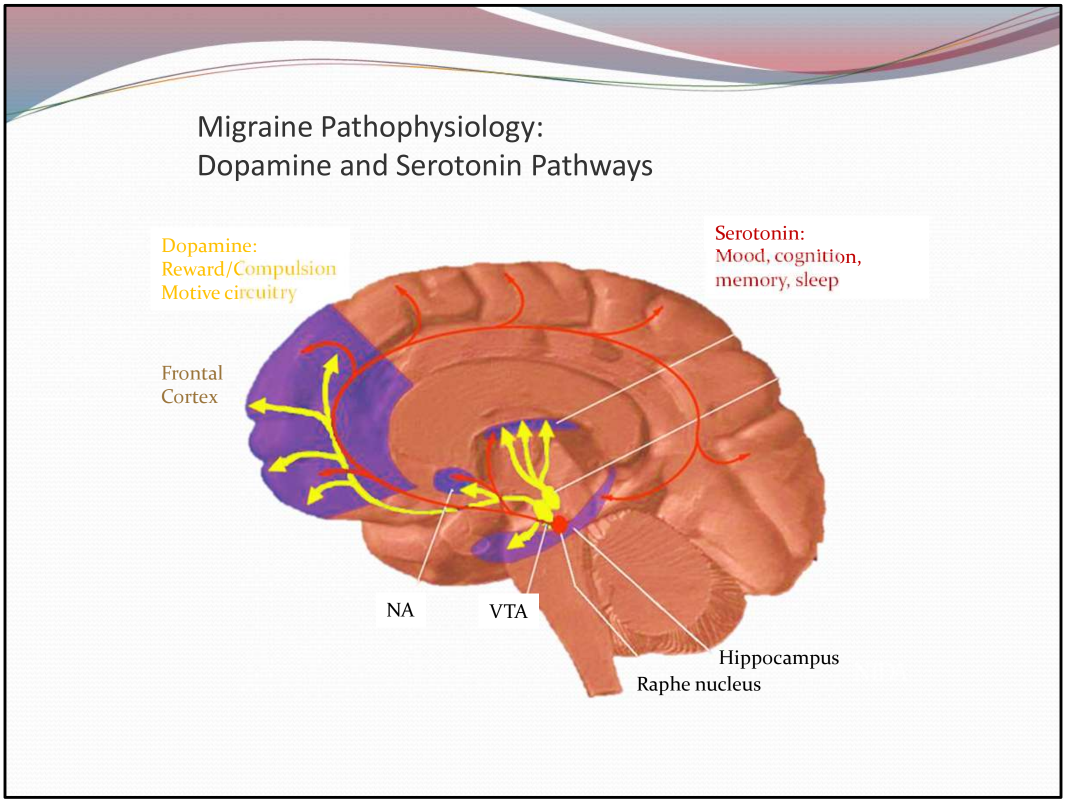
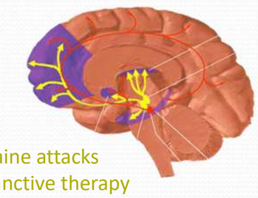


The symptoms of the migraineur are likely to involve both the serotonergic and dopaminergic pathways in the brain.

Dopamine neurons (shown in yellow) influence pleasure, motivation, motor function and saliency of stimuli or events. Serotonin (shown in red) plays a role in learning, memory, sleep and mood.

Migraine Pathophysiology: Dopamine and Serotonin Pathways

Dopamine:
Reward/Compulsion
Motive circuitry



Serotonin:
Mood, cognition,
memory, sleep

- Dopamine agonist can stimulate migraine attacks
- Dopamine antagonist are used as adjunctive therapy for acute attacks
- Prodromal symptomatology (mood changes, yawning, drowsiness, food craving), accompanying symptoms (nausea, vomiting, hypotension) and postdromal symptoms (mood changes, drowsiness, tiredness) may be related to dopaminergic activation. The dopaminergic system could also play a role in the headache phase, either by taking part in nociception mechanisms, or by regulating cerebral blood flow.

- Serotonin (5HT)
  - Selective agonist to subtypes of 5HT receptors can stop an acute attack
- The symptoms that involve mood and appetite may be related to the alteration in central serotonergic neurotransmission.

The symptoms of the migraineur are likely to involve both the serotonergic and dopaminergic pathways in the brain.

Dopamine neurons (shown in yellow) influence pleasure, motivation, motor function and saliency of stimuli or events. Serotonin (shown in red) plays a role in learning, memory, sleep and mood.

This slide is intended to introduce the role of both dopamine and serotonin in the pathophysiology of migraine headaches.

Peripheral serotonergic pathways that involve the vasculature are important in the cascade of events that produce migraine attacks. In addition, Serotonin, released from brainstem serotonergic nuclei, also plays an important role in the pathogenesis of migraine. The symptoms that involve mood and appetite may be related to the alteration in central serotonergic neurotransmission.

This is probably mediated via its direct action upon the cranial vasculature, through its role in central pain control pathways and through cerebral cortical projections of brainstem serotonergic nuclei.

Data also support a role for dopamine in the pathophysiology of certain subtypes of migraine. Most migraine sx can be induced by dopaminergic stimulation. Moreover, there is dopamine receptor hypersensitivity in migraineurs, as demonstrated by the induction of yawning, N/V, hypotension, and other sx of a migraine attack by dopaminergic agonists at doses that do not affect nonmigraine sufferers.

Dopamine receptor antagonists are effective therapeutic agents in migraine, especially when given parenterally or concurrently with other antimigraine agents.

Funct Neurol. 2000;15 Suppl 3:171-81.

Dopamine involvement in the migraine attack.

Fanciullacci M, Alessandri M, Del Rosso A.

Source

Department of Internal Medicine, Headache Centre, University of Florence, Italy.

Abstract

Clinical evidence and recent genetic findings seem to indicate an involvement of dopamine in the pathophysiology of the migraine attack. Prodromal symptomatology (mood changes, yawning, drowsiness, food craving), accompanying symptoms (nausea, vomiting, hypotension) and postdromal symptoms (mood changes, drowsiness, tiredness) may be related to dopaminergic activation. The dopaminergic system could also play a role in the headache phase, either by taking part in nociception mechanisms, or by regulating cerebral blood flow. A body of pharmacological findings seems to support this involvement. Migraine patients, between attacks, show a higher responsiveness to acute administration of dopaminergic agents. Apomorphine administration induces in migraineurs more yawns as well other dopaminergic symptoms e.g. nausea, vomiting, dizziness. Migraine has been associated with hypotension and, occasionally, with syncope. Bromocriptine causes severe orthostatic syndrome in migraine patients. Both piribedil and apomorphine markedly increase cerebral blood flow of migraine patients, thus indicating enhanced responsiveness of dopamine receptors which are involved in cerebral blood flow regulation. Interictal dopaminergic hypersensitivity has also been demonstrated by means of neuroendocrine tests. Altered dopaminergic control of prolactin secretion exists in migrainous women. L-deprenyl, a MAO-B inhibitor, is significantly more effective in reducing prolactin levels in migraineurs than in controls. Taken together, these findings support the view that hypersensitivity of peripheral and central dopaminergic receptors is a specific migraine trait. Finally, a high density of lymphocytic D5 receptors has been found in migraine sufferers, thus suggesting their upregulation. Therefore, the hypothesis that dopaminergic activation is a primary pathophysiological component in certain subtypes of migraine, namely those characterized by marked dopaminergic symptomatology, has been advanced. From this perspective, a blockade of

dopaminergic hyperresponsive receptors can be considered as a rationale for the therapy of migraine.

Headache Work-up

- Video 6: Dr. Kinnevey

Migraine Trigger Identification

Avoid/be aware of triggers:

- Menses
- Sleep deprivation
- Oversleeping
- Skipping meals
- Poor diet
- Foods with nitrites (nitrites (processed meats, canned, smoked, aged meats)
- ETOH
- Bright lights
- Altitude
- Smoking
- Red Wine
- Caffeine
- Foods high in tyramine (chocolate, aged cheeses, yogurt, sour cream, soy sauce, banana, avocado, nuts, yeast extract (beer)

Aug/2017	START TIME	FOOD	MEDICATIONS	ACTIVITY	SYMPTOMS
1	7:00 PM	glass of wine			nausea
2					
3	8:30 PM	glass of wine			irritability
4	7:30 PM	coffee			nausea
5					
6					
7	12:30 PM	glass of wine			spots in vision
8					
9					
10	9:30 AM			jogging outside	nausea

Headache diary

- Foods with MSG
- Aspartame
- New medications
- Strong emotions (stress, anger, anxiety, depression, fear)

https://www.webmd.com/migraines-headaches/migraine-triggers-your-personal-checklist

Evidence for MSG as trigger is low but suggests possibility: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4870486/

Labs

- ESR
  - If suspected temporal arteritis
- CBC
  - If suspected infection
- CMP or other labs
  - If suspected systemic illness

Imaging

American College of Radiology ACR Appropriateness Criteria

https://acsearch.acr.org/docs/69482/Narrative/

- **No need for imaging[5]**
  - Age<50, features of a primary headache, history of similar headaches, no abnormal neurologic findings, no concerning change in headache pattern, no high-risk comorbid conditions, no new or concerning findings on history or PE
- **Urgent Imaging[3,5]**
  - Thunderclap headache (a few seconds to a minute)
  - Headache with altered mental status or seizure
  - Prior intervention (suspect deficits)
  - Acute or subacute neck pain with horner syndrome and neurologic deficit (cervical artery dissection)
  - Headache with suspected meningitis or encephalitis (fever, altered mental status, nuchal rigidity)
  - Headache with orbital or periorbital symptoms (visual impairment, pain)
- **Routine Imaging[3,5]**
  - Change in headache characteristics (severity, side-shift, worsening)
  - Accompanied by neurologic deficit or abnormality (disequilibrium, pronator drift, weakness, papilledema)
  - Headache in immunocompromised patients and cancer patients

Choosing Wisely

An initiative of the ABIM Foundation

ABIM FOUNDATION

ACR AMERICAN COLLEGE OF RADIOLOGY

Imaging tests for headaches

When you need a CT scan or MRI—and when you don't

CT scans and MRIs are called imaging tests because they take pictures, or images, of the inside of the body. Many people who have very painful headaches want a CT scan or an MRI. They want to find out if their headaches are caused by a serious problem, such as a brain tumor. But most of the time you don't need these tests. Here's why:

Imaging tests rarely help.

Doctors see many patients for headaches. And most of them have migraines or headaches caused by tension. Both kinds of headaches can be very painful. But a CT scan or an MRI rarely shows why the headache occurs. And they do not help you ease the pain.

A doctor can diagnose most headaches during an office visit. The doctor asks you questions about your health and your symptoms. This is called a medical history. Then the doctor does a test of your reflexes, called a neurological exam. If your medical history and exam are normal, imaging tests usually will not show a serious problem.

CT scans have risks.

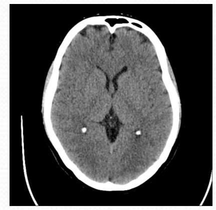
A CT scan of the head uses a low radiation dose. This may slightly increase the risk of harmful effects. Risks from radiation exposure may add up, so it is best to avoid unnecessary radiation.

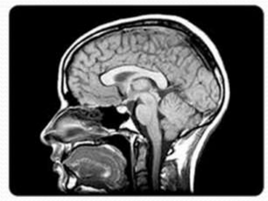
Choosing Wisely:

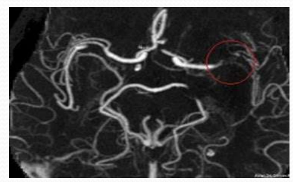
https://www.choosingwisely.org/wp-content/uploads/2018/02/Imaging-Tests-For-Headaches-ACR.pdf

Which Imaging?

- Head CT
  - Best for bone abnormalities and hemorrhage/fluid
  - Quick- used for trauma
  - Good blog for emergent head CT reads
    - https://www.downeastem.org/downeastem/2016/11/19/how-to-read-a-head-ct-andrew-perron-md
    - KNOW difference between epidural hematoma, subarachnoid hemorrhage, and subdural hematoma imaging appearance
- Head MRI
  - With and without contrast- good for looking for brain tumors and details
- CTA/MRA- IV contrast
  - Imaging vessels (arteries, veins or both)

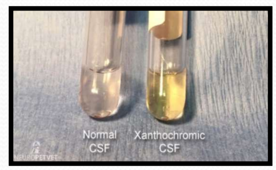






Lumbar Puncture

- Indicated for:
  - Clinical suspicion of subarachnoid hemorrhage with a negative CT, or suspicion of infectious, inflammatory or neoplastic etiology of headache or normopressure hydrocephalus
  - Also, MS, Guillain-Barre, paraneoplastic syndromes
- **Contraindications: raised intracranial pressure, thrombocytopenia or bleeding disorders, suspected spinal epidural abscess**
- If suspect elevated intracranial pressure- CT first
- Looking for:
  - Opening pressure
  - Color/turbidity
  - Xanthochromia (yellow color)
  - Red cells
  - White cells
  - White cell differential count
  - Protein
  - Glucose
  - Venereal disease research laboratory test
  - Gram stain
  - Acid fast bacilli smear
  - Fungal prep
  - Cultures
    - Routine, fungal, mycobacterial
  - Cryptococcal antigen
  - Herpes simplex virus
- Complications:
  - **Headache**, bleeding, cerebral herniation, neurologic symptoms, back pain, infection



https://www.uptodate.com/contents/lumbar-puncture-technique-indications-contraindications-and-complications-in-adults

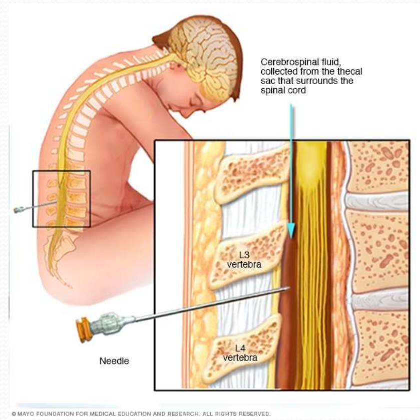
"LP can produce or hasten herniation in all conditions in which it may occur spontaneously. The withdrawal of CSF removes the stopper from below, thus adding to the effects of the compression from above, and increasing the brain shifts already present."

That's why a craniotomy can relieve the pressure above and help prevent worrisome downward herniation that compresses the brainstem. But if you relieve the pressure below, the brain shift could worsen as it tries to move down into the lower pressure area.

When a neuroradiologist looks at a head CT prior to an LP, he/she is looking for signs of brain shift that are already occurring or a bleed or mass that would be causing elevated ICP. A CT can't directly measure ICP. That is only measured by opening pressure on an LP or using catheters that are surgically inserted into one of the lateral ventricles. They are working on developing noninvasive methods of ICP but none are commonly used yet.

Post Dural Puncture Headache

- 24-48 hours after an meningeal puncture - can be as late as 7 days after
- Posturally related- better with lying down, worse with sitting or standing
- Bilateral frontal or occipital +/- neck stiffness, nausea, vomiting, photophobia, diplopia, tinnitus, mental status changes
- 85% resolve within 5 days- rarely can last for months
- Leak occurs faster than production, CSF volume shifts, brain shifts caudally placing traction on structures causing pain and various symptoms
- To decrease risk:
  - Smaller needle size, non cutting needle tip, parallel axis when inserting
- Can be treated with epidural blood patch (injecting patient's own blood into the epidural space to compress the thecal sac and increase CSF pressure)



PDPH related to neuraxial anesthesia is most common in obstetric patients. The reported incidence of PDPH after unintentional dural puncture (UDP) with an epidural needle (which is larger in diameter than a spinal needle) for labor analgesia is between 81 and 88 percent [9,10]. UDP occurs in <1 to as high as 6 percent of epidural placements.
https://www.uptodate.com/contents/post-dural-puncture-headache

Headache, which occurs in 10 to 30 percent of patients, is one of the most common complications following a diagnostic LP procedure.
https://www.uptodate.com/contents/lumbar-puncture-technique-indications-contraindications-and-complications-in-adults

Headache Treatments

Video 7: Dr. Gayer

Primary Headache Key Treatment Principles

- Consider whether treatment needs to be preventative or abortive
- **Focus on prevention and education**
  - Avoiding triggers
  - Reduction of medication overuse
  - Discontinue smoking
  - Regular eating and sleeping patterns
  - Exercise
  - Biofeedback and behavioral treatment
  - Physical therapy
  - OMT
- **Pharmacologic**
- **Interventional treatment**
  - Neuroblockade (nerve, facet, epidural space)
  - Radio-frequency and cryolysis procedures
  - Implantations and stimulation
  - Hospital and rehabilitation programs

Comprehensive Assessment: Osteopathic management of headaches begins with a thorough patient evaluation. The osteopathic physician takes a detailed medical history, including information about the frequency, duration, and characteristics of the headaches. They also assess lifestyle factors, stress levels, posture, and other relevant aspects of the patient's health.

The treatment of primary headaches, such as tension-type headaches, migraines, and cluster headaches, typically involves a combination of lifestyle modifications, acute symptom management, and, in some cases, preventive measures. It's important to note that individual responses to treatments may vary, so it's crucial for individuals with primary headaches to work closely with healthcare professionals to develop a personalized treatment plan. Here are some key treatment principles and expert opinions for primary headaches:

1. **Accurate Diagnosis:** Proper diagnosis is essential. Consult with a healthcare provider who specializes in headache disorders to confirm the type of primary

headache you're experiencing. This will guide treatment choices.

2. **Lifestyle Modifications:** Lifestyle changes can play a significant role in headache management. These include:
    1. **Regular Sleep:** Maintain a consistent sleep schedule and aim for 7-9 hours of quality sleep per night.
    2. **Stress Reduction:** Practice relaxation techniques, such as deep breathing, meditation, or yoga, to manage stress.
    3. **Diet:** Identify and avoid trigger foods or drinks that can exacerbate headaches (common triggers include alcohol, caffeine, processed foods, and certain additives).
    4. **Hydration:** Ensure you are adequately hydrated throughout the day.
    5. **Regular Exercise:** Engage in regular physical activity as it can help reduce the frequency and intensity of headaches.

3.Acute Symptom Management: When a headache occurs, it's important to have effective medications on hand. Over-the-counter pain relievers, such as ibuprofen or acetaminophen, may provide relief for tension-type headaches. For migraines, prescription medications such as triptans or gepants are often more effective. Consult your healthcare provider for the best acute treatment option for your specific headache type.

4. **Osteopathic Manipulative Treatment (OMT):** OMT is a cornerstone of osteopathic care. It involves using the hands to diagnose, treat, and prevent various conditions, including headaches. OMT techniques may include:
    - **Myofascial release:** Gentle pressure and stretching of soft tissues to relieve muscle tension and improve blood flow.
    - **Cranial osteopathy:** Manipulation of the bones of the skull to address restrictions in cranial motion, which some believe can affect headache patterns.
    - **Muscle energy techniques:** Active engagement of the patient's muscles to release tension and improve joint mobility.
    - **Counterstrain:** Gentle positional release techniques to alleviate pain and muscle spasm.

5.Preventive Medications: If your headaches are frequent or severe, your healthcare provider may recommend preventive medications. These medications are taken daily to reduce the frequency and severity of headaches. Common preventive medications include beta-blockers, antiepileptic drugs, and certain antidepressants. The choice of medication should be tailored to your specific needs and health profile.

6.Behavioral Therapies: Cognitive-behavioral therapy (CBT) and biofeedback techniques can be helpful in managing primary headaches, especially migraines. These therapies can help you learn to recognize and manage triggers and develop coping strategies.

7.Alternative Therapies: Some individuals find relief through complementary and alternative therapies, such as acupuncture, massage therapy, or herbal remedies. It's important to consult with your healthcare provider before trying these approaches to ensure they are safe and appropriate for your condition.

8. **Regular Follow-Up:** Headache management should be an ongoing process. Regularly follow up with your healthcare provider to assess the effectiveness of your treatment plan and make necessary adjustments.

9. **Patient Education:** Understanding your headache triggers and patterns is essential. Keep a headache diary to track your symptoms, potential triggers, and medication use. This information can help you and your healthcare provider make informed decisions about treatment.

10. **Avoid Medication Overuse:** Be cautious about using acute headache medications too frequently, as this can lead to medication-overuse headaches. Follow your healthcare provider's recommendations for medication use.

11. **Emergency Plan:** If you have cluster headaches or other severe forms of primary headache, work with your healthcare provider to develop an emergency plan for rapid relief during attacks.

Remember that what works best for managing primary headaches can vary from person to person. Therefore, it's crucial to collaborate closely with a healthcare provider experienced in headache management to develop a tailored treatment plan that addresses your specific needs and goals.

Interventional treatments are a subset of medical procedures that aim to alleviate pain or manage various medical conditions by directly targeting specific anatomical structures or neural pathways. These treatments can be particularly beneficial for individuals with chronic pain or neurological conditions that are not adequately managed through conservative therapies alone. Below, I'll explain and provide expert opinions using practice guidelines for several interventional treatment modalities:

1.Neuroblockade (Nerve, Facet, Epidural Space):

1. **Explanation:** Neuroblockade involves the injection of medications, such as local anesthetics or steroids, near or directly into nerves, nerve roots, facet joints, or the epidural space. This procedure is used to block or reduce pain signals from these areas, providing relief for various conditions, including chronic back pain, sciatica, and facet joint arthropathy.
2. **Expert Opinion and Practice Guidelines:** The American Society of Interventional Pain Physicians (ASIPP) provides guidelines and recommendations for neuroblockade procedures. These guidelines emphasize the importance of proper patient selection, informed consent, precise injection techniques, and the use of fluoroscopic or ultrasound guidance for accurate needle placement. They also stress the need for post-procedure monitoring and evaluation of outcomes to assess the efficacy of the intervention.

2.Radio-Frequency and Cryolysis Procedures:

1. **Explanation:** Radio-frequency (RF) ablation and cryolysis procedures involve the use of heat (RF) or extreme cold (cryolysis) to selectively destroy nerves responsible for transmitting pain signals. These procedures are commonly used for conditions like chronic joint pain, sacroiliac joint pain, and facet joint pain.
2. **Expert Opinion and Practice Guidelines:** Expert consensus, including guidance from organizations like the North American Neuromodulation Society (NANS), underscores the importance of careful patient evaluation, proper localization of target nerves, and the use of advanced imaging techniques to enhance precision during RF and cryolysis procedures. Long-term pain relief and functional improvement are often seen as success criteria.

1.Implantations and Stimulation:

1. **Explanation:** Implantable devices, such as spinal cord stimulators (SCS) and peripheral nerve stimulators (PNS), are used to modulate pain signals and improve the quality of life for patients with chronic pain conditions, such as failed back surgery syndrome or complex regional pain syndrome (CRPS).
2. **Expert Opinion and Practice Guidelines:** Organizations like the International Neuromodulation Society (INS) provide guidance on the selection of appropriate candidates for implantable devices, as well as the surgical techniques and post-implantation care. Expert opinion and guidelines emphasize careful patient selection, comprehensive preoperative assessments, and ongoing follow-up to monitor device efficacy and address any issues or complications.

2.Hospital and Rehabilitation Programs:

1. **Explanation:** Hospital-based pain management and rehabilitation programs are multidisciplinary approaches that incorporate various treatments, including medication management, physical therapy, psychological counseling, and interventional procedures, to address chronic pain and functional limitations.
2. **Expert Opinion and Practice Guidelines:** The American Chronic Pain Association (ACPA) and the American Pain Society (APS) have published guidelines emphasizing the importance of a biopsychosocial approach to pain management. These programs should include individualized treatment plans, patient education, and a focus on improving daily functioning and quality of life. Collaboration among healthcare providers from different disciplines is essential in these programs.

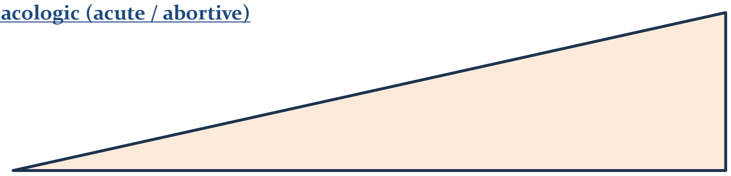
In all interventional treatments, patient safety, informed consent, and comprehensive assessment are paramount. Additionally, expert opinion and practice guidelines continually evolve to reflect advances in medical knowledge and technology. Therefore, healthcare providers performing these interventions should stay current with the latest evidence-based practices and adhere to the recommendations provided by relevant professional societies and organizations to ensure the best possible outcomes for their patients.

Migraine Treatment

Nonpharmacologic

Avoid dehydration, wear sunglasses, sleep, OMM, meditation, biofeedback, psychotherapy, acupuncture

Pharmacologic (acute / abortive)



Mild **Moderate --- Severe** **Status Migrainous**

Pharmacologic (Preventive)

Neuromodulation

Single-pulse transcranial magnetic stimulation (sTMS)
Noninvasive vagus nerve stimulation (nVNS)
Remote electrical neuromodulation
Transcutaneous supraorbital nerve stimulation

Osteopathic Manipulative Treatment (OMT): OMT is a cornerstone of osteopathic care. It involves using the hands to diagnose, treat, and prevent various conditions, including headaches. OMT techniques may include:

- **Myofascial release:** Gentle pressure and stretching of soft tissues to relieve muscle tension and improve blood flow.
- **Cranial osteopathy:** Manipulation of the bones of the skull to address restrictions in cranial motion, which some believe can affect headache patterns.
- **Muscle energy techniques:** Active engagement of the patient's muscles to release tension and improve joint mobility.
- **Counterstrain:** Gentle positional release techniques to alleviate pain and muscle spasm.

Neuromodulation

Single-pulse transcranial magnetic stimulation (sTMS) is FDA approved for the acute treatment of migraine. Two pulses can be applied at the onset of an attack, and this can be repeated. The use of sTMS is safe where there is no cranial metal implant, and offers an option to patients seeking nonpharmaceutical approaches to treatment. Similarly, a noninvasive vagus nerve stimulator (nVNS) is FDA approved for the treatment of migraine attacks in adults. One to two 120-s doses may be applied for attack treatment. Remote electrical neuromodulation using a smartphone app that stimulates the upper arm for 30–45 min is also effective for treatment of acute migraine, as is transcutaneous supraorbital nerve stimulation for 60 min; both are FDA approved.

NEUROMODULATIONLimited data from small randomized controlled trials suggest benefit of various forms of neuromodulation for the treatment of acute migraine pain. These treatments stimulate the central or peripheral nervous system with an electrical current or a magnetic field [144]. They are options for patients who prefer nonpharmacologic treatments or who have an inadequate response, inability to tolerate, or contraindications to drug treatments for migraine.

Transcutaneous supraorbital nerve stimulation — Transcutaneous supraorbital nerve stimulation can reduce migraine pain intensity, as shown in the ACME trial, which randomly assigned 109 subjects having an acute migraine attack of at least three hours duration to a one-hour treatment session with external trigeminal nerve stimulation or sham stimulation [145]. Pain was assessed using a visual analog scale, ranging from 0 (no pain) to 10 (maximum pain). The mean reduction in pain intensity at one hour was greater for the true stimulation group compared with the sham group (-3.46 versus -1.78). There were five minor adverse events and no serious adverse events in the true stimulation group.

The supraorbital transcutaneous trigeminal nerve stimulator device used in this trial is approved for marketing in the United States, Canada, Europe, and several additional countries. The evidence pertaining to migraine prevention is reviewed separately. (See "Preventive treatment of episodic migraine in adults", section on 'Neuromodulation'.)

Remote electrical neuromodulation — Data from several trials suggest that a device applying nonpainful electrical skin stimulation can reduce acute migraine pain [146-148]. The stimulating device, controlled with a smartphone, consists of an armband with rubber electrodes and a power source. The armband is applied, and stimulation started as soon as possible after the onset of a migraine attack. In a sham-controlled crossover pilot trial of 71 patients, the proportion of responders was higher with active stimulation compared with sham stimulation [146].

A later trial of 252 adults with episodic migraine tested randomized patients in a 1:1 ratio to receive 30 to 45 minutes of active stimulation (frequency 100 to 120 Hz, pulse width 400 microseconds, and output current up to 40 mA adjusted by the patient) or sham stimulation (pulse frequency approximately 0.08 Hz, modulated pulse width 40 to 550 microseconds) [147]. Sham stimulation was designed to produce a sensation similar to active stimulation but with a frequency too low to induce pain inhibition. At two hours after treatment, more patients assigned to active stimulation were pain-free (37 percent, versus 18 percent with sham, absolute risk difference [ARD] 19 percent) or had pain relief, defined as an improvement from severe or moderate pain to mild or none, or from mild pain to none (67 versus 39 percent, ARD 28 percent). Mild device-associated adverse events occurred in approximately 4 percent and included a warm sensation, arm or hand numbness, redness, itching, tingling, muscle spasm, arm pain, shoulder pain, and neck pain. There were no serious adverse events.

The remote electrical neuromodulation device used in this trial is approved for marketing in the United States.

Transcranial magnetic stimulation — The efficacy of single-pulse transcranial magnetic stimulation (TMS) was demonstrated in a sham-controlled trial of 201 adults with episodic migraine with aura [149]. The analysis was based upon 164 patients who treated at least one attack of migraine during the aura phase. Pain freedom at two hours post-treatment was significantly greater with the TMS device compared with sham stimulation (39 versus 22 percent, absolute risk reduction 17 percent, 95% CI 3-31 percent). Furthermore, significance for a sustained pain-free response was maintained at both 24 and 48 hours. There were no serious adverse events related to use of the device.

The portable TMS device is available in the United Kingdom and the United States. The TMS device may prove to be useful as a second-line intervention for those who have migraine that does not respond to first-line therapy with triptans or other agents discussed above or who are unable to take these agents because of contraindications or intolerance. TMS should not be used to treat migraine for patients who have epilepsy, since there is theoretical concern that TMS could trigger seizures [150].

Noninvasive vagus nerve stimulation — Noninvasive vagus nerve stimulation (nVNS) may be beneficial for the acute treatment of episodic migraine. The PRESTO trial randomly assigned over 240 participants to treatment with a proprietary nVNS device or a sham device [151]. Subjects were instructed to self-administer stimulation for 120 seconds each to right and left sides of the neck within 20 minutes from the onset of migraine pain, and to repeat stimulation, if not pain-free, at 15 minutes and 120 minutes. More patients achieved pain freedom with nVNS compared with sham, but the difference was not significant at 120 minutes (30 versus 20 percent). The most common adverse effects in the nVNS group were discomfort at the application site and nasopharyngitis. The nVNS device used in this trial is approved for marketing in the United States and European Union.

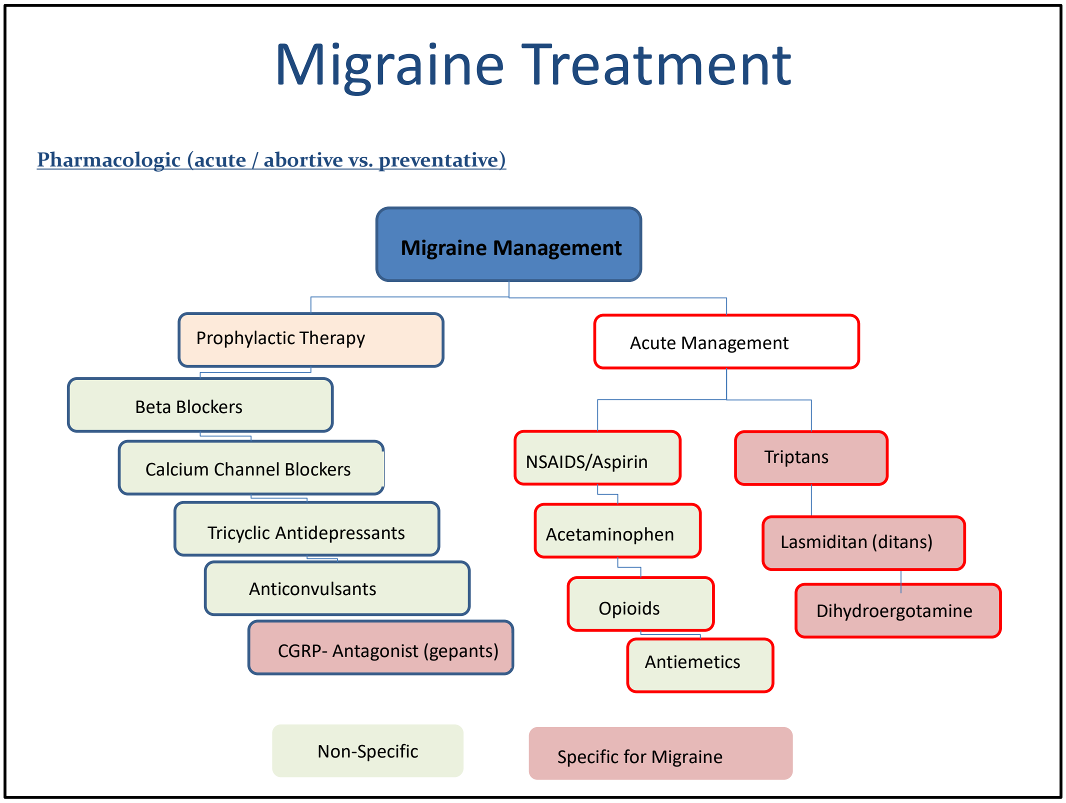
Triptans- 60-80% will have relief with one dose- a second dose can be taken 2-24 hours after first- no more for 24 hours after 2nd dose- they should not be used with other medications that cause vasoconstriction- avoid with ischemic heart disease-uncontrolled hypertension or in patients with complicated auras- not to be used in pregnancy (though some studies have not shown risks in birth defects nor complications)

Isometheptane/ergots/DHE- no in pregnancy and not to be used with basilar migraine

For preventative
Effective first line- beta blockers 60-80 mg day propranolol- side effects can be limiting
Anticonvulsants- valproic acid, carbamazepine
Antidepressants- amitriptyline 10-25mg at bedtime- helps with sleep- side effects from anticholinergic action
CAUTION with triptans and other antidepressants- serotonin syndrome)
There is an association with depression and migraine
Some literature states not enough evidence to use CA channel blockers- though others advocate for them

Caution with butterbur- good evidence it works but not FDA regulated and if you get the kind that is not free of pyrrolizidine alkaloids (PAP can be highly hepatotoxic and may cause cancer.

Information obtained from Dr. Stacey Pierce-Talsma’s 2018 lecture and
https://www.uptodate.com/contents/preventive-treatment-of-migraine-in-adults



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Acute: Symptomatic

- Triptans: Migraine Attacks
  - Naratriptan, Rizatriptan, Sumatriptan ^{OPN}, Frovatriptan, Almotriptan, Eletriptan, Zolmitriptan ^{OPN*}
- Ditan: Lasmiditan
- NSAIDS
  - Acetaminophen, aspirin, caffeine, naproxen, ibuprofen, tofenacin acid, diclofenac, ketorolac ^{P}
- Opioids
  - Meperidine, Butorphanol ^{N}, etc. most covered with opioid lecture
- Dopamine Antagonists
  - Metoclopramide ^{OP}, Prochlorperazine ^{OP}, Chlorpromazine ^{OP}
- Ergot alkaloids
  - Dihydroergotamine ^{OPN}
  - Ergotamine

Chronic: Prophylactic

- Beta-blocker
  - Propranolol
- Tricyclic antidepressant
  - Amitriptyline, nortriptyline (covered antidepressant lecture)
- Anticonvulsants
  - Topiramate, valproate, gabapentin, lamotrigine (covered in anti-seizure lecture)
- Calcium channel blocker
  - Verapamil
- CGRP antagonist
  - Gepants: Rimegepant, Ubrogepant, Zavegepant ^{N}
  - Erenumab. Fremanezumab, galcanezumab

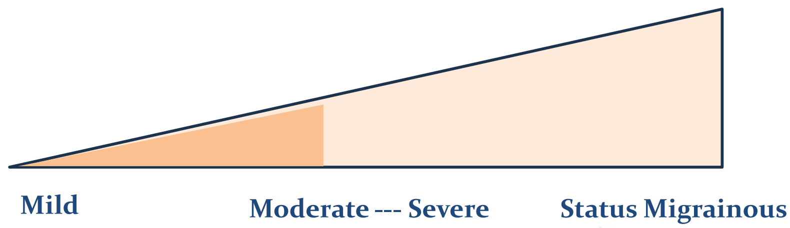
Oral (O), Nasal (N), Parenteral (P)

This represents your drug list for the migraine section of this talk.

- List the drugs used for prevention (beta blockers, tricyclic antidepressants, anticonvulsants, serotonergic drugs) and drugs used to treat acute attacks (triptans, NSAIDs, dopamine antagonist, opioids, ergot alkaloids).

Migraine Treatment

Pharmacologic (acute / abortive)



- NSAIDs or Acetaminophen
- Non- responsive to above then Triptans (multiple classes and forms)- Don't give up after trying just one
- Combinations of above
- Vomiting add antiemitic and avoid oral drugs (use nasal versions of above)
  - Dopamine Antagonist (metoclopramide, Prochlorperazine, chlorpromazine)
  - 5HT3 antagonist (covered separately)

Triptans- 60-80% will have relief with one dose- a second dose can be taken 2-24 hours after first- no more for 24 hours after 2nd dose- they should not be used with other medications that cause vasoconstriction- avoid with ischemic heart disease- uncontrolled hypertension or in patients with complicated auras- not to be used in pregnancy (though some studies have not shown risks in birth defects nor complications)

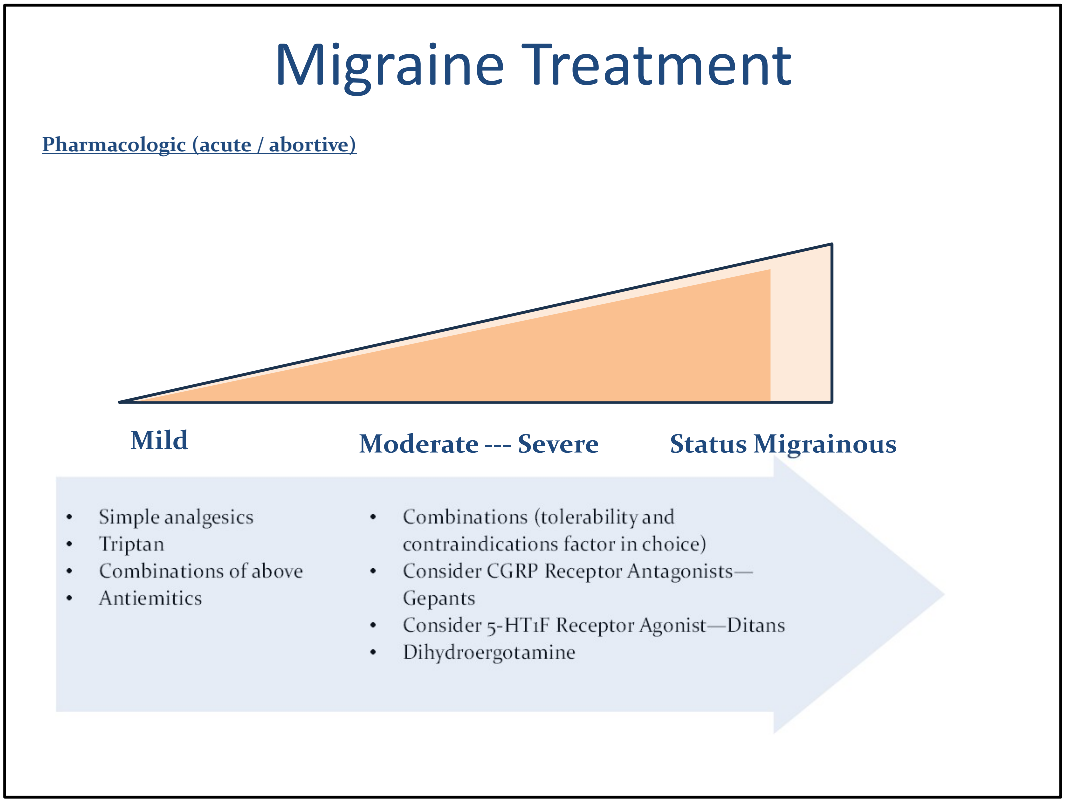
Isometheptane/ergots/DHE- no in pregnancy and not to be used with basilar migraine

For preventative
Effective first line- beta blockers 60-80 mg day propranolol- side effects can be limiting
Anticonvulsants- valproic acid, carbamazepine
Antidepressants- amitriptyline 10-25mg at bedtime- helps with sleep- side effects from anticholinergic action
CAUTION with triptans and other antidepressants- serotonin syndrome)
There is an association with depression and migraine
Some literature states not enough evidence to use CA channel blockers- though others advocate for them

Caution with butterbur- good evidence it works but not FDA regulated and if you get

the kind that is not free of pyrrolizidine alkaloids (PAP can be highly hepatotoxic and may cause cancer.

Information obtained from Dr. Stacey Pierce-Talsma’s 2018 lecture and https://www.uptodate.com/contents/preventive-treatment-of-migraine-in-adults



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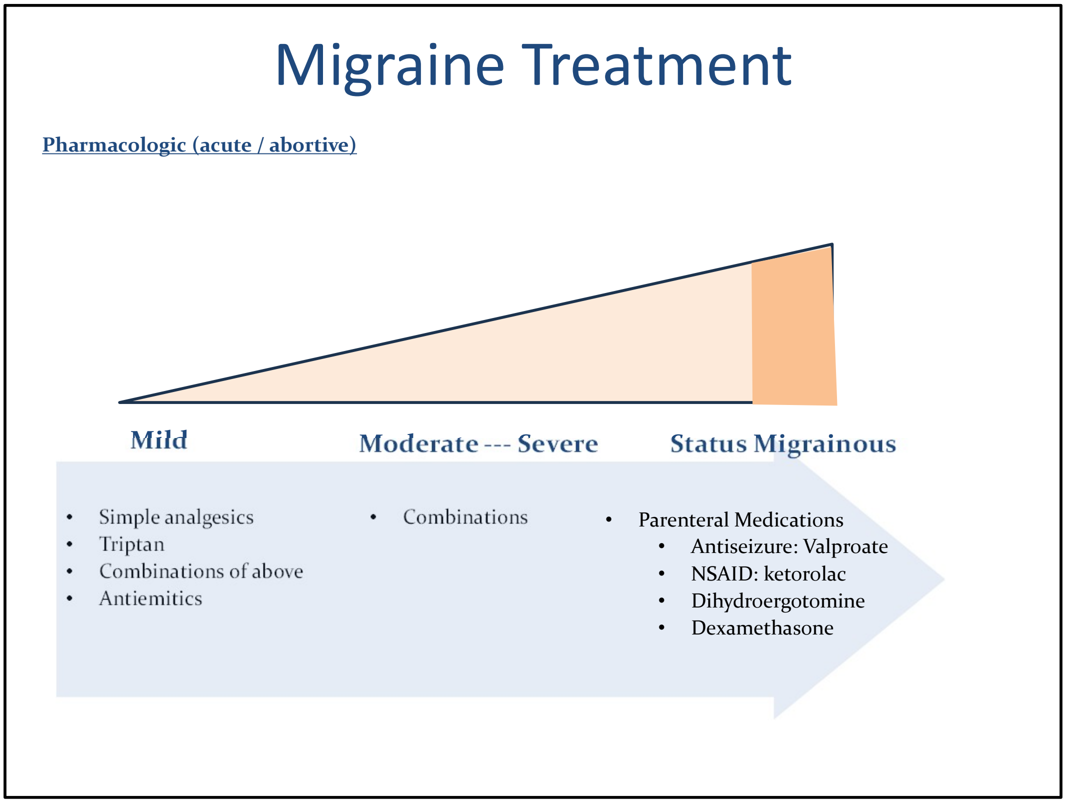

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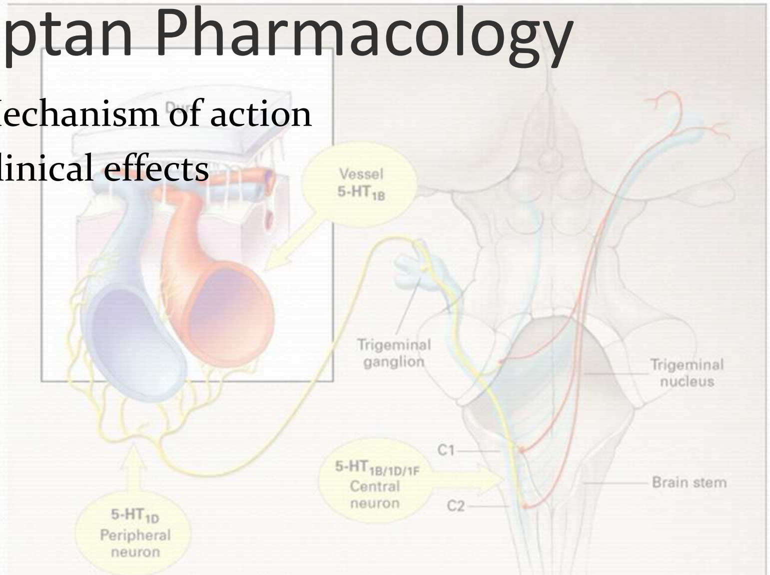
Triptan Pharmacology

- Mechanism of action
- Clinical effects



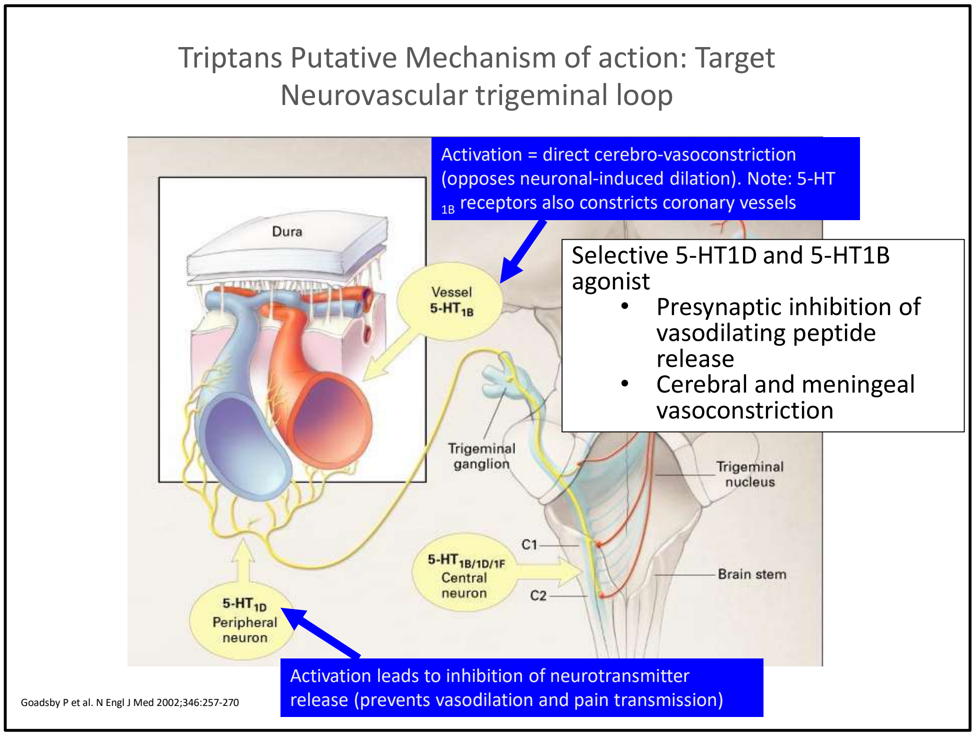
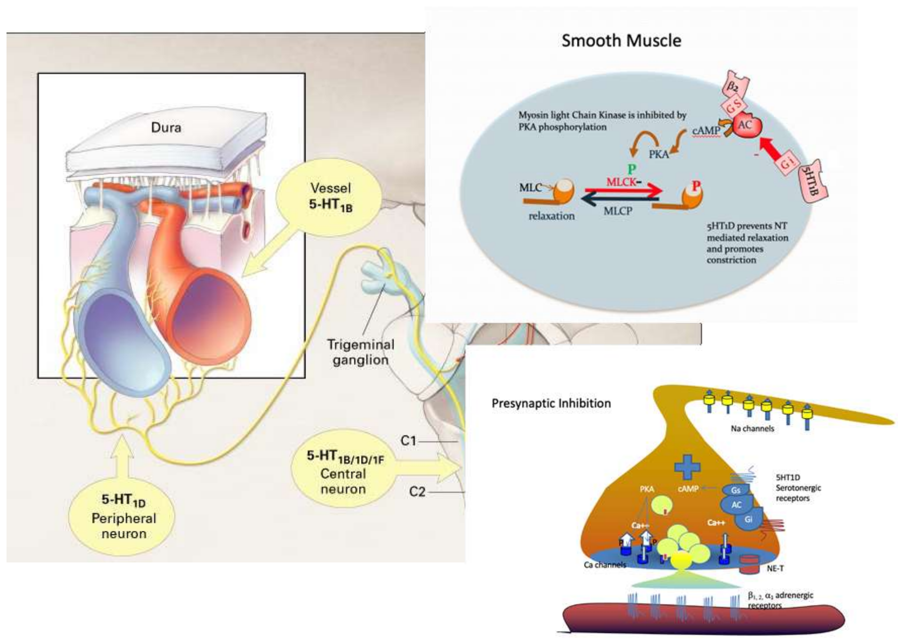


Figure 2. Possible Sites of Action of Triptans in the Trigeminovascular System.

- Design the appropriate therapeutic management for each type of primary headache.



Goadsby P et al. N Engl J Med 2002;346:257-270

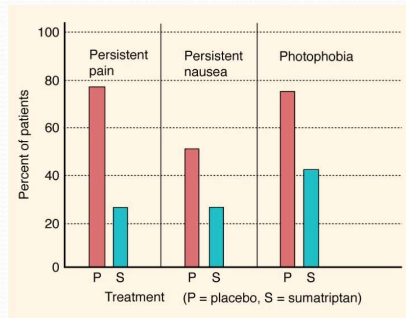
Figure 2. Possible Sites of Action of Triptans in the Trigeminovascular System.

- Design the appropriate therapeutic management for each type of primary headache.

Triptan Pharmacology

Sumatriptan, Almotriptan,
Naratriptan, Rizatriptan, Zolmitriptan

- Half lives 2-3 hours (repeated dosing)
- Efficacy ≥ to ergot alkaloids
- Adverse Effects
  - Mild
    - Tingling, warmth, dizziness, muscle weakness, neck pain, and injection site reactions
  - Anginal-like chest pain
  - Avoid use in patients with hepatic or renal disease or peripheral vascular syndrome

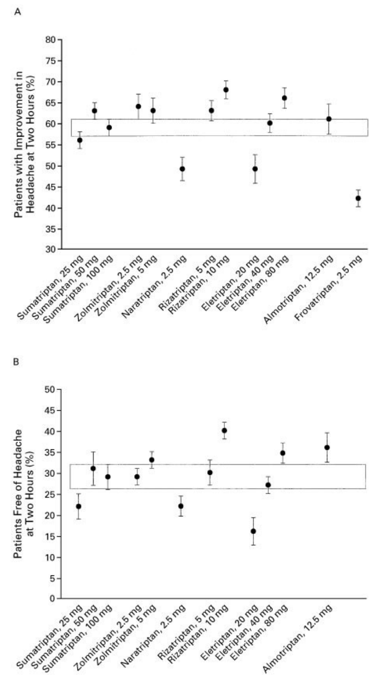


This slide introduces the various triptan drugs. Students should be well aware of the mechanism of action, use and toxicities of this class of drugs. Students should also be able to recognize a triptan by name.

Triptan Pharmacology

Migraine Pharmacotherapy

Sumatriptan,
Almotriptan,
Naratriptan,
Rizatriptan,
Zolmitriptan



This slide is intended to illustrate the differences between the various triptan drugs related to decreased symptoms. The choice between which triptan to use is often empiric depending on the patients response and side affect profile. With all drugs start at the lowest dose and if the desired response is achieved increase the dose.

Nasal Preparations (can not take orally because of vomiting)

Triptans: Migraine Attacks
Sumatriptan OPN, Zolmitriptan OPN*
Ergot alkaloids
Dihydroergotamine OPN

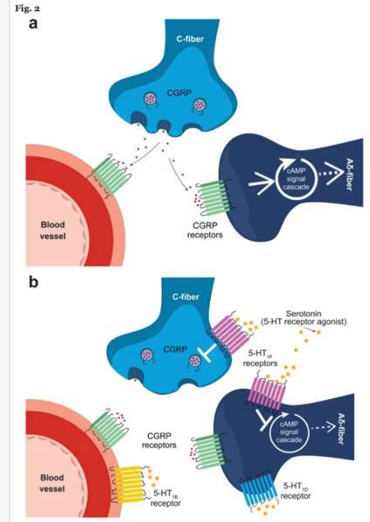
64

New formulation of sumatriptan in a novel delivery format:
ONZETRA Xsail TM Instructions for Use (sumatriptan nasal powder) 11 mg per nosepiece
ONZETRATM Xsail TM is NOT a nasal spray or inhaler.
You blow with your mouth into the device to deliver medication into your nose.
Must pierce both capsules, one at a time, and blow the medication into each nostril separately. Both capsules constitute one dose.
To watch an instructional video on how to use ONZETRA Xsail, text ONZETRA to 313131.

Lasmiditan Pharmacology

- Mechanism of action 5HT1F selective receptor agonist (not associated with blood vessels, still coupled to Gi)
  - More specific,
  - less vasoconstriction (can be used in patient with with relative contraindications to Triptans --CV risk factors)
  - efficacy?
- Clinical effects
  - Beneficial (How much?)
  - Side effects
    - Dizziness (dose dependent)
    - Somnolence (driver impairment?)
    - Fatigue
    - Nausea

Fig. 2



J Headache Pain **21**, 71 (2020). https://doi.org/10.1186/s10194-020-01132-3

From UpToDate

LASMIDITANLasmiditan is a selective serotonin 1F receptor agonist that lacks vasoconstrictor activity and therefore can be used for patients with relative contraindications to triptans due to cardiovascular risk factors [115].

In a randomized trial of over 2200 patients with episodic migraine, the proportion who were headache pain-free at two hours was greater with lasmiditan 200 mg (32 versus 15 percent with placebo; absolute risk difference [ARD] 17 percent; odds ratio [OR] 2.6, 95% CI 2.0-3.6) and lasmiditan 100 mg (28 versus 15 percent; ARD 13 percent; OR 2.2, 95% CI 1.6-3.0) [116]. In a 2021 meta-analysis including five trials and more than 7000 patients, resolution of pain at two hours likelier with lasmiditan than placebo (relative risk [RR] 1.95, 95% CI 1.3-3.0) [109]. Lasmiditan was also effective for treating acute migraine in another randomized placebo-controlled trial of 2310 patients [117].

In October 2019, the US Food and Drug Administration (FDA) approved lasmiditan oral tablets for the acute treatment of migraine in adults [118]. The initial dose of lasmiditan is 50 or 100 mg; there is no benefit with taking a second dose for the same migraine attack. With subsequent attacks, the dose may be increased to 100 or 200 mg as needed, but no more than one dose should be taken in 24 hours [119].

The most common adverse event associated with lasmiditan is dizziness; other relatively frequent adverse events are paresthesia, somnolence, fatigue, and nausea [116,117]. Dizziness with lasmiditan is dose-dependent and largely mild to moderate in severity, with a median duration of 1.5 to 2 hours [120]. The drug may cause driving impairment, and patients should not drive a motor vehicle, operate machinery, or engage in potentially hazardous activities for at least eight hours after each dose of lasmiditan [119].

Ergot Alkaloids

- Dihydroergotamine, and ergotamine
- Mechanism of action
  - Less specific
  - Alpha-adrenergic and 5HT1D and 1B agonist.
- Clinical effects
  - Comes in oral, parenteral and nasal (i.e option for vomiting and status migrainous)
  - Side effects
    - Vasoconstriction (avoid use with CYP inhibitors, other vasoconstrictor drugs, and in patients with CVD)
      - **Note: Safety:** Do **not** use within 24 hours of triptans or another ergotamine preparation. Screen patients for risk factors for coronary artery disease (CAD); perform a cardiovascular evaluation in patients with CAD risk factors being initiated on dihydroergotamine injection or nasal spray (0.5 mg per spray). Perform cardiovascular evaluation regardless of CAD risk factors in patients initiated on nasal spray (0.725 mg per spray). If evaluation reveals coronary artery vasospasm or myocardial ischemia, do not initiate therapy. If evaluation does not reveal coronary artery or ischemic myocardial disease, administer the first dose in an adequately equipped facility, unless the patient has previously received dihydroergotamine without cardiovascular complications. Obtain an ECG immediately following the first dose in patients with CAD risk factors. Limit use to <10 days per month to avoid medication-overuse headache (UpToDate)
      - Serious and life-threatening peripheral ischemia have been associated with the coadministration of dihydroergotamine with potent CYP3A4 inhibitors including protease inhibitors and macrolide antibiotics. Because CYP3A4 inhibition elevates the serum levels of dihydroergotamine, the risk for vasospasm leading to cerebral ischemia and ischemia of the extremities is increased. Hence, concomitant use of these medications is contraindicated. (UpToDate)

ERGOTSA variety of ergotamine preparations, alone and in combination with caffeine and other analgesics, have been used for the abortive treatment of migraine [121]. Both ergotamine and dihydroergotamine bind to 5HT 1b/d receptors, just as triptans do. As the evidence reviewed below suggests, parenteral dihydroergotamine is effective for acute migraine, while the effectiveness of ergotamine is uncertain.
Dihydroergotamine — Dihydroergotamine is an alpha-adrenergic agonist that is a weaker arterial vasoconstrictor and more potent venoconstrictor than ergotamine tartrate. It is also a potent 5-HT 1b/1d receptor agonist. Dihydroergotamine has fewer side effects than ergotamine [7]. It is available for intravenous (IV), intramuscular (IM), subcutaneous, and intranasal use. Dihydroergotamine is often used in combination with an antiemetic drug, and this is always the case when it is given by intravenous administration.
The use of intravenous dihydroergotamine for the treatment of chronic intractable migraine headache or status migrainosus (a debilitating migraine attack lasting for >72 hours) is discussed separately. (See "Medication overuse headache: Treatment and prognosis", section on 'Dihydroergotamine'.)
Parenteral dihydroergotamine (DHE 45) administered with an antiemetic appears to be effective for acute migraine. This conclusion is supported by the findings of a systematic review that analyzed 11 randomized controlled trials of IV or IM dihydroergotamine therapy for acute migraine in adults [122]. The quality of the included studies was variable, and most had relatively small sample sizes. The following observations were reported:

●In two studies, dihydroergotamine alone (without an antiemetic) was less effective on most outcome measures than sumatriptan [123,124], and in one study was less effective on some (but not all) outcome measures than chlorpromazine [125].
●In eight studies, the combination of parenteral dihydroergotamine with an antiemetic (most commonly metoclopramide) was as effective as or more effective than meperidine, valproate, or ketorolac in relieving migraine headache and preventing relapses.

Whether dihydroergotamine contributed additional benefit when used with an antiemetic in these studies is uncertain, since the antiemetic metoclopramide is known to be effective for migraine when used alone (see 'Metoclopramide' above). However, in several studies dihydroergotamine combined with an antiemetic was superior to other agents combined with the same antiemetic, suggesting that dihydroergotamine does have independent efficacy for migraine [125].

Self-administered intranasal dihydroergotamine has been found, in placebo-controlled trials, to be efficacious for the treatment of migraine symptoms [126,127]. In one trial of over 300 patients with migraine, for example, 27 percent of patients who administered 2 mg of intranasal dihydroergotamine had resolution of their headache within 30 minutes [128]. By four hours after treatment, 70 percent of the headaches were resolved and returned within 24 hours in only 14 percent. No serious adverse effects of treatment were observed.

A delivery system targeting the upper nasal cavity may improve systemic availability of dihydroergotamine. In an open-label study of 360 patients with migraine, 52 percent of patients reported resolution of the most bothersome migraine symptom and 66 percent reported pain relief at two hours when using dihydroergotamine mesylate delivered to the upper nasal cavity [127]. The most common treatment-related adverse effects were nasal congestion (15 percent), nausea (7 percent), and nasal discomfort (5 percent).

Subcutaneous dihydroergotamine may be slightly more effective than the intranasal preparation, but it has the disadvantage that it does not come in a preloaded syringe. Nevertheless, patients can be taught to self-inject this agent. One study randomly assigned 295 patients with acute migraine (with or without aura) to receive 1 mg of subcutaneous dihydroergotamine or 6 mg of subcutaneous sumatriptan succinate; a second injection of the same drug was used in two hours if patients did not experience initial relief [123]. Headache relief occurred in 86 and 83 percent by four hours, respectively, a difference that was not statistically significant. However, dihydroergotamine was more likely to produce relief by 24 hours (90 versus 77 percent) and was associated with a lower incidence of recurrence in the 24 hours after therapy (18 versus 45 percent).

Dihydroergotamine should not be used within 24 hours of triptans or other ergot-like agents. Dihydroergotamine is contraindicated in patients with hypertension or ischemic heart disease, and in pregnancy. Dihydroergotamine should not be used in combination with peripheral and central vasoconstrictors or with potent inhibitors of

CYP3A4 (including protease inhibitors, azole antifungals, and some macrolide antibiotics) and in patients with hemiplegic migraine or migraine with brainstem aura.
Ergotamine — It is unclear if it is the ergotamine itself or the other ingredients in the combination drugs that provide the most effect. There are two observations that question the efficacy of ergotamine alone: oral and rectal ergotamine have a very poor bioavailability (2 and 5 percent, respectively) [129], and most placebo-controlled trials of oral ergotamine alone have failed to show efficacy in the relief of migraine [130]. One controlled trial found that rectal suppositories containing ergotamine (2 mg) plus caffeine (100 mg) were as effective as sumatriptan (25 mg) suppositories for migraine headache relief [131,132]. However, there were more side effects among patients treated with ergotamine.
Ergotamine tartrate may be associated with significant side effects and may worsen the nausea and vomiting associated with migraine. In addition, vascular occlusion and rebound headaches have been reported with oral doses exceeding 6 tablets per 24 hours or 10 tablets per week. Years of use also may be associated with valvular heart disease [133].
Ergotamine should not be used within 24 hours of triptans or other ergot-like agents. Ergots should be avoided in patients with coronary artery disease because they cause sustained coronary artery constriction [134], peripheral vascular disease, hypertension, and liver or kidney disease. In addition, ergotamine overuse has been associated with an increased risk of cerebrovascular, cardiovascular, and peripheral ischemic complications, particularly among those using cardiovascular drugs [135]. They also should not be used in patients who have hemiplegic migraine, migraine with brainstem aura, and migraine with prolonged aura because they may reduce cerebral blood flow.
A European consensus panel reviewed the use of ergotamine for the acute treatment of migraine and concluded that ergotamine is the drug of choice in relatively few patients with migraine because of issues of efficacy and side effects [131]. Suitable candidates may be those with prolonged duration of attacks (eg, greater than 48 hours) and possibly frequent headache recurrence.

Dopamine Antagonist

Drug	Use	Toxicity
Metoclopramide	Acute migraine attacks + prokinetic drug used for GERD and is an antiemetic. Adjunctive therapy enhances gastric absorption, decreases nausea/vomiting and restores gastric motility	Drowsiness, fatigue, and other CNS effects, pseudoparkinsonism, diarrhea, hyperprolactinemia (gynecomastia, galactorrhea, impotence)
prochlorperazine	Acute migraine attacks, Anxiety, antiemetic, schizophrenia	Similar to metoclopramide + rare—agranulocytosis. See antipsychotics lecture for more details
chlorpromazine		

UPTODATE: Metoclopramide

10% of patients include drowsiness, fatigue, and lassitude at normal prescribed dosages; however, the incidence rises to 70% in with dosages of 1—2 mg/kg per dose (i.e., for reduction of nausea with chemotherapy). Insomnia, confusion, restlessness, depression, and headache occur less frequently. There are isolated and rare reports of seizures in the literature, relationships to the use of metoclopramide are not established. Other rarely reported events include hallucinations.

Various CNS reactions can occur with metoclopramide, some of which are dose-related. Extrapyramidal and/or acute dystonic reaction are possible during metoclopramide therapy in roughly 0.2% (1/500) of patients treated with 30 to 40 mg/day. Extrapyramidal effects can include akathisia, bulbar type of speech, facial grimacing, oculogyric crisis, torticollis, or trismus. These effects are more likely to occur in teenagers and young adults. The incidence is higher with higher doses, such as those dosages used in the treatment of nausea due to chemotherapy (i.e., incidence roughly 2%) and in those cancer patients not administered diphenhydramine as a prophylactic medication. Akathisia may respond to dosage reductions. In severe cases, stridor and dyspnea may occur or laryngospasm; the administration of diphenhydramine is indicated in cases of severe EPS. Benztropine may also be given. Extrapyramidal effects usually occur within 24—48 hours of treatment and generally disappear within 24 hours of discontinuation.

Pseudoparkinsonism may develop including bradykinesia, tremor, cogwheel rigidity, and mask-like facies. The parkinsonian-like symptoms generally subside within 2—3 months following the discontinuation of metoclopramide. Tardive dyskinesia has been reported with long-term use. Tardive dyskinesia occurs more frequently in elderly female patients and can be irreversible. Although tardive dyskinesia is more likely to occur after prolonged therapy or high dosage, relatively brief therapy has been associated with this effect as well.

The cholinergic-like effects of metoclopramide increase gastric and duodenal motility, which can cause nausea or diarrhea.

Metoclopramide can antagonize dopamine receptors in the pituitary gland and indirectly stimulate prolactin secretion. Hyperprolactinemia can develop in both men and women, causing gynecomastia in males, breast enlargement in women, and galactorrhea. Impotence (erectile dysfunction), may develop in males, and menstrual irregularity in women may occur secondary to hyperprolactinemia and could potentially cause temporary infertility or reduced fertility. These effects are generally reversible upon discontinuation of the drug. Serum prolactin levels return to normal in about 1 week, and adverse effects diminish within several weeks to several months.

Many reported adverse reactions with metoclopramide appear to be infrequent or isolated occurrences. Cardiovascular events have included AV block, hypotension, hypertension, sinus bradycardia, and supraventricular tachycardia (SVT). Transient increases in fluid retention (edema, acute heart failure) may occur secondary to a transient elevation in aldosterone secretion induced by the pharmacologic action of metoclopramide.

Other rare side effects include hypersensitivity reactions like rash (unspecified), angioedema, urticaria or bronchospasm. Some products contain sulfites which may induce such reactions in susceptible individuals, such as those with asthma.

Jaundice has been reported in patients taking metoclopramide who were on medications with known hepatotoxic potential.

Urinary adverse reactions include urinary urgency and urinary incontinence.

Rare occurrences of neuroleptic malignant syndrome (NMS) have been reported during metoclopramide therapy. This potentially fatal syndrome is comprised of the symptom complex of hyperthermia, altered consciousness, muscular rigidity, and autonomic dysfunction (irregular pulse or blood pressure, tachycardia, diaphoresis, and cardiac arrhythmias).

Rarely, withdrawal symptoms such as dizziness, nervousness or headaches may occur after discontinuation of metoclopramide. A slower taper when stopping metoclopramide may be appropriate for some patients, although this is not typically recommended.[5688]

A few cases of leukopenia, neutropenia, and agranulocytosis has been reported in patients receiving metoclopramide. However, a causal relationship to metoclopramide is unclear and sufficient documentation of these events in the medical literature is lacking. Only one case report that recurred with drug rechallenge has been published.
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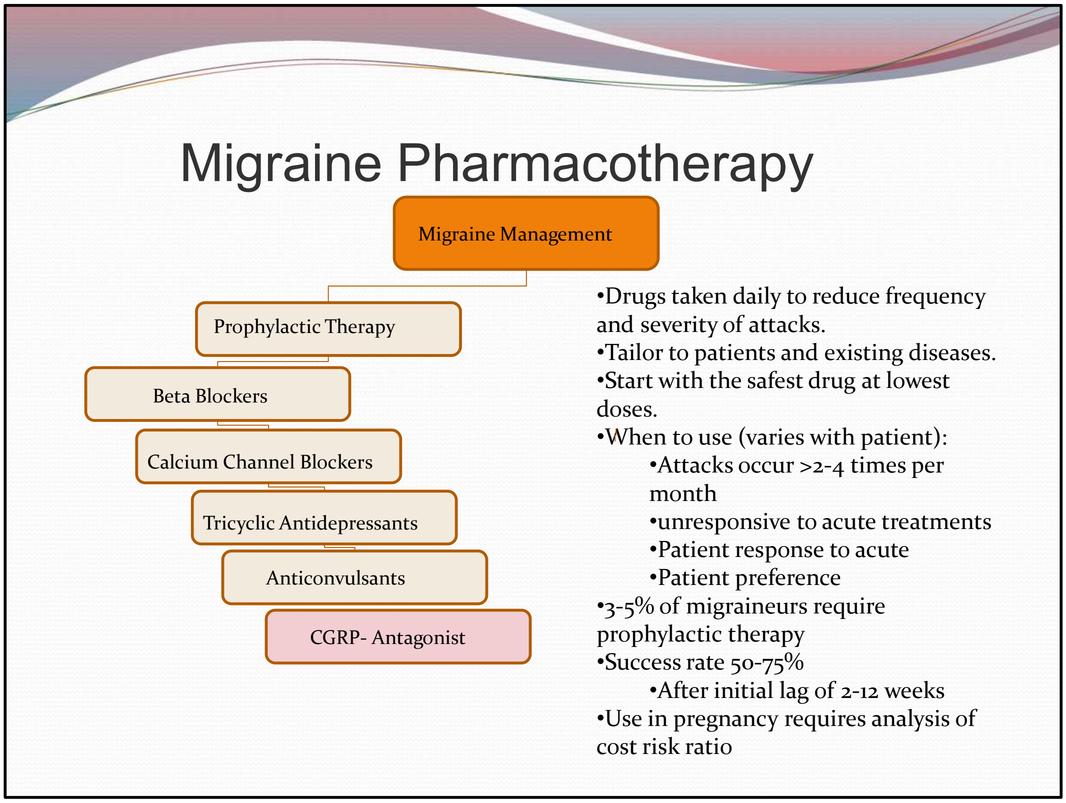


- Differentiate between preventative and acute migraine headache management.
- Apply knowledge regarding mechanism, use and toxicity to design the appropriate pharmacotherapy for prevention of future headaches.
- List the drugs used for prevention (beta blockers, tricyclic antidepressants, anticonvulsants, serotonergic drugs) and drugs used to treat acute attacks (triptans, NSAIDs, dopamine antagonist, opioids, ergot alkaloids).
- Balance the use vs toxicity of each drug

There are two main approaches to treatment of migraine H/As, preventative therapy and treatment of acute migraine attacks.

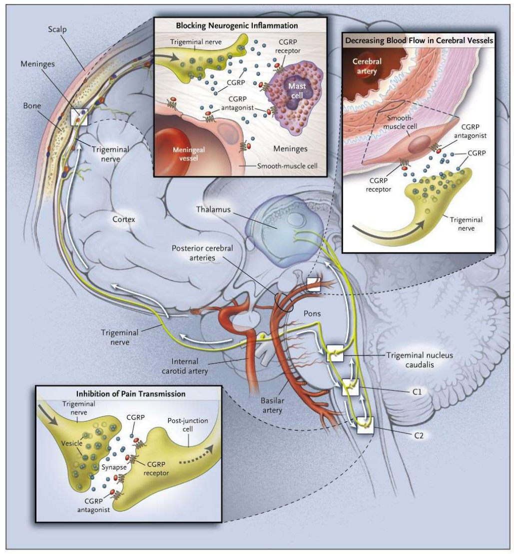
AAN: Evidence Based Guidelines—US Headache Consortium recommendations:
Preventative therapy may be more appropriately guided by the following.

1) Recurring migraines that, in the patients' opinion, significantly interfere with their daily routines, despite acute treatment
2) Frequent headaches

3) Contraindication to, failure of, or overuse of acute therapies
4) Adverse events with acute therapies
5) The cost of both acute and preventative therapies
6) Patient preference
7) The presence of uncommon migraine conditions, including hemiplegic migraine, basilar migraine, migraine with prolonged aura, or migrainous infarction (to prevent neurologic damage)

CGRP Antagonist

- Mechanism of action prevent the activation of trigeminovacular pain transmission and the vasodilatory component of neurogenic inflammation.
- Types:
  - Oral CGRP receptor antagonist: Rimegepant, Ubrogepant, Zavegepant [N]
  - MAB to CGRP: Fremanezumab, Galcanezumab,
- Clinical effects
  - Moderate to severe
  - Cardio protective?
  - Long term side effects unclear



Durham P. N Engl J Med 2004;350:1073-1075

Figure. Possible Sites of Action of the Nonpeptide CGRP-Receptor Antagonist BIBN 4096 BS. The binding of the calcitonin gene-related peptide (CGRP) antagonist to receptors on major cerebral vessels would decrease blood flow to the brain. Binding to meningeal blood vessels and mast cells would inhibit neurogenic inflammation, and binding to receptors on second-order neurons would inhibit the transmission of pain.

UpToDate

CGRP ANTAGONISTSPharmacologic modulation of calcitonin-gene related peptide (CGRP) activity appears to mediate trigeminovascular pain transmission in migraine (see "Pathophysiology, clinical manifestations, and diagnosis of migraine in adults"). Monoclonal antibodies directed against the CGRP receptor or ligand are given by injection for migraine prevention. Small-molecule CGRP antagonists (also termed "gepants") are oral options available for the acute treatment of migraine in patients with either insufficient response or contraindication (eg, coronary artery disease) to treatment with triptans. Medications and typical dosing for acute migraine treatment include:

●Rimegepant 75 mg as a single oral dose
●Ubrogepant 50 to 100 mg orally, may repeat dose in two hours; maximum dose 200 mg per 24 hours
●Zavegepant 10 mg given intranasally as a single spray

Several acute treatment trials demonstrated benefit with either rimegepant or ubrogepant [104-108]. As an example, one trial randomly assigned adult patients with episodic migraine in a 1:1:1 ratio to one tablet of ubrogepant 100 mg, ubrogepant 50 mg, or placebo taken within four hours of a single migraine attack; an optional second dose or the patient's own rescue medication was permitted 2 to 48 hours after the first dose [108]. With efficacy data for 1327 patients, the proportion with pain freedom at two hours after treatment was greater for those assigned to the 100 mg and 50 mg doses of ubrogepant compared with placebo (21.2, 19.2, and 11.8 percent, respectively). Also, the proportion who reported absence of their most bothersome migraine-associated symptom (ie, photophobia, phonophobia, or nausea) at two hours after treatment was greater for those assigned to ubrogepant 100 and 50 mg compared with placebo (37.7, 38.6, and 27.8 percent). The most common adverse events were nausea, somnolence, and dry mouth.

A 2021 meta-analysis of available trials found that pain resolution by two hours was likelier with rimegepant (relative risk [RR] 1.8, 95% CI 1.5-2.1) or ubrogepant (RR 1.6, 95% CI 1.3-1.9) than placebo [109]. Ubrogepant received US Food and Drug Administration (FDA) approval for the treatment of acute migraine in adults in December 2019 [110], and rimegepant received similar FDA approval in February 2020 [111]. However, given the short-term, single-attack design of these trials [105-108], long-term data are needed to determine safety and tolerability. Rimegepant has been found to be effective in some patients who did not respond previously to a triptan [112], but the efficacy of CGRP antagonists compared with triptans is not yet known.

Zavegepant is a small-molecule CGRP-receptor antagonist for acute treatment that is administered by intranasal route. In a trial of 1405 adult patients with migraine, those assigned to zavegepant 10 mg were likelier to be pain free at two hours than patients assigned to placebo (24 versus 15 percent; risk difference 9 percent, 95% CI 4.5-13.1) [113]. In addition, resolution of the most bothersome acute symptom (eg, photophobia, nausea) was more common with zavegepant (40 versus 31 percent). Adverse events were transient and mild and included dysgeusia and nasal discomfort. Nasal administration of a CGRP-receptor antagonist provides rapid absorption and effect and may be preferred for patients with nausea and/or vomiting who are unable to tolerate oral options. Zavegepant was approved for acute migraine by the US FDA in 2023 [114].

The safety of using a CGRP antagonist within two to four hours of a triptan or ergotamine agent is not well established.

Several CGRP antagonists are approved for migraine prevention, as discussed separately. (See "Preventive treatment of episodic migraine in adults", section on 'CGRP antagonists'.)

CGRP antagonists — The calcitonin gene-related peptide (CGRP) is a therapeutic target in migraine because of its hypothesized role in mediating trigeminovascular pain transmission and the vasodilatory component of neurogenic inflammation (see "Pathophysiology, clinical manifestations, and diagnosis of migraine in adults", section on 'Role of calcitonin gene-related peptide'). In 2018, the US Food and Drug Administration (FDA) approved the CGRP antagonists erenumab [36], fremanezumab [37], and galcanezumab [38] for migraine prevention.

Although probably unrelated to treatment, there have been three deaths in clinical trial subjects who received CGRP monoclonal antibodies [39-41]. Additional concerns have been raised about the potential for adverse cardiovascular, pulmonary, and psychiatric effects with CGRP antagonists [42], even though these were not observed in randomized trials.

There is little evidence to guide the use of the CGRP antagonists in specific populations (eg, children, older adults, and women during pregnancy and lactation). Some experts recommend avoiding the use of CGRP antagonists for women who are pregnant or likely to become pregnant, given the potential risks of reducing the action of CGRP during pregnancy, and avoiding CGRP antagonists for individuals with or at high risk for cardiovascular disease, since CGRP has theoretical cardioprotective and vasodilatory effects [11,43]. More data are needed regarding the long-term safety of this class of medications [42].

Erenumab — Erenumab, a human monoclonal antibody that binds to and inhibits the CGRP receptor, was modestly effective for migraine prevention in a placebo-controlled trial of over 900 adults with episodic migraine [44]. The trial randomly assigned subjects in a 1:1:1 ratio to subcutaneous injections of erenumab 70 mg, erenumab 140 mg, or placebo monthly for six months. At baseline, the mean number of migraine days per month was 8.3 in the overall population. At months four through six, the number of migraine days per month was reduced by 3.2 in the 70 mg erenumab group, 3.7 in the 140 mg erenumab group, and 1.8 in the placebo group. The rates of adverse events were similar between the erenumab and placebo groups.

Another trial evaluated 246 subjects with treatment-resistant episodic migraine who had failed to respond to or did not tolerate between two and four previous migraine preventive medications [12]. Subjects were randomly assigned to receive erenumab 140 mg or placebo by subcutaneous injection every four weeks. At 12 weeks, the proportion of patients who had a 50 percent or more reduction in the mean number of monthly migraine days was greater for the group assigned to erenumab compared with the group assigned to placebo (30 versus 17 percent, odds ratio 2.7, 95% CI 1.4-5.2, absolute risk difference 13 percent).

The recommended dose of erenumab is 70 mg subcutaneous injection once monthly in the abdomen, thigh, or upper arm. The dose can be increased to 140 mg once monthly if needed. Injection site reactions and constipation are the most common adverse effects.

Fremanezumab — Fremanezumab, a human monoclonal antibody that binds to both isoforms of the CGRP ligand, appears to be modestly effective for prevention of episodic migraine [13,39]. One trial randomly assigned 875 adults with episodic migraine in a 1:1:1 ratio to subcutaneous fremanezumab 225 mg monthly for three months, a single dose of fremanezumab 675 mg (intended to support a quarterly dose regimen), or placebo [39]. At three months compared with placebo, the mean number of migraine days per month decreased by 1.5 in the fremanezumab monthly dose group and by 1.3 days in the single higher dose group. The proportion of patients with at least a 50 percent reduction in the mean number of monthly migraine days was 48 percent in the fremanezumab monthly dose group and 44 percent in the fremanezumab single higher dose group, compared with 28 percent for the placebo group.

The recommended dose of fremanezumab is 225 mg once monthly or 675 mg (given as three consecutive injections of 225 mg each) every three months administered subcutaneously in the abdomen, thigh, or upper arm. The most common adverse effects are injection site reactions.

Galcanezumab — Galcanezumab, a human monoclonal antibody that binds to the CGRP ligand, is also modestly effective for episodic migraine prevention, as demonstrated in several placebo-controlled trials [45,46]. As an example, one of these trials randomly assigned 858 patients with episodic migraine to monthly subcutaneous galcanezumab 120 mg, galcanezumab 240 mg, or placebo in a 1:1:2 ratio [47]. At six months, the mean number of migraine days per month decreased by 4.7 and 4.6 for the galcanezumab 120 and 240 mg groups, respectively, compared with 2.8 days for the placebo group.

Galcanezumab treatment is started with a loading dose of 240 mg, given as two consecutive doses of 120 mg each, followed by monthly doses of 120 mg, administered subcutaneously in the thigh, upper arm, or buttocks. Injection site reactions were the most common adverse events in clinical trials.

Other agents — A number of other agents have been studied for migraine prevention. Guidelines issued in 2012 from the AAN concluded that Petasites (butterbur) is effective for migraine prevention [48]. In addition, the AAN noted that feverfew, magnesium, certain nonsteroidal anti-inflammatory drugs, and riboflavin are probably effective.

Preventative Therapy

Drug		Selected side effects	Evidence
b-blocker	propranolol	↓ energy, depression, contraindicated with COPD and Asthma	A
TCAs	amitriptyline, doxepin, nortriptyline	Many see antidepressant lecture	A,C,C
Anti-convulsants	topiramate	Paresthesias, cognitive symptoms, weight loss, glaucoma, caution with nephrolithiasis	A
	valproate	Drowsiness, fetal abnormalities, hematologic or liver abnormalities	A
	gabapentin	Dizziness, sedation	B
CGRP antagonist	Gepants MABs	Constipation, injection site reactions	?
Herbal	Petasites (butterbur)	unclear	A, 2 Class 1 studies

- Differentiate between preventative and acute migraine headache management.
- Apply knowledge regarding mechanism, use and toxicity to design the appropriate pharmacotherapy for prevention of future headaches.
- List the drugs used for prevention (beta blockers, tricyclic antidepressants, anticonvulsants, serotonergic drugs) and drugs used to treat acute attacks (triptans, NSAIDs, dopamine antagonist, opioids, ergot alkaloids).
- Balance the use vs toxicity of each drug

This table organizes the choices of medications we have to prevent migraine headaches.

The beta blockers, tricyclic antidepressants, and anticonvulsants are frequently employed. These medications must be taken on a daily basis at doses sufficient to reduce the frequency of H/As. Patients must be willing to put up with side effects that are associated with the different classes of drugs utilized.
AAN: Divalproex sodium and sodium valproate are established as effective in migraine prevention (multiple Class I studies).

Data are insufficient to determine the effectiveness of gabapentin (1 Class III study).
Lamotrigine is established as ineffective for migraine prevention (2 Class I studies).
Oxcarbazepine is possibly ineffective for migraine prevention (1 Class II study).
Topiramate is established as effective for migraine prevention (4 Class I studies, multiple Class II studies; 1 negative Class II study).
Topiramate is probably as effective for migraine prevention as propranolol (1 Class I study), sodium valproate (1 Class I study), and amitriptyline (2 Class II studies).
There is conflicting Class II evidence for use of fluoxetine.
Venlafaxine is probably effective for migraine prevention (1 Class I study) and is possibly as effective as amitriptyline in migraine prevention (1 Class II study).
Amitriptyline is probably effective for migraine prevention (multiple Class II studies); it is probably as effective as topiramate (2 Class II studies) and possibly as effective as venlafaxine (1 Class II study) for migraine prevention.

Back to Pat

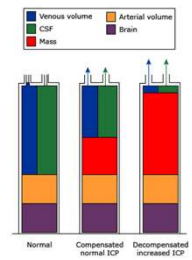
Approach to ICP Management

Video 8: Dr. Gayer

In adults, the intracranial compartment is protected by the skull, a rigid structure with a fixed internal volume of 1400 to 1700 mL

- Brain parenchyma – 80 percent
- Cerebrospinal fluid (CSF) – 10 percent
- Blood – 10 percent

Intracranial compensation for mass

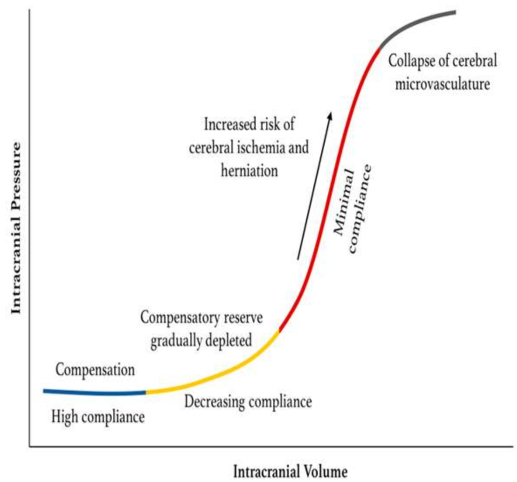


Normally, the intracranial components are in equilibrium as shown in chamber 1. Initially, the volume of a space-occupying lesion is compensated for by displacement of blood and CSF and ICP remains normal (chamber 2). When the limits of this compensation is reached; any additional increase in the volume of the mass lesion is accompanied by a corresponding increase in ICP (chamber 3, decompensated phase).

Data from Pathophysiology and management of the intracranial vault. In: Textbook of Pediatric Intensive Care, 3rd ed, Rogers, MC (Ed), Williams and Wilkins 1996. p. 646; figure 18.1.

UpToDate

Intracranial Pressure



- The brain has some intrinsic compensatory mechanisms to maintain stable mean ICP (modification in brain venous blood pool is the primary mechanism and movement of CSF into the extracranial space to degree)
- Compensatory reserve is finite. When reserve is depleted, even small elevations in volume can lead to potentially dangerous elevations in ICP

Sensors 2018 journal article:
https://www.mdpi.com/1424-8220/18/2/465

Causes of intracranial hypertension*

Traumatic brain injury/intracranial hemorrhage
Subdural, epidural, or intraparenchymal hemorrhage
Ruptured aneurysm
Diffuse axonal injury
Arteriovenous malformation or other vascular anomalies
Central nervous system infections (eg, encephalitis, meningitis, abscess)
Ischemic stroke
Neoplasm
Vasculitis
Hydrocephalus
Hypertensive encephalopathy
Idiopathic intracranial hypertension (pseudotumor cerebri)

* For further information on clinical manifestations, diagnosis, or treatment of these conditions, refer to specific UpToDate topics.

UpToDate®

Elevated Intracranial Pressure

- **Symptoms:**
  - Headache (pressure on CN V via dura blood vessels), decreased consciousness, nausea, vomiting.
- **Signs:**
  - Papilledema
  - Periorbital bruising
  - Bradycardia, respiratory depression, and hypertension
  - Herniation syndrome

Papilledema

Papilledema, characterized by blurring of the optic disc margins, loss of physiologic cupping, hyperemia, and fullness of the veins, in a 5-year-old girl with intracranial hypertension due to vitamin A intoxication.

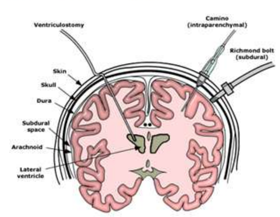
Courtesy of Gerald Striph, MD. UpToDate®

https://www.uptodate.com/contents/evaluation-and-management-of-elevated-intracranial-pressure-in-adults

Elevated ICP Management

- Monitor (risk present, Glascow coma scale <8)
  - ICP device (Monitoring (intraventricular "gold standard", intraparenchymal, subarachnoid, epidural)
  - CT scans
- **General Management**
  - Surgical address cause
  - Fluid Management (avoid drinking water, use isotonic fluids)
  - Sedation
  - Blood pressure control
  - Aggressive fever treatment
  - Head elevation
- Specific Therapies
  - Hypertonic Saline Bolus
  - Intravenous Mannitol
  - Hyperventilation to PCO_2 of 26 to 30 mmHG
  - Barbiturates
  - Removal of CSF
  - Surgical- decompressive craniectomy

Intracranial pressure monitors



Ventriculostomy allows both ICP monitoring and therapeutic drainage of cerebrospinal fluid (CSF). Subdural and intraparenchymal monitors cannot be used to drain CSF.

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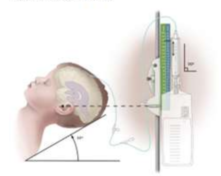
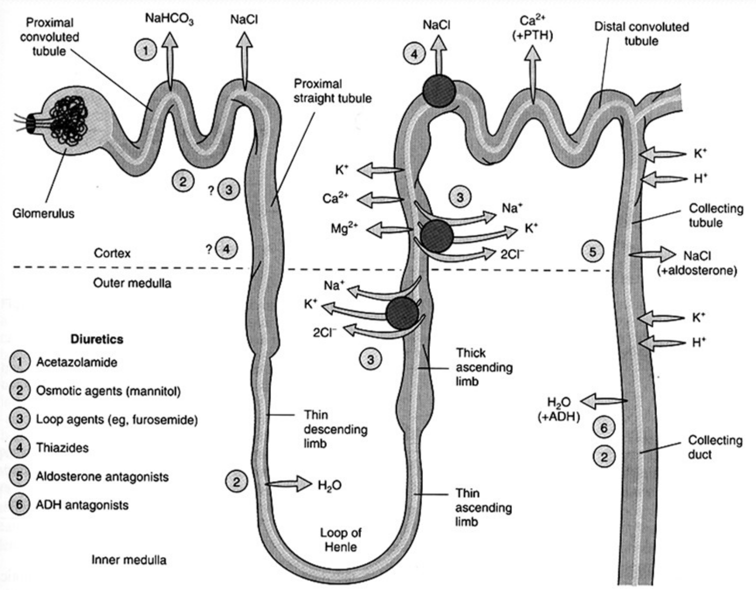
External ventricular drain



An external ventricular drain (EVD) is a small catheter inserted through the skull usually into the lateral ventricle, which is typically connected to a closed collecting device to allow for drainage of cerebrospinal fluid. The EVD can also be connected to a transducer that records intracranial pressure.

UpToDate

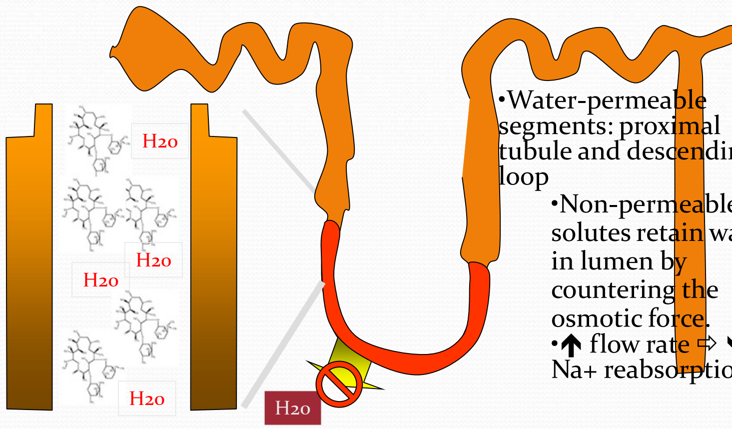
There are three ways to monitor pressure in the skull (intracranial pressure):
A thin, flexible tube threaded into one of the two cavities, called lateral ventricles, of the brain (intraventricular catheter)
A screw or bolt placed just through the skull in the space between the arachnoid membrane and cerebral cortex (subarachnoid screw or bolt)
A sensor placed into the epidural space beneath the skull (epidural sensor)
The intraventricular catheter is thought to be the most accurate method, but if immediate access is needed, a subarachnoid bolt is typically used. If no qualified brain surgeon (neurosurgeon) is available to place a bolt, then an epidural sensor will probably be used.
To insert an intraventricular catheter, a burr hole is drilled through the skull and the catheter is inserted through the brain matter into the laterallateral ventricle, which normally contains liquid (cerebrospinal fluid or CSF) that protects the brain and spinal cord. Not only can the intracranial pressureintracranial pressure (ICP) be monitored, but it can be lowered by draining cerebral spinal fluid (CSF) out through the catheter. This catheter may be difficult to get in place when there is increased intracranial pressure, since the ventricles change shape under increased pressure and are often quite small because the brain expands around them from injury and swelling.
A subarachnoid screw or bolt is a hollow screw that is inserted through a hole drilled in the skull and through a hole cut in the outermost membrane protecting the brain and spinal cord (dura mater).
The epidural sensor is placed through a burr hole drilled in the skull, just over the epidural covering. Since no hole is made in the epidural lining, this procedure is less invasiveinvasive than other methods, but it cannot remove excess CSF.



Define the mechanism of action, reason for use, and toxicity of Mannitol

Osmotic Diuretics: Effects

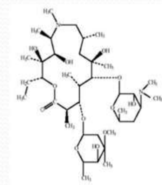
H2o and sodium reabsorption in Loop of Henle



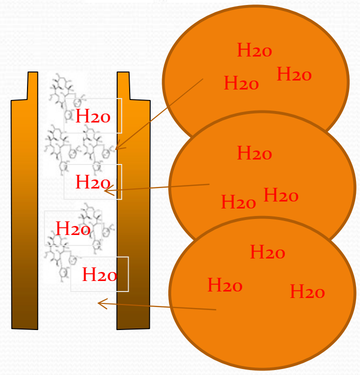
- Water-permeable segments: proximal tubule and descending loop
  - Non-permeable solutes retain water in lumen by countering the osmotic force.
  - ↑ flow rate ⇨ ↓ Na+ reabsorption.

Define the mechanism of action, reason for use, and toxicity of Mannitol

Mannitol



- Mechanism
  - Osmotically draws water into vascular space out of the parenchyma—removes water from cells thus reducing intracellular volume and intracranial pressure
  - Effects presents within minutes
- Toxicity (excessive use)
  - Prior to diuresis or with renal impairment
    - Hyponatremia—headache, nausea, and vomiting
  - Without water intake
    - Dehydration
      - Hypernatremia
      - Hyperkalemia (water leaves cells leads to cellular loss of K+)



Define the mechanism of action, reason for use, and toxicity of Mannitol

Readings

- **Required Readings**:
  Harrison's OnLine Principles of Internal Medicine 19th ed. Chapter 21 Headache
  Katzung's Chapter 16, section on serotonin (pgs 281-293)
  Supplemental Readings:
  - Interactive medical case--A Pain in the Brain: April 11, 2013,Redig A.J.,Vaidya A., Bienfang D.C., Milligan T., N Engl J Med 2013; 368:e20
  - *Clinical effectiveness of osteopathic treatment in chronic migraine: 3-Armed randomized controlled trial*

    Cerritelli, Ginevri, et al.

    Available Online 21 January 2015
    Science Direct ; www.elsevierhealth.com/journals/ctim
  - **UpToDate:**
    - Pathophysiology, clinical manifestations, and diagnosis of migraine in adults
      Literature review current through: Oct 2016. | **This topic last updated:** Jun 10, 2016.

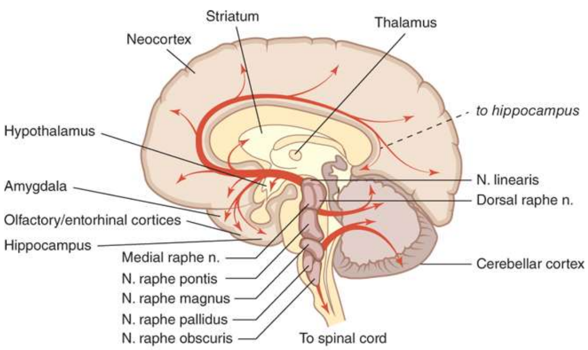
Serotonin | Ergot Alkaloids | Migraine | Elevated ICP

Objectives are listed on BB and at the end of the PPT

Supplemental

- Video 9: Serotonin Pharmacology and Ergot Alkaloids

5-hydroxytryptamine or Serotonin



- **CNS: neurotransmitter**
  - Located cell body of midbrain and pons project to all levels
  - Mood, sleep/arousal, temperature regulation, libido, vomiting, and appetite
- **Periphery: autocoid**
  - GI enterochromaffin cells (unknown)
  - GI enteric nervous system (peristalsis-Ach release)
  - Platelets (clotting)
  - Peripheral nerves (nociception)
  - chemoreceptor reflex (vagal nerve activation)
  - Lungs (bronchoconstriction, direct and via Ach release—observed in carcinoid tumors, CT)
  - Blood vessels (vasoconstriction),
  - Chornic exposure (CT) valvular fibroplasia
  - skeletal muscle (hyperthermia during serotonin syndrome)

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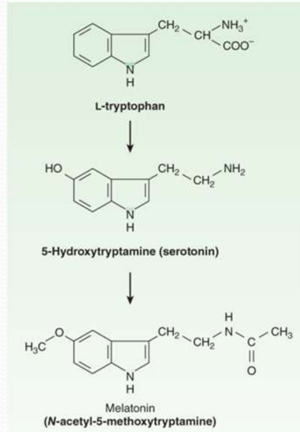
B. Tissue and Organ System Effects
1. Nervous system—Serotonin is present in a variety of sites in the brain. Its role as a neurotransmitter and its relation to the actions of drugs acting in the central nervous system are discussed in Chapters 21 and 30. Serotonin is also a precursor of melatonin in the pineal gland (Figure 16–2; see Box: Melatonin Pharmacology). **Repinotan,** a 5-HT_{1A} agonist currently in clinical trials, appears to have some antinociceptive action at higher doses while reversing opioid-induced respiratory depression.
Melatonin Pharmacology
Melatonin is *N*-acetyl-5-methoxytryptamine (Figure 16–2), a simple methoxylated and *N*-acetylated product of serotonin found in the pineal gland. It is produced and released primarily at night and has long been suspected of playing a role in diurnal cycles of animals and the sleep-wake behavior of humans. Melatonin receptors have been characterized in the central nervous system and several peripheral tissues. In the brain, MT_1 and MT_2 receptors are found in membranes of neurons in the suprachiasmatic nucleus of the hypothalamus, an area associated—from lesioning experiments—with circadian rhythm. MT_1 and MT_2 are seven-transmembrane G_i protein-coupled receptors. The result of receptor binding is inhibition of adenylyl cyclase. A third receptor, MT_3, is an enzyme; binding to this site has a poorly defined physiologic role, possibly related to intraocular pressure. Activation of the MT_1 receptor results in sleepiness, whereas the MT_2 receptor may be related to the light-dark synchronization of the biologic circadian clock. Melatonin has also been implicated in energy metabolism and obesity, and administration of the agent reduces body weight in certain animal models. However, its role in these processes is poorly understood, and there is no evidence that melatonin itself is of any value in obesity in humans. Other studies suggest that melatonin has antiapoptotic effects in experimental models. Recent research implicates melatonin receptors in depressive disorders. Insomnia associated with autism spectrum disorder may respond to melatonin.
Although serotonergic neurons are not found below the site of injury to the adult spinal cord, constitutive activity of 5-HT receptors may play a role following such a lesion—administration of 5-HT_2 blockers appears to reduce skeletal muscle spasm following this type of injury.
2. Respiratory system—Serotonin has a small direct stimulant effect on bronchiolar smooth muscle in normal humans, probably via 5-HT_{2A} receptors. It also appears to facilitate acetylcholine release from bronchial vagal nerve endings. In patients with carcinoid syndrome, episodes of bronchoconstriction occur in response to elevated levels of the amine or peptides released from the tumor. Serotonin may also cause hyperventilation as a result of the chemoreceptor reflex or stimulation of bronchial sensory nerve endings.
3. Cardiovascular system—Serotonin directly causes the contraction of vascular smooth muscle, mainly through 5-HT_2 receptors. In humans, serotonin is a powerful vasoconstrictor except in skeletal muscle and the heart, where it dilates blood vessels.
At least part of the 5-HT-induced vasodilation requires the presence of vascular endothelial cells. When the endothelium is damaged, coronary vessels are constricted by 5-HT. As noted previously, serotonin can also elicit reflex bradycardia by activation of 5-HT_3 receptors on chemoreceptor nerve endings. A triphasic blood pressure response is often seen following injection of serotonin in experimental animals. Initially, there is a decrease in heart rate, cardiac output, and blood pressure caused by the chemoreceptor response. After this decrease, blood pressure increases as a result of vasoconstriction. The third phase is again a decrease in blood pressure attributed to vasodilation in vessels supplying skeletal muscle. In contrast, pulmonary and renal vessels seem very sensitive to the vasoconstrictor action of serotonin.
Studies in knockout mice suggest that 5-HT, acting on 5-HT_{1A}, 5-HT_2, and 5-HT_4 receptors, is needed for normal cardiac development in the fetus. On the other hand, chronic exposure of adults to 5-HT_{2B} agonists is associated with valvulopathy and adult mice lacking the 5-HT_{2B} receptor gene are protected from cardiac hypertrophy. Preliminary studies suggest that 5-HT_{2B} antagonists can prevent development of pulmonary hypertension in animal models.
Serotonin also constricts veins, and venoconstriction with increased capillary filling appears to be responsible for the flush that is observed after serotonin administration or release from a carcinoid tumor. Serotonin has small direct positive chronotropic and inotropic effects on the heart, which are probably of no clinical significance. However, prolonged elevation of the blood level of serotonin (which occurs in carcinoid syndrome) is associated with pathologic alterations in the endocardium (subendocardial fibroplasia), which may result in valvular or electrical malfunction.
Serotonin causes blood platelets to aggregate by activating 5-HT_2 receptors. This response, in contrast to aggregation induced during normal clot formation, is not accompanied by the release of serotonin stored in the platelets. The physiologic role of this effect is unclear.
4. Gastrointestinal tract—Serotonin is a powerful stimulant of gastrointestinal smooth muscle, increasing tone and facilitating peristalsis. This action is caused by the direct action of serotonin on 5-HT_2 smooth muscle receptors plus a stimulating action on ganglion cells located in the enteric nervous system (see Chapter 6). 5-HT_{1A} and 5-HT_7 receptors may also be involved. Activation of 5-HT_4 receptors in the enteric nervous system causes increased acetylcholine release and thereby mediates a motility-enhancing or "prokinetic" effect of selective serotonin agonists such as cisapride. These agents are useful in several gastrointestinal disorders (see Chapter 62). Overproduction of serotonin (and other substances) in carcinoid tumor is associated with severe diarrhea. Serotonin has little effect on gastrointestinal secretions, and what effects it has are generally inhibitory.
5. Skeletal muscle and the eye—5-HT_2 receptors are present on skeletal muscle membranes, but their physiologic role is not understood. As discussed in the box, **serotonin syndrome** is associated with skeletal muscle contractions and precipitated when MAO inhibitors are given with serotonin agonists, especially antidepressants of the selective serotonin reuptake inhibitor class (SSRIs; see Chapter 30). Although the hyperthermia of serotonin syndrome results from excessive muscle contraction, serotonin syndrome is probably caused by a central nervous system effect of these drugs (Table 16–4 and Box: Serotonin Syndrome and Similar Syndromes).
Serotonin Syndrome and Similar Syndromes
Excess synaptic serotonin causes a serious, potentially fatal syndrome that is diagnosed on the basis of a history of administration of a serotonergic drug within recent weeks and physical findings. It has some characteristics in common with neuroleptic malignant syndrome (NMS) and malignant hyperthermia (MH), but its pathophysiology and management are quite different (Table 16–4).
As suggested by the drugs that precipitate it, serotonin syndrome occurs when overdose with a single drug, or concurrent use of several drugs, results in excess serotonergic activity in the central nervous system. It is predictable and not idiosyncratic, but milder forms may easily be misdiagnosed. In experimental animal models, many of the signs of the syndrome can be reversed by administration of 5-HT_2 antagonists; however, other 5-HT receptors may be involved as well. Dantrolene is of no value, unlike the treatment of MH.
NMS is idiosyncratic rather than predictable and appears to be associated with hypersensitivity to the parkinsonism-inducing effects of D_2-blocking antipsychotics in certain individuals. MH is associated with a genetic defect in the RyR1 calcium channel of skeletal muscle sarcoplasmic reticulum that permits uncontrolled calcium release from the sarcoplasmic reticulum when precipitating drugs are given (see Chapter 27).
Studies in animal models of glaucoma indicate that 5-HT_{2A} agonists reduce intraocular pressure. This action can be blocked by ketanserin and similar 5-HT_2 antagonists.

Serotonin Pharmacology

- Biosynthesis: L-tryptophan → 5 hydroxytryptamine (serotonin)
  - →pineal gland (melatonin)
- Metabolism: Monoamine oxidase



Source: Bertram G. Katzung: Basic & Clinical Pharmacology, Fourteenth Edition

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This slide introduces the biosynthesis of serotonin and melatonin as well as the various receptors that are activated by serotonin. Most receptors are GPCR receptors except for 5HT3, which is a cation channel.

It is important to note: this later distinction because outliers make for good board questions.

The following is for reference only:

From Katzung Text

Melatonin Pharmacology

Melatonin is *N*-acetyl-5-methoxytryptamine (Figure 16–2), a simple methoxylated and *N*-acetylated product of serotonin found in the pineal gland. It is produced and released primarily at night and has long been suspected of playing a role in diurnal cycles of animals and the sleep-wake behavior of humans. Melatonin receptors have been characterized in the central nervous system and several peripheral tissues. In the brain, MT_1 and MT_2 receptors are found in membranes of neurons in the suprachiasmatic nucleus of the hypothalamus, an area associated—from lesioning experiments—with circadian rhythm. MT_1 and MT_2 are seven-transmembrane G_i protein-coupled receptors. The result of receptor binding is inhibition of adenylyl cyclase. A third receptor, MT_3, is an enzyme; binding to this site has a poorly defined physiologic role, possibly related to intraocular pressure. Activation of the MT_1 receptor results in sleepiness, whereas the MT_2 receptor may be related to the light-dark synchronization of the biologic circadian clock. Melatonin has also been implicated in energy metabolism and obesity, and administration of the agent reduces body weight in certain animal models. However, its role in these processes is poorly understood, and there is no evidence that melatonin itself is of any value in obesity in humans. Other studies suggest that melatonin has antiapoptotic effects in experimental models. Recent research implicates melatonin receptors in depressive disorders. Insomnia associated with autism spectrum disorder may respond to melatonin.

Melatonin is promoted commercially as a sleep aid by the food supplement industry (see Chapter 64). There is an extensive literature supporting its use in ameliorating jet lag. It is used in oral doses of 0.5–5 mg, usually administered at the destination bedtime. **Ramelteon** is a selective MT_1 and MT_2 agonist that is approved for the medical treatment of insomnia. This drug has no addiction liability (it is not a controlled substance), and it appears to be distinctly more efficacious than melatonin (but less efficacious than benzodiazepines) as a hypnotic. It is metabolized by P450 enzymes and should not be used in individuals taking CYP1A2 inhibitors. It has a half-life of 1–3 hours and an active metabolite with a half-life of up to 5 hours. Ramelteon may increase prolactin levels. **Tasimelteon** is a newer MT_1 and MT_2 agonist that is approved for non-24-hour sleep-wake disorder. **Agomelatine,** an MT_1 and MT_2 agonist and a 5-HT_{2C} antagonist, is approved in Europe for use in major depressive disorder.

Serotonin Receptor Pharmacology

- Receptors: 5-HT (14 so far)

Sub type	effector	G prot	LOCALIZATION	FUNCTION
$5HT_{1A}$	↓ AC	Gi	Raphe nuclei, cortex, hippocampus	Somatodendritic autoreceptor
$5HT_{1B}$	↓ AC	Gi	Subiculum, globus pallidus, substantia nigra	Presynaptic autoreceptor
$5HT_{1D}$	↓ AC	Gi	Cranial vessels, globus pallidus, substantia nigra	Presynaptic autoreceptor, vasoconstriction
$5HT_{1E}$	↓ AC	Gi	Cortex, striatum	—
$5HT_{1F}$	↓ AC	Gi	Dorsal raphe, hippocampus, periphery	—
$5HT_{2A}$	↑ PLC, PLA_2	Gq	Platelets, smooth muscle, cerebral cortex	Aggregation, contraction, neuronal excitation
$5HT_{2B}$	↑ PLC	Gq	Stomach fundus	Smooth muscle contraction
$5HT_{2C}$	↑ PLC, PLA_2	Gq	Choroid plexus, substantia nigra, basal ganglia	CSF production, neuronal excitation
$5HT_3$	Cations	channel	Parasympathetic nerves, solitary tract, area postrema, GI tract	Neuronal excitation
$5HT_4$	↑ AC	Gs	Hippocampus, striatum, GI tract	Neuronal excitation
$5HT_{5A}$	↓ AC	Gi	Cortex, hippocampus	Unknown
$5HT_{5B}$	Unknow n		—	Pseudogene in humans
$5HT_6$	↑ AC	Gs	Hippocampus, striatum, nucleus accumbens	Neuronal excitation
$5HT_7$	↑ AC	Gs	Hypothalamus, hippocampus, GI tract	Smooth muscle relaxation

This slide introduces the biosynthesis of serotonin and melatonin as well as the various receptors that are activated by serotonin. Most receptors are GPCR receptors except for 5HT3, which is a cation channel.

It is important to note: this later distinction because outliers make for good board questions.

Serotonin Receptor Pharmacology

- Receptors: 5-HT (14 so far)

Sub type	effector	G prot	LOCALIZATION	AGONISTS	ANTAGONIST	USE
$5HT_{1A}$	↓ AC	Gi	Raphe nuclei, cortex, hippocampus	Buspirone		anxiolytic
$5HT_{1B}$	↓ AC	Gi	Subiculum, globus pallidus, substantia nigra	Triptans		
$5HT_{1D}$	↓ AC	Gi	Cranial vessels, globus pallidus, substantia nigra	Triptans		
$5HT_{1E}$	↓ AC	Gi	Cortex, striatum			
$5HT_{1F}$	↓ AC	Gi	Dorsal raphe, hippocampus, periphery	Lasmiditan		
$5HT_{2A}$	↑ PLC, PLA_2	Gq	Platelets, smooth muscle, cerebral cortex		Cyproheptadine (H1), Ketanserin (α1) **Ritanserin**	Carcinoid tumors, htn, Antiplatelet
$5HT_{2B}$	↑ PLC	Gq	Stomach fundus			
$5HT_{2C}$	↑ PLC, PLA_2	Gq	Choroid plexus, substantia nigra, basal ganglia	Lorcaserin (withdrawn)		Appetite suppressant
$5HT_3$	Cations	channel	Parasympathetic nerves, solitary tract, area postrema, GI tract		Ondansetron, tropisetron	Antiemetic
$5HT_4$	↑ AC	Gs	Hippocampus, striatum, GI tract		Cisapride, tegaserod	GI reflux, motility disorders
$5HT_{5A}$	↓ AC	Gi	Cortex, hippocampus			
$5HT_{5B}$	Unknown		—			
$5HT_6$	↑ AC	Gs	Hippocampus, striatum, nucleus accumbens			
$5HT_7$	↑ AC	Gs	Hypothalamus, hippocampus, GI tract			

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Carcinoid tumors (most common GI neuroendocrine tumors). Enterochromaffin cells in origin. The excrete--Serotonin, norepinephrine, dopamine, histamine, many peptide hormones, and prostaglandins. Symptoms include cutaneous flushing, venous telangiectasia, diarrhea, bronchospasm, cardiac valvular lesions.

From Katzung text
SEROTONIN-RECEPTOR ANTAGONISTS
A wide variety of drugs with actions at other receptors (eg, α adrenoceptors, H_1-histamine receptors) also have serotonin receptor-blocking effects.
Phenoxybenzamine (see Chapter 10) has a long-lasting blocking action at 5-HT_2 receptors. In addition, the ergot alkaloids discussed in the last portion of this chapter are partial agonists at serotonin receptors.
Cyproheptadine resembles the phenothiazine antihistaminic agents in chemical structure and has potent H_1-receptor-blocking as well as 5-HT_2-blocking actions. The actions of cyproheptadine are predictable from its H_1 histamine and 5-HT receptor affinities. It prevents the smooth muscle effects of both amines but has no effect on the gastric secretion stimulated by histamine. It also has significant antimuscarinic effects and causes sedation.
The major clinical applications of cyproheptadine are in the treatment of the smooth muscle manifestations of **carcinoid tumor** and in **cold-induced urticaria**. The usual dosage in adults is 12–16 mg/d orally in three or four divided doses. It is of some value in serotonin syndrome, but because it is available only in tablet form,

cyproheptadine must be crushed and administered by stomach tube in unconscious patients. The drug also appears to reduce muscle spasms following spinal cord injury, in which constitutive activity of 5-HT_{2C} receptors is associated with increases in Ca^{2+} currents leading to spasms. Anecdotal evidence suggests some efficacy as an appetite *stimulant* in cancer patients, but controlled trials have yielded mixed results.

Ketanserin blocks 5-HT_2 receptors on smooth muscle and other tissues and has little or no reported antagonist activity at other 5-HT or H_1 receptors. However, this drug potently blocks vascular α_1 adrenoceptors. The drug blocks 5-HT_2 receptors on platelets and antagonizes platelet aggregation promoted by serotonin. The mechanism involved in ketanserin's hypotensive action probably involves α_1 adrenoceptor blockade more than 5-HT_2 receptor blockade. Ketanserin is available in Europe for the treatment of hypertension and vasospastic conditions but has not been approved in the USA. **Ritanserin,** another 5-HT_2 antagonist, has little or no α-blocking action. It has been reported to alter bleeding time and to reduce thromboxane formation, presumably by altering platelet function.

Ondansetron is the prototypical 5-HT_3 antagonist. This drug and its analogs are very important in the prevention of nausea and vomiting associated with surgery and cancer chemotherapy. They are discussed in Chapter 62.

Considering the diverse effects attributed to serotonin and the heterogeneous nature of 5-HT receptors, other selective 5-HT antagonists may prove to be clinically useful.

Mixed Agents

Specific agonist	Receptor subtype	Use
Dexfenfluramine	↑serotonin release, ↓ uptake	Appetite suppressant
SSRIs	Block reuptake	Depression
Flibanserin	5-$HT_{1A\ agonist}$ 5-HT_{2A} antagonist	Treat hypoactive sexual desire disorder in women

From Katzun Text
Serotonin Agonists
Serotonin has no clinical applications as a drug. However, several receptor subtype-selective agonists have proved to be of value. **Buspirone,** a 5-HT_{1A} agonist, has received attention as an effective nonbenzodiazepine anxiolytic (see Chapter 22). Appetite suppression appears to be associated with agonist action at 5-HT_{2C} receptors in the central nervous system, and **dexfenfluramine,** a selective 5-HT agonist, was widely used as an appetite suppressant but was withdrawn because of cardiac valvulopathy. **Lorcaserin,** a 5-HT_{2C} agonist, is approved by the FDA for use as a weight-loss medication (see Box: Treatment of Obesity).

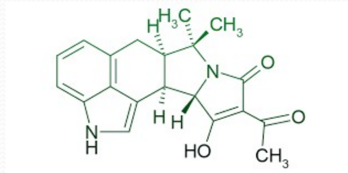
Other Serotonin Agonists in Clinical Use
Flibanserin, a 5-HT_{1a} agonist and 5-HT_{2A} antagonist, is approved for treatment of hypoactive sexual desire disorder in women. Due to inadequate evidence of efficacy, it was refused approval in 2010 and 2013. The clinical trials that led to its approval in 2015 showed a very small but significant increase in satisfactory sexual desire and activities over several weeks of daily oral administration. Consumption of alcohol is contraindicated due to increased risk of severe hypotension. Other adverse effects include syncope, nausea, fatigue, dizziness, and somnolence.

Cisapride, a 5-HT_4 agonist, was used in the treatment of gastroesophageal reflux and motility disorders. Because of toxicity, it is now available only for compassionate use in the USA. **Tegaserod**, a 5-HT_4 partial agonist, is used for irritable bowel syndrome with constipation (see Chapter 62).
Compounds such as **fluoxetine** and other SSRIs, which modulate serotonergic transmission by blocking reuptake of the transmitter, are among the most widely prescribed drugs for the management of depression and similar disorders. These drugs are discussed in Chapter 30.

Ergot Alkaloids

Bromocriptine,
Ergonovine,
Ergotamine,
Methysergide

- Migraine treatment
- Hyperprolactinemia
- Postpartum Hemorrhage
- Diagnosis of variant Angina
- Treatment of Alzheimer's dementia



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History of Ergot Poisoning

- *Claviceps purea*, rye fungus
  - histamine, acetylcholine, tyramine, and ergot alkaloids
- Ergotism (St. Anthony's Fire)
  - demetia with florid hallucinations
  - prolonged vasospasm (gangrene)
  - uterine contraction (abortion)
- Note: imagine societal repercussions from bouts of gangrene, spontaneous abortions, hallucinations and convulsions in the meddle ages

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Ergotism: The Satan Loosed in Salem?
Convulsive ergotism may have been a physiological basis for the Salem witchcraft crisis in 1692.

Linnda R. Caporael

From *Science* Vol. 192 (2 April 1976)

Numerous hypotheses have been devised to explain the occurrence of the Salem witchcraft trials in 1692, yet a sense of bewilderment and doubt pervades most of the historical perspectives on the subject. The physical afflictions of the accusing girls and the imagery of the testimony, therefore, is dismissed as imaginary in foundation. One avenue of understanding that has yet to be sufficiently explored is that a physiological condition, unrecognized at the time, may have been a factor in the Salem incident. Assuming that the content of the court records is basically an honest account of the deponents' experiences, the evidence suggests that convulsive ergotism, a disorder resulting from the ingestion of grain contaminated with ergot, may have initiated the witchcraft delusion.

Suggestions of physical origins of the afflicted girls' behavior have been dismissed without research into the matter. In looking back, the complexity of the psychological and social factors in the community obscured the potential existence of physical pathology, suffered not only by the afflicted children, but also by a number of other community members. The value of such an explanation, however, is clear. Winfield S. Nevins best reveals the implicit uncertainties of contemporary historians (1: 2, p. 235).

. . . I must confess to a measure of doubt as to the moving causes in this terrible tragedy. It seems impossible to believe a tithe of the statements which were made at the trials. And yet it is equally difficult to say that nine out of

every ten of the men, women, and children who testified upon their oaths, intentionally and wilfully falsified. Nor does it seem possible that they did, or could invent all these marvelous tales, fictions rivaling the imaginative genius of Haggard or Jules Verne.
The possibility of a physiological condition fitting the known circumstances and events would provide a comprehensible framework for understanding the witchcraft delusion in Salem.

Background

Prior to the Salem witchcraft trials, only five executions on the charge of witchcraft are known to have occurred in Massachusetts (3, 4). Such trials were held periodically, but the outcomes generally favored the accused. In 1652, a man charged with witchcraft was convicted of simply having told a lie and was fined. Another man, who confessed to talking to the devil, was given counsel and dismissed by the court because of the inconsistencies in his testimony. A bad reputation in the community combined with the accusation of witchcraft did not necessarily insure conviction. The case against John Godfrey of Andover, a notorious character consistently involved in litigation, was dismissed. In fact, soon after the proceedings, Godfrey sued his accusers for defamation and slander and won the case.

The supposed witchcraft at Salem Village was not initially identified as such. In late December 1691, about eight girls, including the niece and daughter of the minister, Samuel Parris, were afflicted with unknown "distempers" (1, 4-6). Their behavior was characterized by disorderly speech, odd postures and gestures, and convulsive fits (7). Physicians called in to examine the girls could find no explanation for their illness, and in February one doctor suggested the girls might be bewitched. Parris seemed loath to accept this explanation at the time and resorted to private fasting and prayer. At a meeting at Parris's home, ministers from neighboring parishes advised him to "sit still and wait upon the Providence of God to see what time might discover" (6, p.25).

A neighbor, however, took it upon herself to direct Parris's Barbados slave, Tituba, in the concocting of a "witch cake" in order to determine it witchcraft was present. Shortly thereafter, the girls made an accusation of witchcraft against Tituba and two elderly women of general ill repute in Salem Village, Sarah Good and Sarah Osborn. The three women were taken into custody on 29 February 1692. The afflictions of the girls did not cease, and in March they accused Martha Corey and Rebecca Nurse. Both of these women were well respected in the village and were covenanting members of the church. Further accusations by the children followed.

Examinations of the accused were conducted in Salem Village until 11 April by two magistrates from Salem Town. At that time, the examination were moved from the outlying farming area to the town and were heard by Deputy Governor Danforth and six of the ablest magistrates in the colony, including Samuel Sewall. This council had no authority to try accused witches, however, because the colony had no legal government--a state of affairs that had existed for 2 years. By the time Sir William Phips, the new governor, arrived from England with the charter establishing the government of Massachusetts Bay Colony, the jails as far away from Salem as Boston were crowded with prisoners from Salem awaiting trial. Phips appointed a special Court of Oyer and Terminer, which heard it first case on 2 June. The proceedings resulted in conviction, and the first condemned witch was hanged on 10 June.

Before the next sitting of the court, clergymen in the Boston area were consulted for their opinion on the issues pending. In an answer composed by Cotton Mather, the ministers advised "critical and exquisite caution" and wished "that there may be as little as possible of such noise, company and openness as may too hastily expose them that are examined" (2, p.83). The ministers also concluded that spectral evidence (the appearance of the accused's apparition to an accuser) and the test of touch (the sudden cessation of a fit after being touched by the accused witch) were insufficient evidence for proof of witchcraft.

The court seemed insensitive to the advice of the ministers, and the trials and executions in Salem continued. By 22 September, 19 men and women had been sent to the gallows, and one, Giles Corey, had been pressed to death, an ordeal calculated to force him to enter a plea to the court so that he could be tried. The evidence used to obtain the convictions was the test of touch and spectral evidence. The afflicted girls were present at the examinations and trials, often creating such pandemonium that the proceedings were interrupted. The accused witches were, for the most part, persons of good reputation in the community; one was even a former minister in the village. Several notable individuals were "cried out" upon, including John Alden and Lady Phips. All the men and women who were hanged had consistently maintained their innocence; not one confessor to the crime was executed. It had become obvious early in the course of the proceedings that those who confessed would not be executed.

On 17 September 1692, the Court of Oyer and Terminer adjourned the witchcraft trials until 2 November; however, it never met again to try that crime. In January 1693 the Superior Court of Judicature, consisting of the magistrates on the Court of Oyer and Terminer, met. Of 50 indictments handed in to the Superior Court by the grand jury, 20 persons were brought to trial. Three were condemned but never executed and the rest were acquitted. In May Governor Phips ordered a general reprieve, and about 150 accused witches were released. The end of the witchcraft crisis was singularly abrupt (2, 4, 8).

Tituba and the Origin Tradition

Repeated attempts to place the occurrences at Salem within a consistent framework have failed. Outright fraud, political factionalism, Freudian psychodynamics, sensation seeking, clinical hysteria, even the existence of witchcraft itself, have been proposed as explanatory devices. The problem is primarily one of complexity. No single explanation can ever account for the delusion; an interaction of them all must be assumed. Combinations of interpretations, however, seem insufficient without some reasonable justification for the initially afflicted girls' behavior. No mental derangement or fraud seems adequate in understanding how eight girls, raised in the soul-searching Puritan tradition, simultaneously exhibited the same symptoms or conspired together for widespread notoriety.

All modern accounts of the beginnings of Salem witchcraft begin with Parris's Barbados slave, Tituba. The tradition is that she instructed the minister's daughter and niece, as well as some other girls in the neighborhood, in magic tricks and incantations at secret meetings held in the patronage kitchen (2, 4, 8, 9). The odd behavior of the girls, whether real or fraudulent, was a consequence of these experiments.

The basis for the tradition seems two-fold. In a warning against divination, John Hale wrote in 1702 that he was informed that one afflicted girl had tried to see the future with an egg and glass and subsequently was followed by a "diabolical molestation" and died (6). The egg and glass (an improvised crystal ball) was an English method of divination. Hale gives no indication that Tituba was involved, or for that matter, that a group of girls was involved. I have been unable to locate any reference that any of the afflicted girls died prior to Hale's publications.

The other basis for the tradition implicating Tituba seems to be simply the fact that she was from the West Indies. The Puritans believed the American Indians worshiped the devil, most often described as a black man (4). Curiously, however, Tituba was not questioned at her examination about activities as a witch in her birthplace. Historians seem bewitched themselves by fantasies of voodoo and black magic in the tropics, and the unfounded supposition that Tituba would inevitably be familiar with malefic arts of the Caribbean has survived.

Calef (7) reports that Tituba's confession was obtained under duress. She at first denied knowing the devil and suggested the girls were possessed. Although Tituba ultimately became quite voluble, her confession was rather pedestrian in comparison with the other testimony offered at the examination and trials. There is no element of West Indian magic, and her descriptions of the black man, the hairy imp, and witches flying through the sky on sticks reflect an elementary acquaintance with the common English superstitions of the time (9-11).

Current Interpretations

1) *Fraud*. Various interpretations of the girls' behavior diverge after the discussion of its origins. The currently accepted view is that the children's symptoms of affliction were fraudulent (4, 8, 12). The girls may have perpetrated fraud simply to gain notoriety or to protect themselves from punishment by adults as their magic experiments became the topic of rumor (2). One author supposed that the accusing girls craved "Dionysiac mysteries" and that some were "no more seriously possessed that a pack of bobby-soxers on the loose" (8, p.29). The major difficulty in accepting the explanation of purposeful fraud is the gravity of the girls' symptoms; all the eyewitness accounts agree to the severity of the affliction (6, 10, 11, 13).

Upham (4) appears to accept the contemporaneous descriptions and ascribes to the afflicted children the skills of a sophisticated necromancer. He proposes that they were able ventriloquists, highly accomplished actresses, and by "long practice" could "bring the blood to the face, and send it back again" (4, vol.2, p.395). These abilities and more, he assumes, the girls learned from Tituba. As discussed above, however, there is little evidence that Tituba had any practical knowledge of witchcraft. Most colonists, with the exception of some of the accused and their defenders, did not appear even to consider pretense as an explanation for the girls' behavior. The general conclusion of the New Englanders after the tragedy was that the girls suffered from demonic possession (2, 6, 9).

2) *Hysteria*. The advent of psychiatry provided new tools for describing and interpreting the events oat Salem. The term hysteria has been used with varying degrees of license (2, 8, 9, 14), and the accounts of hysteria always begin in the kitchen with Tituba practicing magic. Starkey (8) uses the term in the loosest sense: the girls were hysterical, that is, overexcited, and committed sensational fraud in a community that subsequently fell ill to "mass

hysteria." Hansen (9) proposes the use of the word in a stricter, clinical sense of being mentally ill. He insists that witchcraft really was practiced in Salem and that several of the executed were practicing witches. The girls' symptoms were psychogenic, occasioned by guilt at practicing fortune-telling at their secret meetings. He states that the mental illness was catching and that the witnesses and majority of the confessors became hysterics as a consequence of their fear of witchcraft. However, if the girls were not practicing divination, and if they did indeed develop true hysteria, then they must all have developed hysteria simultaneously -- hardly a credible supposition. Furthermore, previous witchcraft accusations in other Puritan communities in New England had never brought on mass hysteria.

Psychiatric disorder is used un a slightly different sense in the argument that the witchcraft crisis was a consequence of two party (pro-Parris and anti-Parris) factionalism in Salem Village (14). In this account, the girls are unimportant factors in the entire incident. Their behavior "served as a kind of Rorschach test into which adults read their own concerns and expectations" (14, p.30). The difficulty with linking factionalism to the witch trials is that supporters of Parris were also prosecuted while some non-supporters were among the most vociferous accusers (2, 14). Thus, it becomes necessary to resort to projection, transference, individual psychoanalysis, and numerous psychiatric disorders to explain the behavior of the adults in the community who were using the afflicted children as pawns to resolve their own personal and political differences.

Of course, there was fraud and mental illness at Salem. The records clearly indicate both. Some depositions are simply fanciful renditions of local gossip or cases of malice aforethought. There is also testimony based on exaggerations of nightmares and inebriated adventures. However, not all the records are thus accountable.

3) *Physiological explanations*. The possibility that the girls' behavior had a physiological basis has rarely arisen, although the villagers themselves first proposed physical illness as an explanation. Before the accusations of witchcraft began, Parris called in a number of physicians (6, 7). In an early history of the colony, Thomas Hutchinson wrote that "there are a great number of persons who are willing to suppose the accusers to have been under bodily disorders which affected their imagination" (12, vol. 2, p.47). A modern historian reports a journalist's suggestion that Tituba had been dosing the girls with preparations of jimsonweed, a poisonous plant brought to New England from the West Indies in the early 1600's (8, footnote on p.284). However, because the Puritans identified no physiological cause, later historians have failed to investigate such a possibility.

Ergot

Interest in ergot (*Claviceps purpura*) was generated by epidemics or ergotism that periodically occurred in Europe. Only a few years before the Salem witchcraft trials the first medical scientific report on ergot was made (15). Denis Dodart reported the relation between ergotized rye and bread poisoning in a letter to the French Royal Academie des Sciences in 1676. John Ray's mention of ergot in 1677 was the first in English. There is no reference to ergot in the United States before an 1807 letter by Dr. John Stearns recommending powdered ergot sclerotia to a medical colleague as a therapeutic agent in childbirth. Stearns is generally credited with the "discovery" of ergot; certainly his use prompted scientific research on the substance. Until the mid-19th century, however, ergot was not known as a parasitic fungus, but was thought to be sunbaked kernels of grains (15-17).

Ergot grows on a large variety of cereal grains--especially rye--in a slightly curved, fusiform shape with sclerotia replacing individual grains on the host plant. The sclerotia contain a large number of potent pharmacologic agents, the ergot alkaloids. One of the most powerful is isoergine (lysergic acid amide). This alkaloid, with 10 percent of the activity of a D-LSD (lysergic acid diethylamide), is also found in ololiuqui (morning glory seeds), the ritual hallucinogenic drugs used by the Aztecs (15, 16).

Warm, damp, rainy springs and summers favor ergot infestations. Summer rye is more prone to the development of the sclerotia than winter rye, and one field may be heavily ergotized while the adjacent field is not. The fungus may dangerously parasitize a crop one year and not reappear again for many years. Contamination of the grain may occur in varying concentrations. Modern agriculturalists advise farmers not to feed their cattle grain containing more than one to three sclerotia per thousand kernels of grain, since ergot has deleterious effects on cattle as well as on humans (16, 18).

Ergotism, or long-term ergot poisoning, was once a common condition resulting from eating contaminated rye bred. In some epidemics it appears that females were more liable to the disease than males (19). Children and pregnant women are most likely to be affected by the condition, and individual susceptibility varies widely. It takes 2 years for ergot in powdered form to reach 50 percent deterioration, and the effects are cumulative (18, 20). There are two types of ergotism--gangrenous and convulsive. As the name implies, gangrenous ergotism is characterized by dry gangrene of the extremities followed by the falling away of the affected portions of the body. The condition occurred in epidemic proportions in the Middle Ages and was known by a number of names, including *ignis sacer*, the holy fire.

Convulsive ergotism is characterized by a number of symptoms. These include crawling sensations in the skin, tingling in the fingers, vertigo, tinnitus aurium, headaches, disturbances in sensation, hallucination, painful muscular contractions leading to epileptiform convulsions, vomiting, and diarrhea (16, 18, 21). The involuntary muscular fibers such as the myocardium and gastric and intestinal muscular coat are stimulated. There are mental disturbances such as mania, melancholia, psychosis, and delirium. All of these symptoms are alluded to in the Salem witchcraft records.

Evidence for Ergotism in Salem

It is one thing to suggest convulsive ergot poisoning as an initiating factor in the witchcraft episode, and quite another to generate convincing evidence that it is more that a mere possibility. A jigsaw of details pertinent to growing conditions, the timing of events in Salem, and symptomology must fit together to create a reasonable case. From these details, a picture emerges of a community stricken with an unrecognized physiological disorder affecting their minds as well as their bodies.

1) *Growing conditions*. The common grass along the Atlantic Coast from Virginia to Newfoundland was and is wild rye, a host plant for ergot. Early colonists were dissatisfied with it as forage for their cattle and reported that it often made the cattle ill with unknown diseases (22). Presumably, then, ergot grew in the New World before the Puritans arrived. The potential source for infection was already present, regardless of the possibility that it was imported with the English rye.

Rye was the most reliable of the Old World grains (22) and by the 1640's ot was a well-established New England crop. Spring sowing was the rule; the bitter winters made fall sowing less successful. Seed time for the rye was April and the harvesting took place in August (23). However, the grain was stored in barns and often waited months before being threshed when the weather turned cold. The timing of Salem events fits this cycle. Threshing probably occurred shortly before Thanksgiving, the only holiday the Puritans observed. The children's symptoms appeared in December 1691. Late the next fall, 1692, the witchcraft crisis ended abruptly and there is no further mention of the girls or anyone else in Salem being afflicted (4, 9).

To some degree or another all rye was probably infected with ergot. It is a matter of the extent of the infection and the period of time over which the ergot is consumed rather than the mere existence of ergot that determines the potential for ergotism. In his 1807 letter written from upstate New York, Stearns (15, p. 274) advised his medical colleague that, "On examining a granary where rye is stored, you will be able to procure a sufficient quantity [of ergot sclerotia] from among that grain." Agricultural practice had not advanced, even by Stearns's time, to widespread use of methods to clean or eliminate the fungus from the rye crop. In all probability, the infestation of the 1691 summer rye crop was fairly light; not everyone in the village or even in the same families showed symptoms.

Certain climatic conditions, that is, warm, rainy springs and summers, promote heavier than usual fungus infestation. The pattern of the weather in 1691 and 1692 is apparent from brief comments in Samuel Sewall's diary (24). Early rains and warm weather in the spring progressed to a hot and stormy summer in 1691. There was a drought the next year, 1692, thus no contamination of the grain that year would be expected.

2) *Localization*. "Rye," continues Stearns (15, p.274), "which grows in low, wet ground yields [ergot] in greatest abundance." Now, one of the most notorious of the accusing children in Salem was Thomas Putnam's 12-year-old daughter, Ann. Her mother also displayed symptoms of the affliction and psychological historians have credited the senior Ann with attempting to resolve her own neurotic complaints through her daughter (8, 9, 14). Two other afflicted girls also lived in the Putnam residence. Putnam had inherited one of the largest landholdings in the village. His father's will indicates that a large measure of the land, which was located in the western sector of Salem Village, consisted of swampy meadows (25) that were valued farmland to the colonists (22). Accordingly, the western acreage of Salem Village, may have been an area of contamination. This contention is further substantiated by the pattern of residence of the accusers, the accused, and the defenders of the accused living within the boundaries of Salem Village (Fig. 1). Excluding the afflicted girls, 30 of 32 adult accusers lived in the western section and 12 of the 14 accused witches lived in the eastern section, as did 24 of the 29 defenders (14). The general pattern of residence, in combination with the well-documented factionalism of the eastern and western sectors, contributed to the progress of the witchcraft crisis.

The initially afflicted girls show a slightly different residence pattern. Careful examination reveals plausible explanations for contamination in six of the eight cases.

Three of the girls, as mentioned above, lived in the Putnam residence. If this were the source of ergotism, their exposure to ergotized grain would be natural. Two afflicted girls, the daughter and niece of Samuel Parris, lived in the parsonage almost exactly in the center of the village. Their exposure to contaminated grain from western land is also explicable. Two-thirds of Parris's salary was paid in provisions; the villagers were taxed proportionately to their landholding (4). Since Putnam was one of the largest landholders and an avid supporter of Parris in the minister's community disagreements, an ample store of ergotized grain would be anticipated in Parris's larder. Putnam was also Parris's closest neighbor with afflicted children in residence.

The three remaining afflicted girls lived outside the village boundaries to the east. One, Elizabeth Hubbard, was a servant in the home of Dr. Griggs. It seems plausible that the doctor, like Parris, had Putnam grain, since Griggs was a professional man, not a farmer. As the only doctor in town, he probably had many occasions to treat Ann Putnam Sr., a woman known to have much ill health (2, 4). Griggs may have traded his services for provisions or bought food from the Putnams.

Another of the afflicted, Sarah Churchill, was a servant in the house of a well-off farmer (25). The farm lay along that Wooleston River and may have offered good growing conditions for ergot. It seems probable, however, that Sarah's affliction was a fraud. She did not become involved in the witchcraft persecutions until May, several months after the other girls were afflicted, and she testified in only two cases, the first against her master. One deponent claimed that Sarah later admitted to belying herself and others (11).

How Mary Warren, a servant in the Proctor household, would gain access to grain contaminated with ergot is something of a mystery. Proctor had a substantial farm to the southeast of Salem and would have had no need to buy or trade for food. Both he and his wife were accused of witchcraft and condemned. None of the Proctor children showed any sign of the affliction: in fact, three were accused and imprisoned. One document offered as evidence against Proctor indicated that Mary stayed overnight in the village (11). How often she stayed or with whom is unknown.

Mary's role in the trials is particularly curious. She began as an afflicted person, was accused of witchcraft by the other afflicted girls, and then became afflicted again. Two depositions filed against her strongly suggest, however, that at least her first affliction may have been a consequence of ergot poisoning. Four witnesses attested that she believed she had been "distempered" and during the time of her affliction had thought she had seen numerous apparitions. However, when Mary was well again, she could not say that she had seen any specters (11). Her second affliction may have been the result of intense pressure during her examination for witchcraft crimes.

Ergotism and the Testimony

The utmost caution is necessary in assessing the physical and mental states of people dead for hundreds of years. Only the sketchiest accounts of their lives remain in public records. In the case of ergot, a substance that affects mental as well as physical states, recognition of the social atmosphere of Salem in early spring 1692 is basic to understanding the directions the crisis took. The Puritans' belief in witchcraft was a totally accepted part of their religious tenets. The malicious workings of Satan and his cohorts were just as real to the early colonists as their belief in God. Yet, the low incidence of witchcraft trials in New England prior to 1692 suggests that the Puritans did not always resort to accusations of black magic to deal with irreconcilable differences or inexplicable events.

The afflicted girls' behavior seemed to be no secret in early spring. Apparently it was the great consternation that some villagers felt that induced Mary Sibley to direct the making of the witch cake of rye meal and the urine of the afflicted. This concoction was fed to a dog, ostensibly in the belief that the dog's subsequent behavior would indicate the action of any malefic magic (14). The fate of the dog is unknown; it is quite plausible that it did have convulsions, indicating to the observers that there was witchcraft involved in the girls' afflictions. Thus, the experiments with the witch cake, rather than any magic tricks of Tituba, initiated succeeding events.

The importance of the witch cake incident has generally been overlooked. Parris's denouncement of his neighbor's action is recorded in his church records. He clearly stated that, until the making of the cake, there was no suspicion of witchcraft and no reports of torturing apparitions (4). Once a community member had gone "to the Devil for help against the Devil," as Parris put it, the climate for the trials had been established. The afflicted girls, who had made no previous mention of witchcraft, seized upon a cause for their behavior--as did the rest of the community. The girls named three persons as witches and their afflictions thereby became a matter for the legal authorities rather than the medical authorities or the families of the girls.

The trial records indicate numerous interruptions during the proceedings. Outbursts by the afflicted girls describing the activities of invisible specters and "familiars" (agents of the devil in animal form) in the meeting house were common. The girls were often stricken with violent fits that were attributed to torture by apparitions. The spectral evidence of the trials appears to be the hallucinogenic symptoms and perceptual disturbance accompanying ergotism. The convulsions appear to be epileptiform (6, 13).

Accusations of choking, pinching, pricking with pins, and biting by the specter of the accused formed the standard testimony of the afflicted in almost all the examinations and trials (26). the choking suggests the involvement of the involuntary muscular fibers that is typical of ergot poisoning; the biting, pinching, and pricking may allude to the crawling and tingling sensations under the skin experienced by ergotism victims. Complaints of vomiting and "bowels almost pulled out" are common in the depositions of the accusers. The physical symptoms of the afflicted and any of the other accusers are those induced by convulsive ergot poisoning.

When examined in the light of a physiological hypothesis, the content of so-called delusional testimony, previously dismissed as imaginary by historians, can be reinterpreted as evidence of ergotism. After being choked and strangled by the apparition of a witch sitting on his chest, John Londer testified that a black thing came through the window and stood before his face. "The body of it looked like a monkey, only the feet were like cock's feet, with claws, and the face somewhat more like a man's than a monkey . . . the thing spoke to me . . . " (25, p.45).

Joseph Bayley lived out of town in Newbury. According to Upham (4), the Bayleys, en route to Boston, probably spent the night at the Thomas Putnam residence. As the Bayleys left the village, they passed the Proctor house and Joseph reported receiving a "very hard blow" on the chest, but no one was near him. He saw the Proctors, who were imprisoned in Boston at the time, but his wife told him that she saw only a "little maid." He received another blow on the chest, so strong that he dismounted from his horse and subsequently saw a woman coming toward him. His wife told him she saw nothing. When he mounted his horse again, he saw only a cow where he had seen the woman. The rest of Bayley's trip was uneventful, but when he returned home, he was "pinched and nipped by something invisible for some time" (11). It is a moot point, of course, what or how much Bayley ate at the Putnams', or that he even really stayed there. Nevertheless, the testimony suggests ergot. Bayley had the crawling sensations in the skin, disturbances in sensations, and muscular contractions symptomatic of ergotism. Apparently his wife had none of the symptoms and Bayley was quite candid in so reporting.

A brief but tantalizing bit of testimony comes from a man who experienced visions that he attributed to the evil eye cast on him by an accused witch. He reported seeing about a dozen "strange things" appear in his chimney in a dark room. They appeared to be something like jelly and quavered with a strange motion. Shortly, they disappeared and a light the size of a hand appeared in the chimney and quivered and shook with an upward motion (27). As in Bayley's experience, this man's wife saw nothing. The testimony is strongly reminiscent of the undulating objects and lights reported in experiences induced by LSD (28).

By the time the witchcraft episode ended in the late fall 1692, 20 persons had been executed and at least two had died in prison. All the convictions were obtained on the basis of the controversial spectral evidence (2). One of the commonly expressed observations about the Salem Village witchcraft episode is that it ended unexpectedly for no apparent reason (2, 4). No new circumstances to cast spectral evidence in doubt occurred. Increase Mather's sermon on 3 October 1692, which urged more conclusive evidence than invisible apparitions or the test of touch, was just a stronger reiteration of the clergy's 15 June advice to the court (2). The grounds fro dismissing the spectral evidence had been consistently brought up by the accused and many of their defenders throughout the examinations. There had always been a strong undercurrent of opposition to the trials and the most vocal individuals were not always accused. In fact, there was virtually no support in the colonies for the trials, even from Boston, only 15 miles away. The most influential clergymen lent their support guardedly at best; most were opposed. The Salem witchcraft episode was an event localized in both time and space.

How far the ergotized grain may have been distributed is impossible to determine clearly. Salem Village was the source of Salem Town's food supply. It was in the town that the convictions and orders for executions were obtained. Maybe the thought processes of the magistrates, responsible and respected men in the Colony, were altered. In the following years, nearly all of them publicly admitted to errors of judgment (2). These posttrial documents are as suggestive as the court proceedings.

In 1696, Samuel Sewall made a public acknowledgment of personal guilt because of the unsafe principles the court followed (2). In a public apology, the 12 jurymen stated (9, p.210), "We confess that we ourselves were not capable to understand nor able to withstand the mysterious delusion of the Powers of Darkness and Prince of the Air . . . [we] do hereby declare that we justly fear that we were sadly deluded and mistaken . . . " John Hale, a minister involved in the trials from the beginning, wrote (6, p.167), "such was the darkness of the day . . . that we walked in the clouds and could not see our way."

Finally, Ann Putnam, Jr., who testified in 21 cases, made a public confession in 1706 (2, p.250).

I justly fear I have been instrumental with others though ignorantly and unwittingly, to bring upon myself and this land the guilt of innocent blood; though what was said or done by me against any person I can truly and uprightly say before God and man. I did it not for any anger, malice or ill will to any person, for I ahd no such things against one of them, but what I did was ignorantly, being deluded of Satan.
One Satan in Salem may well have been convulsive ergotism.

Conclusion

One could reasonably ask whether, if ergot was implicated in Salem, it could have been implicated in other witchcraft incidents. The most cursory examination of the Old World witchcraft suggests an affirmative answer. The district of Lorraine suffered outbreaks of both ergotism (15) and witchcraft persecutions (4) periodically throughout the Middle Ages until the 17th century. As late as the 1700's, the clergy of Saxony debated whether convulsive ergotism was symptomatic of disease or demonic possession (17). Kittredge (3), an authority on English witchcraft, reports what he calls "a typical case" of the early 1600's. The malicious magic of Alice Trevisard, an accused witch, backfired and the witness reported that Alice's hands, fingers, and toes "rotted and consumed away." The sickness sounds suspiciously like gangrenous ergotism. Years later, in 1762, one family in a small English village was stricken with gangrenous ergotism. The Royal Society determined the diagnosis. The head of the family, however, attributed the condition to witchcraft because of the suddenness of the calamity (29).

Of course, there can never be hard proof for the presence of ergot in Salem, but a circumstantial case is demonstrable. The growing conditions and the pattern of agricultural practices fit the timing of the 1692 crisis. The physical manifestations of the condition are apparent from the trial records and contemporaneous documents. While the fact of perceptual distortions may have been generated by ergotism, other psychological and sociological factors are not thereby rendered irrelevant; rather, these factors gave substance and meaning to the symptoms. The content of hallucinations and other perceptual disturbance would have been greatly influenced by the state of mind, mood, and expectations of the individual (30). Prior to the witch cake episode, there is no clue as to the nature of the girls' hallucinations. Afterward, however, a delusional system, based on witchcraft, was generated to explain the content of the sensory data (31, p.137). Valins and Nisbett (31, p.141), in a discussion of delusional explanations of abnormal sensory data, write, "The intelligence of the particular patient determines the structural coherence and internal consistency of the explanation. The cultural experiences of the patient determine the content--political, religious, or scientific--of the explanation." Without knowledge of ergotism and confronted by convulsions, mental disturbance, and perceptual distortions, the New England Puritans seized upon witchcraft as the best explanation for the phenomena.

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20. C. M. Gruber. *The Cyclopedia of Medicine, Surgery, Specialties* (Davis, Philadelphia, 1950), vol. 5, pp.245-248.
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32. I thank C. F. Paul and M. B. Brewer for their helpful comments on the manuscript

Ergot Alkaloids: Pharmacodynamics

Ergot Alkaloid	a-R	D-R	$5\text{-}HT_2$	Uterine M Stimulation
Bromocriptine Cabergoline pergolide	-	+++	-	o
Ergonovine	++	PA	+++	
Ergotamine	PA	o	PA	+++
Lysergic acid diethylamide (LSD)	o	++	-- (peripheral) ++(central)	+
Methysergide	+/o	+/o	PA	+/o

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Ergot Alkaloids: Clinical Pharmacology

Ergot Alkaloid	Uterine M Stimulation	Toxicity
Bromocriptine Cabergoline pergolide	Amenorrhea, infertility in woman and galactorrhea in both sexes due to hyperprolactinemia (D2)	GI (diarrhea (5HT), nausea, vomiting (D2)), drowsiness
Ergonovine	Postpartum hemorrhage (alternative to oxytocin), Diagnosis of variant angina	GI (above), Prolonged vasospasm
Ergotamine Dihydroergotamine	Acute migraine headache	Same as above
Methysergide	Migraine prophylaxis (withdrawn in US)	GI, connective tissue proliferation (retroperitoneal space, pleural cavity, endocardial, ureter), hallucinations, leg cramps, Hair loss

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From Katzung's text

BASIC PHARMACOLOGY OF ERGOT ALKALOIDS

Chemistry & Pharmacokinetics

Two major families of compounds that incorporate the tetracyclic **ergoline** nucleus may be identified; the amine alkaloids and the peptide alkaloids (Table 16–7). Drugs of therapeutic and toxicologic importance are found in both groups.

TABLE 16–7[1]Major ergoline derivatives (ergot alkaloids).

View Table|

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The ergot alkaloids are variably absorbed from the gastrointestinal tract. The oral dose of ergotamine is about 10 times larger than the intramuscular dose, but the speed of absorption and peak blood levels after oral administration can be improved by administration with caffeine (see below). The amine alkaloids are also absorbed from the rectum and the buccal cavity and after administration by aerosol inhaler. Absorption after intramuscular injection is slow but usually reliable. Semisynthetic analogs such as bromocriptine and cabergoline are well absorbed from the gastrointestinal tract.

The ergot alkaloids are extensively metabolized in the body. The primary metabolites are hydroxylated in the A ring, and peptide alkaloids are also modified in the peptide moiety.

Pharmacodynamics

A. Mechanism of Action

The ergot alkaloids act on several types of receptors. As shown by the color outlines in Table 16–7, the nuclei of both catecholamines (phenylethylamine, left panel) and 5-HT (indolethylamine, right panel) can be discerned in the ergoline nucleus. Their effects include agonist, partial agonist, and antagonist actions at α adrenoceptors and serotonin receptors (especially 5-HT_{1A} and 5-HT_{1D}; less for 5-HT_2 and 5-HT_3); and agonist or partial agonist actions at central nervous system dopamine receptors (Table 16–8). Furthermore, some members of the ergot family have a high affinity for presynaptic receptors, whereas others are more selective for postjunctional receptors. There is a powerful stimulant effect on the uterus that seems to be most closely associated with agonist or partial agonist effects at 5-HT_2 receptors. Structural variations increase the selectivity of certain members of the family for specific receptor types.

TABLE 16–8[1]Effects of ergot alkaloids at several receptors.[1]

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B. Organ System Effects

1. Central nervous system—As indicated by traditional descriptions of ergotism, certain of the naturally occurring alkaloids are powerful hallucinogens. **Lysergic acid diethylamide** (**LSD**; “acid”) is a synthetic ergot compound that clearly demonstrates this action. The drug has been used in the laboratory as a potent peripheral 5-HT_2 *antagonist*, but good evidence suggests that its behavioral effects are mediated by *agonist* effects at prejunctional or postjunctional 5-HT_2 receptors in the central nervous system. In spite of extensive research, no clinical value has been discovered for LSD’s dramatic central nervous system effects. Abuse of this drug has waxed and waned but is still widespread. It is discussed in Chapter 32.

Dopamine receptors in the central nervous system play important roles in extrapyramidal motor control and the regulation of pituitary prolactin release. The actions of the peptide ergoline **bromocriptine** on the extrapyramidal system are discussed in Chapter 28. Of all the currently available ergot derivatives, bromocriptine, **cabergoline**, and **pergolide** have the highest selectivity for the pituitary dopamine receptors. These drugs directly suppress prolactin secretion from pituitary cells by activating regulatory dopamine receptors (see Chapter 37). They compete for binding to these sites with dopamine itself and with other dopamine agonists such as apomorphine. They bind with high affinity and dissociate slowly.

2. Vascular smooth muscle—The actions of ergot alkaloids on vascular smooth muscle are drug-, species-, and vessel-dependent, so few generalizations are possible. In humans, **ergotamine** and similar compounds constrict most vessels in nanomolar concentrations (Figure 16–4). The vasospasm is prolonged. This response is partially blocked by conventional α-blocking agents. However, ergotamine’s effect is also associated with “epinephrine reversal” (see Chapter 10) and with *blockade* of the response to other α agonists. This dual effect reflects the drug’s partial agonist action (Table 16–7). Because ergotamine dissociates very slowly from the α receptor, it produces very long-lasting agonist and antagonist effects at this receptor. There is little or no effect at β adrenoceptors.

Although much of the vasoconstriction elicited by ergot alkaloids can be ascribed to partial agonist effects at α adrenoceptors, some may be the result of effects at 5-HT receptors. Ergotamine, ergonovine, and methysergide all have partial agonist effects at 5-HT_2 vascular receptors. The remarkably selective antimigraine action of the ergot derivatives was originally thought to be related to their actions on vascular serotonin receptors. Current hypotheses, however, emphasize their action on prejunctional neuronal 5-HT receptors.

After overdosage with ergotamine and similar agents, vasospasm is severe and prolonged (see Toxicity, below). This vasospasm is not easily reversed by α antagonists, serotonin antagonists, or combinations of both.

Ergotamine is typical of the ergot alkaloids that have a strong vasoconstrictor action. The hydrogenation of ergot alkaloids at the 9 and 10 positions (Table 16–6) yields dihydro derivatives that have reduced serotonin partial agonist and vasoconstrictor effects and increased selective α-receptor-blocking actions.

3. Uterine smooth muscle—The stimulant action of ergot alkaloids on the uterus, as on vascular smooth muscle, appears to combine α agonist, serotonin agonist, and other effects. Furthermore, the sensitivity of the uterus to the stimulant effects of ergot increases dramatically during pregnancy, perhaps because of increasing dominance of α_1 receptors of the uterus as pregnancy progresses. As a result, the uterus at term is more sensitive to ergot than earlier in pregnancy and far more sensitive than the nonpregnant organ.

In very small doses, ergot preparations can evoke rhythmic contraction and relaxation of the uterus. At higher concentrations, these drugs induce powerful and prolonged contracture. **Ergonovine** is more selective than other ergot alkaloids in affecting the uterus and is an agent of choice in obstetric applications of the ergot drugs although oxytocin, the peptide hormone, is preferred in most cases.

4. Other smooth muscle organs—In most patients, the ergot alkaloids have little or no effect on bronchiolar or urinary smooth muscle. The gastrointestinal tract, on the other hand, is quite sensitive. In some patients, nausea, vomiting, and diarrhea may be induced even by low doses. The effect is consistent with action on the central nervous system emetic center and on gastrointestinal serotonin receptors.

Thank You

- Good Luck